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Life Cycle Inventories of Electric and Electronic Equipment: Production, Use and Disposal

Data v2.0 (2007)



Roland Hischier, Mischa Classen, Martin Lehmann,
Wolfram Scharnhorst
EMPA, St. Gallen / Dübendorf

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Commissioners:	Swiss Centre for Life Cycle Inventories, Dübendorf Swiss Federal Office for the Environment (BAFU - FOEN), Bern Swiss Federal Office for Energy (BFE), Bern Swiss Federal Office for Agriculture (BLW), Bern
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Software Support:	ifu Hamburg GmbH
Project leader:	Rolf Frischknecht, ecoinvent Centre, Empa, Dübendorf
Marketing and Sales:	Annette Köhler, ecoinvent Centre, Empa, Dübendorf

Life Cycle Inventories of Electric and Electronic Equipment

Project leader: Roland Hischier, Empa

Authors: Mischa Classen, Empa
Roland Hischier, Empa
Martin Lehmann, Empa
Wolfram Scharnhorst, Empa

Review: Rolf Frischknecht, ESU-services Ltd
Jürgen Sutter, ETHZ

Contact address: Empa
P.O. Box
CH-9014 St. Gallen
<http://www.ecoinvent.org/>
empa.sg@ecoinvent.org

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Summary

This report describes datasets in the electronics sector – starting from single components (part I) up to IT services (part IV) along the complete value chain, including the dismantling and disposal/recycling of the devices (part V). On the level of services, the data are restricted to the use of desktop computer, laptop computer and printer – the preceding chapters contain then the devices (part III), modules (part II) resp. components (part I) that are required to supply the IT services..

Components (part I): Datasets for Integrated Circuits (ICs), transistors, diodes, capacitors, inductors, resistors, transformers, connectors and switches are reported. Apart from a few exceptions, all components are modelled on a per kg basis. Nevertheless, this report (including the meta information) refers to the used weight per piece in order to allow the use of these data also in those cases where the number of components is known (instead of the weight). The datasets of the subcategory “components” do not contain the disposal part, as the components are usually disposed of as integrated parts of a complete device. However, the infrastructure is included in all the component datasets.

Modules (part II): Datasets for printed wiring boards (PWB), cables, batteries, toner modules, drives (Harddisk and DVD/CD-ROM) and the power supply unit (PSU) are reported. Thereof, PWB are considered as the most important modules. They are modelled either as bare or unmounted boards (on a per m² basis) or as various types of mounted PWB (on a per kg basis). Pb-free soldering, Pb-containing soldering as well as an unspecified (mix of Pb-free and Pb-containing soldering) datasets are available for each single PWB inventoried within ecoinvent. Cables are modelled per length and unit of plugs. Again, the weight per length is reported in order to apply the data in all those cases where the cable weight instead of the length is known. Batteries, Toner modules, Drives and PSU are either reported on a per kg basis or on a per unit basis – depending on the practicability of either modelling strategy. Similar to the components, no disposal is included, as modules are usually disposed of as parts of complete devices.

Devices (part III): Datasets for a desktop computer, a laptop computer, a cathode ray tube (CRT) screen, a liquid crystal display (LCD) screen, and a laser printer are reported. All these devices are modelled per unit – i.e. one laptop computer. Unlike most ecoinvent datasets, each of these datasets contains its individual, state-of-the-art disposal process – allowing the user the inclusion of the complete life cycle (without the use phase) by just one single process.

IT-Services (part IV): The printing activity is modelled in two ways – per hour of printing and per kg of printed paper. For the use of desktop and laptop computer – modelled per hour of use –, datasets distinguishing between home and office use are available. It is highly recommended to use the datasets “... , office use” resp. “... , home use” in order to completely take into account the production of the devices with the three distinguished use modes active, standby/sleep and off.

Disposal (part V): The dismantling, recycling and disposal activities of the two Swiss WEEE recycling systems of SWICO and SENS are used as starting point. All reported datasets “disposal,, to WEEE treatment” represent the complete chain of the respective treatment activities. As far as possible, it is recommended to use these datasets. All recycling processes are established as multi-output processes with a 100% allocation to the “disposal activity”..

Preface

Since the beginning of the 1990's an increase in the research activities developing environmental assessment methodologies and models for ICT products can be observed. A variety of studies has been performed. However, a limited availability of process specific data restricted the published LCI studies of ICT components to little and often outdated data. According to Andrae (2005) the steps of producing integrated circuits (IC's) and printed wiring boards (PWB) dominate the overall environmental load of the whole telecommunication. These findings are supported by other recent studies. Nevertheless, the data availability in this area is quite poor and only very few articles can be found with quantitative information about these main production efforts – i.e. PWB and chip production (see e.g. Plepys (2004), Schischke et al. (2001), Williams (2000), Williams (2003) or Williams et al. (2002)).

Hence, within the further extension of the database ecoinvent from version 1.1 to 2.0, Empa decided to take care of the integration of the electronics sector. The project setup, research and dataset establishment base on Empa's own funding possibilities. The resulting report comprises all the different individual efforts at Empa to integrate the electronics sector into the database ecoinvent.

The structure of this report follows the approach used for the establishment of the datasets. Thereby a distinction between the three most important life cycle stages “production”, “use” and “disposal” was applied. As the electronics sector is a completely new part within the ecoinvent database, additional categories of datasets were developed, especially for the production and the use phase (left side of Fig. 0.1).

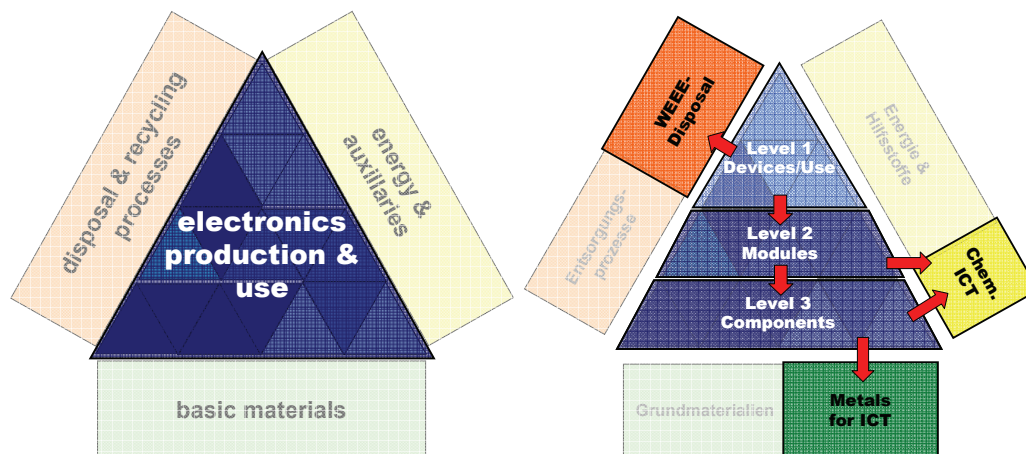


Fig. 0.1 Overall integration of the electronics sector into the existing framework of ecoinvent (left), and detailed project structure (right) including links to additional data work in the framework of the project ecoinvent data v2.0

Within this new part of the database ecoinvent (electronics production and use), an interrelation between three distinguished “levels” of aggregation has been identified – as shown in the right part of Fig. 0.1. The first of these three levels – “devices/use” – covers the datasets that present an added value to the majority of the ecoinvent users. As it is the topmost aggregation of all underlying datasets, this part comprises only a relatively small number of datasets. The two additional levels – “modules” and “components” – on the other side cover a wide range of materials, components and parts (e.g. transistors, printed wiring board, etc.) that have so far not been included in ecoinvent.

Furthermore, as shown in Fig. 0.1 (right side), there are relations between the electronics part and the already existing parts of the database ecoinvent. Then, the establishment of all these new electronics datasets in accordance with the methodological requirements of ecoinvent (described in Frischknecht et al. (2007)) results in additional requirements for basic materials in the area of precious metals, and pure chemicals, as well as specific disposal processes. To cope with this fact, three sub-projects for additional basic metals, chemicals and disposal processes have been set up, partly in cooperation with other institutions of the ecoinvent Centre. As a result, additional chemical datasets developed by ETH Zurich are reported in a new ecoinvent report, dealing with highly pure chemicals (Sutter (2007)). The new metal data – developed at Empa – are reported in the extended metals report (i.e. in Classen et al. (2007)) while the disposal side is integrated here – i.e. in part V of this report.

Within the three distinguished levels for “electronics production & use” (as shown in the right part of Fig. 0.1), a further fragmentation into a total of 10 work packages has been made. This additional structure conducted to a final distinction of four levels – each of them described in a separate part of this report. Hence, this report here distinguishes between the following parts:

- Part I “Electronic components”
- Part II “Modules”
- Part III “Devices”
- Part IV “Use”
- Part V “Disposal of Electric and Electronic Equipment (e-Waste)”

In each of these parts, all those datasets are described that belong to this specific sub-category of processes within the database ecoinvent. The authors of all these parts were careful to describe clearly what the various datasets are representing and where the respective numbers are taken from. In the end, a total of more than 200 new datasets have been established in order to cover the electronics sector in a transparent and modular manner – a modularity that gives a high degree of flexibility to the user when applying data to adopt default datasets or establish specific datasets from scratch.

Part I

Electronic Components

Authors: Roland Hischier, Empa St. Gallen
Martin Lehmann, Empa St. Gallen

Data: v2.0 (2007)

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1 Introduction

An electronic component is a basic electronic element usually packaged in a discrete form with two or more connecting leads or metallic pads¹. Components are intended to be connected together, usually by soldering to a printed wiring board (PWB), to create an electronic circuit with a particular function (for example an amplifier or a radio receiver). Components may be packaged singly (-> resistor, capacitor, transistor, diode, etc.) or in more or less complex groups as integrated circuits (-> resistor array, logic gate, etc.). In this part of the report, singly packed components as well as integrated circuits are described.

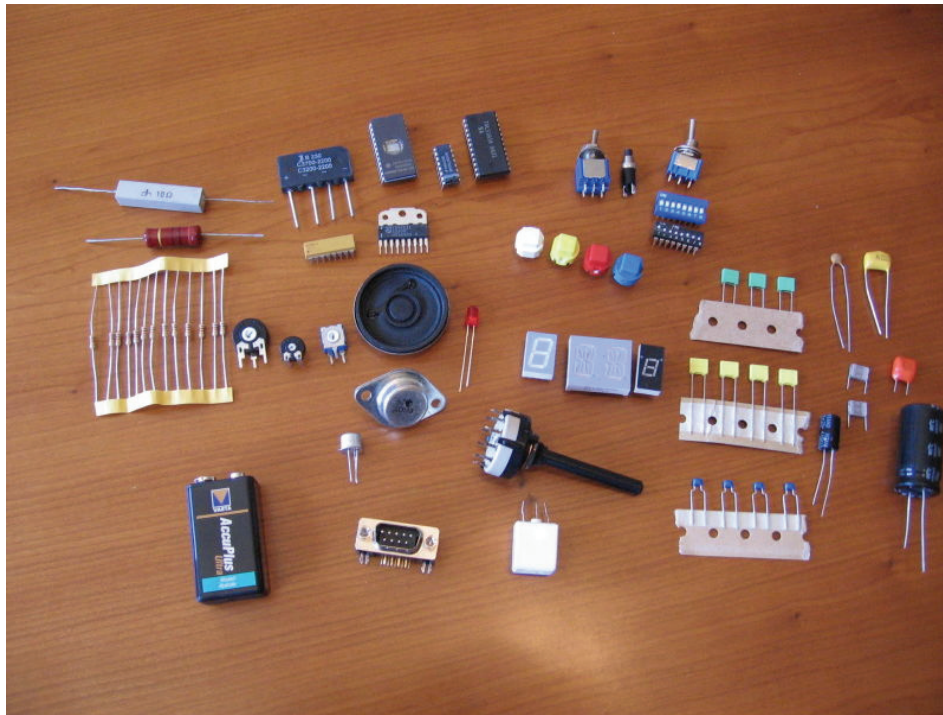


Fig. 1.1 Various electronic components (picture from Wikipedia)

Within all these here described components, two types can be distinguished – the **passive** and the **active** components. While the latter one increases the power of the signal on which it acts – passive components don't. Hence, all passive components draw the energy they use for their own operation directly from the signal on which they are operating, while the energy an active component adds to the signal must be drawn from a power source other than the signal on which it is operating². In detail, this chapter here describes the following electronic components:

(ii) Active components:

- Integrated Circuit (logic type) chapter 3
- Transistors (wired type, SMD type) chapter 4
- Diodes (glass type, SMD type, light emitting diodes) chapter 5

(i) Passive components:

¹ Wikipedia: electronic components (http://en.wikipedia.org/wiki/Electronic_components), accessed March 12, 2007.

² Wikipedia: passive components (http://en.wikipedia.org/wiki/Passive_component), accessed March 12, 2007.

- | | |
|---|------------|
| - Capacitors (film type, SMD type, Tantalum-, electrolyt-) | chapter 6 |
| - Inductors (ring core choke, MRFI type, LMCI type) | chapter 7 |
| - Resistors (metal film type, wirewound type, SMD type, potentiometer) | chapter 8 |
| - Transformers (low voltage, high voltage) | chapter 9 |
| - Connectors (PCI slot, clamp connection, USB) & switches (toggle type) | chapter 10 |

In all those cases where several datasets of the same type of component are established (e.g. 4 different types of capacitors will be shown and described), an additional dataset – expressed as “..., unspecified, at plant” – will be established, representing an unweighted average of this type of component. Such unspecified datasets are established for transistors (described in chapter 4.7), diodes (in chapter 5.7), capacitors (in chapter 6.8), inductors (in chapter 7.7), and for resistors (in chapter 8.8).

Furthermore, the two datasets “electronic component, passive, unspecified, at plant” resp. “electronic component, active, unspecified, at plant” are established, representing an unweighted average from ALL here described active resp. passive components. These two datasets are further described in chapter 11.

However, for the production of all these different components, a respective and adequate infrastructure is requested. Due to the extremely limited availability of data, only one single dataset for an average component production site requiring clean room conditions is established within the framework of the here described active and passive components. This plant together with the most important equipment parts is described in all its details in the next chapter (chapter 2).

A more simple plant, for all those components and/or modules that are not supposed to be produced under clean room conditions, is described in chapter 2 / part II in the dataset “printed wiring board mounting plant”.

2 Infrastructure for Electronic Components Production

Within the electronics industry, a huge variety of different types of plants – responsible for different production steps; and thus supposed to contain different equipment – can be distinguished. However, very few quantitative information is available about the equipment as well as the size of such production plants.

As a consequence, only one dataset requiring clean room conditions is established within the framework of this project – a dataset used for all electronic components (described in the various part of the report). The different parts of this electronic component plant considered and described here, are shown in Fig. 2.1.

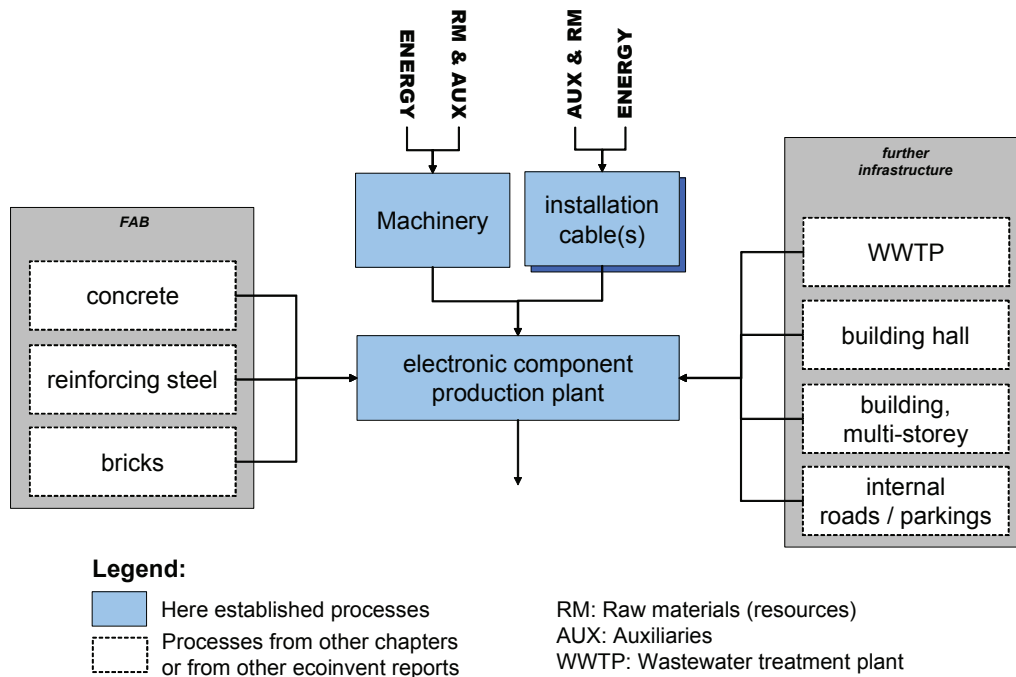


Fig. 2.1 The different parts of an electronic component plant

The information for this plant is mainly based on information from AMD's Fab36 in Dresden, Saxony (Germany) – supported/extended by literature and/or assumptions. Despite the fact that AMD in Dresden is producing microprocessors, this production site is taken as representative for all production plants due to the fact, that also "simpler" active and passive components (e.g. resistors, transistors, diodes, etc.) have to be produced in clean rooms. In case of components that are not supposed to be produced under clean room conditions, the dataset "printed wiring board mounting plant" (described in chapter 2 / part III of this report) is used.

The meta information of the various datasets of the infrastructure for the production of the various electronic components is shown in Tab. 2.1.

Tab. 2.1 ecoinvent meta information of the datasets of the electronic component production infrastructure (plant, machinery, ...) within the database.

Name	electronic component production plant	electronic component machinery, unspecified
Location	GLO	GLO
Infrastructure Process	1	1
Unit	unit	unit
Dataset Version	2.0	2.0
Included Processes	This module includes land use, buildings and facilities (machinery, cables, etc.) as well as the respective disposal activities of a microprocessor production site, used as example for a electronic component plant.	This module includes material, production efforts (transportation, energy, process chemicals, ...) and infrastructure for the production of a machinery used in the production line for electronic components. No information about emissions (air, water) and waste are included.
Amount	1	1
Local Name	Elektronische Bauteile-Fabrik	Maschine für Produktion elektronischer Bauteile, unspezifisch
Synonyms		
General Comment	The functional unit represents a production site for the production of electronic components. Described is AMD's site in Dresden (Germany), comprising 2 Fabs for the production of Microprocessors. Process data is taken mainly from literature, plus from own estimations. The described plant has an output of 2 Mio pieces per year over a time span of 25 years.	The data represent an average of three exemplary machineries used in the production of electronic components (a wafer marking system, a laser blind-via drilling system and a IC trim system). Material data are taken from producer information of these machineries. Infrastructure and production efforts are based on own assumptions. The dataset represents a machinery with an average weight of 2'500 kg.
Start Date	2005	2005
End Date	2005	2005
Data valid for entire period	1	1
Geography Text	This module relates to the global situation.	This module relates to the global situation.
Technology Text	The AMD site in Saxony is taken as representative of a typical electronic components plant	Average of three current machineries sold for the production of electronic components (see general comments).
Representativeness		
Production Volume	about 1 Mio. pieces per year	-
Sampling Procedure	Literature & own estimations	Literature & own estimations
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	None	None

Tab. 2.1 (Cont.) ecoinvent meta information of the datasets of the electronic component production infrastructure (plant, machinery, ...) within the database.

Name	cable, three-conductor cable, at plant	cable, data cable in infrastructure, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	m	m
Dataset Version	2.0	2.0
Included Processes	This module includes material, an estimation of the production efforts (transportation, energy, process chemicals, ...) and of the infrastructure for the production of a three-conductor cable used in the construction of new buildings. No information about emissions (air, water) and waste are included.	This module includes material, an estimation of the production efforts (transportation, energy, process chemicals, ...) and of the infrastructure for the production of a three-conductor cable used in the construction of new buildings. No information about emissions (air, water) and waste are included.
Amount	1	1
Local Name	Kabel, Dreipolkabel, ab Werk	Kabel, Datenkabel in Infrastruktur, ab Werk
Synonyms		
General Comment	The data represent a 1 kV low voltage cable with copper wire sheath, type GKN (Ceander / waveconal cable). Material data are taken from datasheet of a producer of such a cable. Infrastructure and production efforts are based on own assumptions. The dataset represents a cable with a meter-weight of 1.04 kg/m.	The data represent a mixture (50:50) of metallic and metal-free glasfibre cable for the transfer of data. Material data are taken from datasheet of a producer of such cables. Infrastructure and production efforts are based on own assumptions. The dataset is an average of cables with a meter-weight of 0.03 resp. 0.079 kg/m.
Start Date	2005	2005
End Date	2005	2005
Data valid for entire period	1	1
Geography Text	Data from one Swiss producer, used as representative for the global situation.	Data from one Swiss producer, used as representative for the global situation.
Technology Text	Data from one producer of such type of cable.	Mix of two types of cable. Data from one producer of such cables.
Representativeness		
Production Volume	-	-
Sampling Procedure	Literature & own estimations	Literature & own estimations
Extrapolations	see Geography and Technology	see Geography and Technology
Uncertainty Adjustments	None	None

2.1 Dataset “electronic component production plant”

2.1.1 System characterisation

For the infrastructure of the electronic component plant, modules for the construction and dismantling of the different buildings as well as the technical equipment are used. Further the land use of the plant site is also included. In the module no emissions or wastes are included. The module represents AMD's Dresden site comprising two Fabs (Fab36 and Fab38) that are supposed to produce annually about 1 Mio. units each Fab in the future. A total life-span of 25 years is assumed.

2.1.2 Life Cycle Inventory

The AMD site in Saxony has a total surface of 90 acres, i.e. about 364'200 m² (AMD (2007)). The occupation of the area before the occupation by AMD is not known (transformation from unknown). 10% of the total area is – according to several pictures of production site in Saxony – currently still green resp. unoccupied. Third parameter concerning land use is the time that is used to construct and to dismantle the whole plant. Based on the information about AMD's Fab36, 18 month construction time are assumed.

For the built or occupied area, it was estimated – based on the above mentioned pictures for AMD site in Saxony – that about 10% are buildings for the administration – represented by the module „building, multi story“ reported in Kellenberger et al. (2007) – 25% are storage and preparation facilities for the two fabs, 10% are occupied by the energy production unit (combined heat power plant) – both represented by the module „building hall“ from the same report – and another 20% are internal streets and parking areas – represented by the module „roads, company, internal“ described in Spielmann et al. (2007). Further it was assumed that the administration buildings have an average height of 10 m. With his assumptions $1.09 \cdot 10^6$ m³ of multiple story building volume and $8.56 \cdot 10^4$ m² of building hall area was calculated for the whole production plant. The internal streets and parkings account for a total surface of 8.2 ha. The remaining 30% of the occupied area are the two Fabs (Fab36 and Fab38). For the representation of these two units, none of the above mentioned modules is used due to the very specific architecture (big surfaces of clean rooms) and the available information from AMD for Fab36. According to AMD (2007), Fab36 consists of 13'000 t of construction steel (i.e. the structural area), 17'000 m² of masonry, 81'500 m³ of site-placed concrete, 75'000 m³ of building components and 26'400 m³ of building cubic components. It is assumed that Fab38 requests the same amount of construction materials.

Furthermore, it was assumed that the life time of all buildings, the street and parking surface will be, similar like the total assumed occupation time, about 25 years. Hence, no replacement occurs during this here assumed occupation time (production time). The final dismantling of the buildings is already included into the used datasets, while for the streets it is assumed that all material can be recycled and therefore no dismantling module is shown for this part of the infrastructure.

Concerning the equipment within the various buildings – similar to other industrial infrastructure in the ecoinvent database (see e.g. paper mill in Hischer (2007) or the chemical plant in Althaus et al. (2007)) –, we distinguish between the equipment for component production, the supporting parts (e.g. pipelines for process media, data cables, etc.), the internal waste water treatment plant as well as the energy station for the energy supply. All numbers are multiplied by 2 due to the fact that AMD's Saxony site has two Fabs (Fab36 and Fab38).

- **Equipment for component production.** A huge amount of different equipment is requested for the different process steps in the production chain of semiconductors and other electronic components. A conservative estimation, based on a scheme of the Fab36 inside found in Csatory & Kümmerle (2002), leads to about 100 units that are requested. All these units are assumed to have a life-span of 10 years – i.e. 2.5 times the amounts estimated here is needed in the dataset. Hence, for this dataset here, an input of 250 times the dataset “electronic component machinery, unspecified” (see chapter 2.2) is taken into account per Fab, as well as the disposal of a similar amount, by using the dataset “Disposal, industrial devices to WEEE treatment” (see chapter 3 in part V).
- **Supporting parts.** In AMD (2007), the total amount of pipelines (in km) and the total amount of cables (in km) installed in Fab36 are reported. For the process media pipes it is assumed due to the high requirements of the electronics industry, that they are 100% made out of high alloyed steel. The pipelines for heating/cooling are assumed to be made out of low-alloyed steel. For both types of pipeline, due to a lack of respective information, an average weight of 24.6 kg/m is assumed (similar value as for the chemical plant, described in Althaus et al. (2007)). In case of the cables, the below described cable types “three-conductor cable” (see chapter 2.3) and “data cable” (see chapter 2.4) are used. All these supporting parts are assumed to have a life-span of 10 years – i.e.

2.5 times the amounts reported in AMD (2007) are needed in the dataset. Concerning the disposal, it is assumed that the pipes are going to recycling, while for all cables the dataset “disposal, treatment of cables” is used as a proxy.

- Data describing the internal **waste water treatment plant (WWTP)** are taken from the municipal waste water treatment plant, type 5, described in Doka (2007). It is the smallest unit within this project - having a capacity of around 160'000 m³ water per day. Such an installation has an average life-time of about 30 years – therefore, the here reported electronic components plant needs 0.833 units of “wastewater treatment plant, class 5”.
- Data for the **energy station (steam & electricity production)** are already included in the various heat modules used in the datasets of the different components – hence, no data for this infrastructure is taken into account here – as otherwise a double counting of this equipment is done.

2.1.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is only a rough estimation of the actual infrastructure used for the production of all the different electronic components, based on information about one single site (Fab36 of AMD in Dresden). Hence, the respective uncertainty values are accordingly high.

The resulting unit process “electronic component production plant” is shown in Tab. 2.2 on the next page.

Tab. 2.2 Dataset “electronic component production plant”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
land use (industrial area)	electronic component production plant			resource	land			Occupation, industrial area, built up	8.19E+06	m2a	Total area from AMD (2007) & assumption		2.05	(13.2,3,158); data from AMD Saxony site in Dresden
land use (green area)				resource	land			Occupation, industrial area, vegetation	9.11E+05	m2a	Total area from AMD (2007) & assumption		2.05	(13.2,3,158); data from AMD Saxony site in Dresden
land transformation (built area)				resource	land			Transformation, to industrial area, built up	3.28E+05	m2	Total area from AMD (2007) & assumption		1.57	(13.2,3,157); data from AMD Saxony site in Dresden
land transformation (green area)				resource	land			Transformation, to industrial area, vegetation	3.64E+04	m2	Total area from AMD (2007) & assumption		1.57	(13.2,3,157); data from AMD Saxony site in Dresden
land transformation (original land)				resource	land			Transformation, from unknown	3.64E+05	m2	Total area from AMD (2007) & assumption		1.57	(13.2,3,157); data from AMD Saxony site in Dresden
land use (construction / disposal time)				resource	land			Occupation, construction site	109E+06	m2a	Total area from AMD (2007) & assumption		2.05	(13.2,3,158); data from AMD Saxony site in Dresden
storage and preparation facilities				construction processes	buildings	Yes	CH	building, hall	8.19E+04	m2	Total area from AMD (2007) & assumption		3.05	(13.2,3,159); data from AMD Saxony site in Dresden
energy production				construction processes	buildings	Yes	CH	building, hall	3.64E+03	m2	Total area from AMD (2007) & assumption		3.05	(13.2,3,159); data from AMD Saxony site in Dresden
internal road & parking places				transport systems	road	Yes	CH	roads, company, internal	2.05E+06	m2a	Total area from AMD (2007) & assumption		3.05	(13.2,3,159); data from AMD Saxony site in Dresden
Masonry (fabs)				construction materials	bricks	No	RER	brick, at plant	6.80E+05	kg	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,159); data from AMD Saxony site in Dresden
structural area (fabs)				metals	extraction	No	RER	reinforcing steel, at plant	2.60E+07	kg	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,159); data from AMD Saxony site in Dresden
site-placed concrete (fabs)				construction materials	concrete	No	CH	concrete, sole plate and foundation, at plant	163E+05	m3	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,159); data from AMD Saxony site in Dresden
building components (fabs)				construction materials	concrete	No	CH	concrete, exacting, at plant	1.50E+05	m3	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,159); data from AMD Saxony site in Dresden
building cubic components (fabs)				construction materials	concrete	No	CH	concrete, exacting, at plant	5.28E+04	m3	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,159); data from AMD Saxony site in Dresden

continued ▼

Tab. 2.2 (Cont.) Dataset “electronic component production plant”

General Flow information				Representation in ecoinvent						Uncertainty information						
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment		
Equipment for production	electronic component production plant (Part II)	continued		electronics	component	Yes	GLO	electronic component machinery, unspecified	5.00E+02	unit	estimation for Fab36 from Fig.4 in Fab Tech (2002) x 2 (as there are 2 fabs)		3.11	(4.4,2.3,15.9); estimation, based on literature information		
process media pipelines			metals	extraction	No	RER	chromium steel 18/8, at plant	4.43E+06	kg	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,15.9); data from AMD Saxony site in Dresden			
heating/cooling pipelines			metals	extraction	No	RER	steel, low-alloyed, at plant	3.94E+06	kg	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,15.9); data from AMD Saxony site in Dresden			
production of pipelines			metals	processing	No	RER	drawing of pipes, steel	8.36E+06	kg	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,15.9); data from AMD Saxony site in Dresden			
three-conductor cable			electronics	component	No	GLO	cable, three-conductor cable, at plant	150E+06	m	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,15.9); data from AMD Saxony site in Dresden			
data cable			electronics	component	No	GLO	cable, data cable in infrastructure, at plant	7.50E+06	m	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,15.9); data from AMD Saxony site in Dresden			
cable facility management			electronics	component	No	GLO	cable, data cable in infrastructure, at plant	125E+06	m	Data from Fab36 of AMD (2007) x 2 (as there are 2 fabs)		3.05	(13.2,3,15.9); data from AMD Saxony site in Dresden			
Internal waste water treatment plant			waste management	wastewater treatment	Yes	CH	wastewater treatment plant, class 5	1.00E+00	unit	rough estimation		3.09	(4.5,nan,nan,nan,9); rough estimation			
						waste management	recycling	No	GLO	disposal, treatment of cables	197E+06	kg	calculated from input of cables		122	(13.2,3,15.6); calculated, from input of respective equipment
						waste management	recycling	No	CH	disposal, industrial devices, to WEEE treatment	125E+06	kg	calculated from input of machineries		133	(4.4,2.3,15.6); calculated, from input of respective equipment
				electronics	component	Yes	GLO	electronic component production plant	1.00E+00	unit						

2.2 Dataset “electronic component machinery, unspecified”

2.2.1 System characterisation

A dataset, representing the various types of electronic component machineries used in the production of semiconductors, printed wiring boards, capacitors, etc. is established. The dataset contains inputs of the main materials used as well as the most important processing steps. No emissions are included.

2.2.2 Life Cycle Inventory

Although you can find in the internet a variety of information about size and functionalities of various equipment used in the production of electronic components – only very few of these sites contain also information about the weight and material content of the respective equipment. Hence, the dataset here is based on the following information:

- **WaferMark SigmaClean 300 FOUR** – a 300mm Automated Wafer Marking system (GSI Group (2005a)) with an average weight of approx. 650 kg;
- **Drill-Star 800** – a fourth generation laser blind-via PCB drilling system (GSI Lumonics (2004)) with an average weight of approx. 4200 kg (thereof 200 kg plastic and 4'000 kg metals);
- **WaferTrim M350** – an active IC trim system (GSI Group (2005b)) with an average weight of approx. 700 kg;



Fig. 2.2 WaferMark SigmaClean 300 FOUR (left), Drill-Star 800 (center) and WaferTrim M350 (right) – three examples of equipment for the electronics component production (pictures from references in text above)

Based on these information sources, the following assumptions and estimations are used in order to establish the dataset “Electronic component machinery, unspecified”:

- **Raw materials:** Based on the above described examples, an average weight of 2'500 kg is assumed for the unspecified machinery here. Thereof, 2'200 kg are metals (assumed are here 80% low-alloyed steel – 20% chromium steel), 200 kg plastics (HDPE) and the remaining 100 kg various electronic elements for controlling (80 kg of mounted wiring boards, 20 kg of cables).
- **Auxiliaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** No information about energy consumption is given in the examined sources. As an indirect mean, the following material processing steps are taken into account within this dataset:
 - 80% of steel processed with dataset “section bar rolling, steel”;
 - 20% of steel processed with dataset “sheet rolling, chromium steel”;

- All plastics processed with dataset “injection moulding”.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used.
- **Infrastructure and land-use:** Due to a lack of specific information, the infrastructure module established for the production of cars (dataset “road vehicle plant” – reported in Spielmann et al. (2007)) is used as a proxy.
- **Emissions to air & water, waste:** Due to a lack of respective information, no such flows are taken into account.

2.2.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is a very rough estimation of an unspecified machinery used for the production of all the different electronic components, based on weight information from several types of equipment for these activities. Hence, the respective uncertainty values are accordingly high.

The resulting unit process “electronic component machinery, unspecified” is shown in Tab. 2.3.

Tab. 2.3 Dataset “electronic component machinery, unspecified”

General Flow information					Representation in ecoinvent						Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
low-alloyed steel	electronic component machinery, unspecified	continued ▶		metals	extraction	No	RER	steel, low-alloyed, at plant	1.76E+03	kg	Estimation - based on three examples of equipment	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines
chromium steel				metals	extraction	No	RER	chromium steel 18/8, at plant	4.40E+02	kg	Estimation - based on three examples of equipment	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines
plastics				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	2.00E+02	kg	Estimation - based on three examples of equipment	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines
electronics				electronics	module	No	GLO	printed wiring board, mixed mounted, unspec., solder mix, at plant	8.00E+01	kg	Estimation - based on three examples of equipment	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines
cables				electronics	component	No	GLO	cable, network cable, category 5, without plugs, at plant	5.56E+02	m	Estimation - based on three examples of equipment	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines
processing of steel				metals	processing	No	RER	section bar rolling, steel	1.76E+03	kg	calculated from estimated amount of low-alloyed steel	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines
processing of chromium steel				metals	processing	No	RER	sheet rolling, chromium steel	4.40E+02	kg	calculated from estimated amount of chromium steel	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines
processing of plastics				plastics	processing	No	RER	injection moulding	2.00E+02	kg	calculated from estimated plastics amount	1	138	(4.2.11.3.5.3); estimation, based on qualitative information from such kind of machines

Tab. 2.3 (Cont.) Dataset “electronic component machinery, unspecified”

General Flow information					Representation in ecoinvent						Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
infrastructure	electronic component machinery, unspecified	continued		transport systems	road	Yes	RER	road vehicle plant	9.09E+07	unit	proxy	1	3.77	(4.5,13.5,4.9); rough estimation of infrastructure
transport, train				transport systems	train	No	RER	transport, freight, rail	5.00E+02 tkm	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nanana,na5); standard distances
transport, road				transport systems	road	No	RER	transport, lorry >16t, fleet average	2.50E+02 tkm	tkm	Standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nanana,na5); standard distances
				electronics	component	Yes	GLO	electronic component machinery, unspecified	1.00E+00 unit	unit				

2.3 Dataset “cable, three-conductor cable, at plant”

2.3.1 System characterisation

A dataset of a three-conductor power cable has been established, representing a typical low voltage power cable, as shown in Fig. 2.3 and as used in distribution networks and in industry (according to Brugg Cables (1996)). The dataset contains inputs of the main materials used as well as the most important processing steps. No emissions are included.



Fig. 2.3 1kV low voltage cable with copper wire sheath, type GKN (picture taken from Brugg Cables (1996))

2.3.2 Life Cycle Inventory

The dataset here represents 1 meter of a **1 kV low voltage cable with copper wire sheath, type GKN (Ceander / waveconal cable)**. The three copper conductors have a diameter of 16 mm each and the complete cable (incl. all isolation layers) has an average weight of 1.04 kg/m (Brugg Cables (1996)). The following assumptions and estimations are used in order to establish the dataset:

- **Raw materials:** The amount of copper is calculated from the profile of each single conductor (i.e. 16 mm²) and the density of copper of 8'920 kg/m³. Furthermore it is assumed that the outer copper line is about 15% of the total of the three conductors. For the non-metal parts – although in the description in Brugg Cables (1996), the various materials are mentioned – a simplification has been made by using 100% HDPE.
- **Auxiliaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** No information about energy consumption is given in the examined sources. As an indirect mean, the dataset “wire drawing, copper” (for the copper amount) and the dataset “extrusion, plastic pipes” (for total HDPE amount) are used.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used.
- **Infrastructure and land-use:** Due to a lack of specific information, the infrastructure module established for rolling of steel (dataset “rolling mill” – reported in Classen et al. (2007)) is used as a proxy.
- **Emissions to air & water, waste:** Due to a lack of respective information, no such flows are taken into account.

2.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is a very rough estimation of a low-voltage power cable, used for distribution networks and in industry. Hence, the respective uncertainty values are accordingly high.

The resulting unit process “cable, three-conductor cable, at plant” is shown in Tab. 2.4.

Tab. 2.4 Dataset “cable, three-conductor cable, at plant”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
copper	cable, three-conductor cable, at plant	cable, three-conductor type		metals	extraction	No	RER	copper, at regional storage	4.92E-01	kg	calculated from the cable profile & the density of copper	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
plastics				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	5.48E-01	kg	calculated - from the total weight and the copper amount	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
processing of copper (wire drawing)				metals	processing	No	RER	wire drawing, copper	4.92E-01	kg	calculated from amount of copper	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
processing of plastics				plastics	processing	No	RER	extrusion, plastic pipes	5.48E-01	kg	calculated from estimated plastics amount	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
infrastructure				metals	processing	Yes	RER	rolling mill	3.38E-11	unit	rough estimation	1	3.77	(4.5,13.5,4.9); rough estimation of infrastructure
transport, train				transport systems	train	No	RER	transport, freight, rail	2.08E-01	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nanana,na5); standard distances
transport, road				transport systems	road	No	RER	transport, lorry >16t, fleet average	10.4E-01	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nanana,na5); standard distances
				electronics	component	No	GLO	cable, three-conductor cable, at plant	100E+00	m				

2.4 Dataset “cable, data cable in infrastructure, at plant”

2.4.1 System characterisation

A dataset, representing an unweighted average of metallic and metal-free data cable, used in structured permis cabling (according to Dätwyler (2006a) and Dätwyler (2006b)) has been established. The dataset contains modules of the main materials used as well as the most important processing steps. No emissions are included.

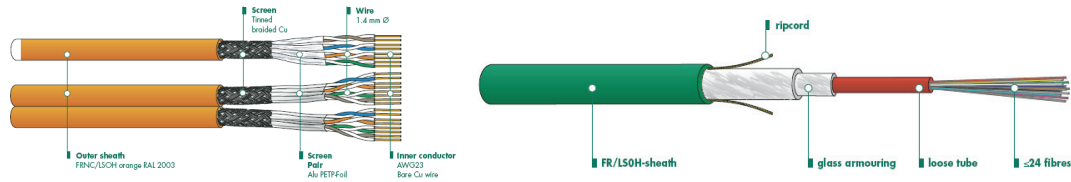


Fig. 2.4 metallic cable (left) and metal-free glasfibre cable (right) (picture taken from Dätwyler (2006a) and Dätwyler (2006b))

2.4.2 Life Cycle Inventory

The dataset here represents 0.5 meter of each of the two above mentioned types of data cables (i.e. 1 meter of cable in total). The following assumptions and estimations are used in order to establish the dataset:

- **Raw materials:** The amount of copper is taken from Dätwyler (2006b). In case of glass fibre cable, it is assumed that the glass fibres represent 50% of the total weight. All remaining material of both cable types is assumed to be HDPE.
- **Auxiliaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** No information about energy consumption is given in the examined sources. As an indirect mean, the dataset “wire drawing, copper” (for copper amount) and the dataset “extrusion, plastic pipes” (for total HDPE amount) are used.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used.
- **Infrastructure and land-use:** Due to a lack of specific information, the infrastructure module established for rolling of steel (dataset “rolling mill” – reported in Classen et al. (2007)) is used as a proxy.
- **Emissions to air & water, waste:** Due to a lack of respective information, no such flows are taken into account.

2.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is a very rough estimation of a data transfer cable, used for distribution networks and in industry, half a traditional copper cable, half a modern glasfibre cable. Hence, the respective uncertainty values are accordingly high.

The resulting unit process “cable, data cable in infrastructure, at plant” is shown in Tab. 2.5.

Tab. 2.5 Dataset “cable, data cable in infrastructure, at plant”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
copper	cable, data cable in infrastructure, at plant	cable, data transfer		metals	extraction	No	RER	copper, at regional storage	1.56E-02	kg	calculated from the cable profile & the density of copper	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
plastics				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	3.42E-02	kg	calculated - from the total weight of the two cable types	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
Glasfaser				glass	construction	No	RER	glass fibre, at plant	1.98E-02	kg	rough assumption	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
processing of copper (wire drawing)				metals	processing	No	RER	wire drawing, copper	1.56E-02	kg	calculated from amount of copper	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
processing of plastics				plastics	processing	No	RER	extrusion, plastic pipes	3.42E-02	kg	calculated from estimated plastics amount	1	1.57	(13.2,3.4,5.3); data from composition of one such cable
infrastructure				metals	processing	Yes	RER	rolling mill	1.77E-12	unit	rough estimation	1	3.77	(4.5,13.5,4.9); rough estimation of infrastructure
transport, train				transport systems	train	No	RER	transport, freight, rail	139E-02	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nananana,5); standard distances
transport, road				transport systems	road	No	RER	transport, lorry >16t, fleet average	6.95E-03	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nananana,5); standard distances
						electronics	component	No	GLO	cable, data cable in infrastructure, at plant	100E+00	m		

3 Microchips

3.1 Introduction

This Chapter describes the wafer fabrication process for semiconductors and two different integrated circuits or microchips, the logic and the memory type.

Chapter 3.2 (“Material Characterisation & Market”) provides information on the history of IC industry, the theory behind Moore’s law, on semiconductor materials and the production stages, about process water, process chemicals and gases. It concludes with the features and functions of two major types of integrated circuits and some future outlook on the semiconductor industry.

The system characterisation (Chapter 3.3) is discussed by explaining a/o functional unit, representativeness and configuration of the datasets. Further information on the use and application of the established datasets completes the part on general information (Chapter 3.4).

The wafer fabrication dataset described in Chapter 3.5 characterises the processes required to prepare and process a high-grade silicon wafer to become an integrated circuit or microchip. Elemental gases, organic and inorganic chemicals, as well as a remarkable amount of high-purity water and energy are being consumed during wafer fabrication. Apart from that, high amounts of air emissions (e.g. VOCs), solid waste and waste water are produced during the different process steps.

The two types of integrated circuits (ICs) pictured in Chapter 3.6 and 3.7 can be differentiated according to their specific functions. The logic type IC performs specified logical operations on the incoming data, it processes data and passes information on. The memory type IC is designed to store and give back data in the same form in which it is entered. For the datasets described here, a Plastic Ball Grid Array package (PBGA) was chosen as logic type representative, whereas for the memory type IC, a Plastic Leaded Chip Carrier (PLCC) was selected. This choice was restricted by the information available.

3.2 Material Characterisation & Market

Most of the text collected for the following Chapter 3.2 stems from Peter van Zant’s book (fifth edition) on Microchip fabrication (van Zant (2004)), parts are extracted from the International Technology Roadmap for Semiconductors (ITRS) published by the Semiconductor Industry Association (SIA (2005)).

3.2.1 General information

The electronics industry in general got its jump start with the discovery of the audion *vacuum tube* in 1906. It enabled the invention of the radio, television and other consumer electronics. It also was the brains of the world’s first electronic computer, named the Electronic Numeric Integrator and Calculator (ENIAC) in 1947. The ENIAC occupied 1500 square feet, weighed 30 tons, generated large quantities of heat, required the services of a small power station, and cost \$400,000 (!) back in 1940. The ENIAC was based on 19,000 vacuum tubes along with thousands of resistors and capacitors (van Zant (2004)).

One of the major drawbacks to the ENIAC and other tube-based computers was a limited operating time due to tube burn-out. These problems were the impetus leading many laboratories around the country/world to seek a replacement for the vacuum tube. That effort came to fruition in 1947, when three Bell Lab scientists demonstrated an electrical amplifier formed from the semiconducting material germanium. This device offered the electrical functioning of a vacuum tube but added the advantages of being solid state, being small and lightweight, and having low power requirements and long lifetime. First named a transfer resistor, the new device soon became known as the *transistor* (van Zant (2004)).

That first transistor was a far distance from the high-density integrated circuits (IC) today. But it was the component that gave birth to the solid-state electronics era with its famous progeny. Besides transistors, solid-state technology is also used to create diodes, resistors, and capacitors (all components which are described in the subsequent chapters of this part of the report here). All such devices, containing only one device per chip, are called *discrete* devices (van Zant (2004)).

The dominance of discrete devices in solid-state circuits came to an end in 1959. In that year, Jack Kilby (Texas Instruments) formed a complete circuit on a single piece of the semi-conducting material germanium. His invention combined several transistors, diodes, and capacitors (five components total). This invention was the *integrated circuit*, the first successful integration of a complete circuit in and on the same piece of a semiconducting substrate. Robert Noyce continued to furnish the technology further, and his integrated circuit became *the* model for all integrated circuits (van Zant (2004)).

Device component size and the number of components in an IC are the two common trackers of IC development. Feature and gate width are further characteristics of semiconductor devices. In today's technology, the most popular transistor is the metal-oxide-semiconductor (MOS) structure. The other major method of building a solid-state transistor is the field effect transistor (FET).

In 1965, Gordon Moore, a founder of Intel, noted that the number of transistors on a chip was doubling every 18 months. He published the observation, which was immediately dubbed as "Moore's law". Industry observers have since used this law to predict the future density of chips (see Tab. 3.1 and Fig. 3.1).

Tab. 3.1 Transistor count on a chip over the years – the Moore's law (Source: Intel Corporation in van Zant (2004))

Year	Transistor count
1978	29,000
1982	275,000
1985	1,200,000
1991	3,100,000
1993	7,500,000
1997	9,500,000
2001	55,000,000

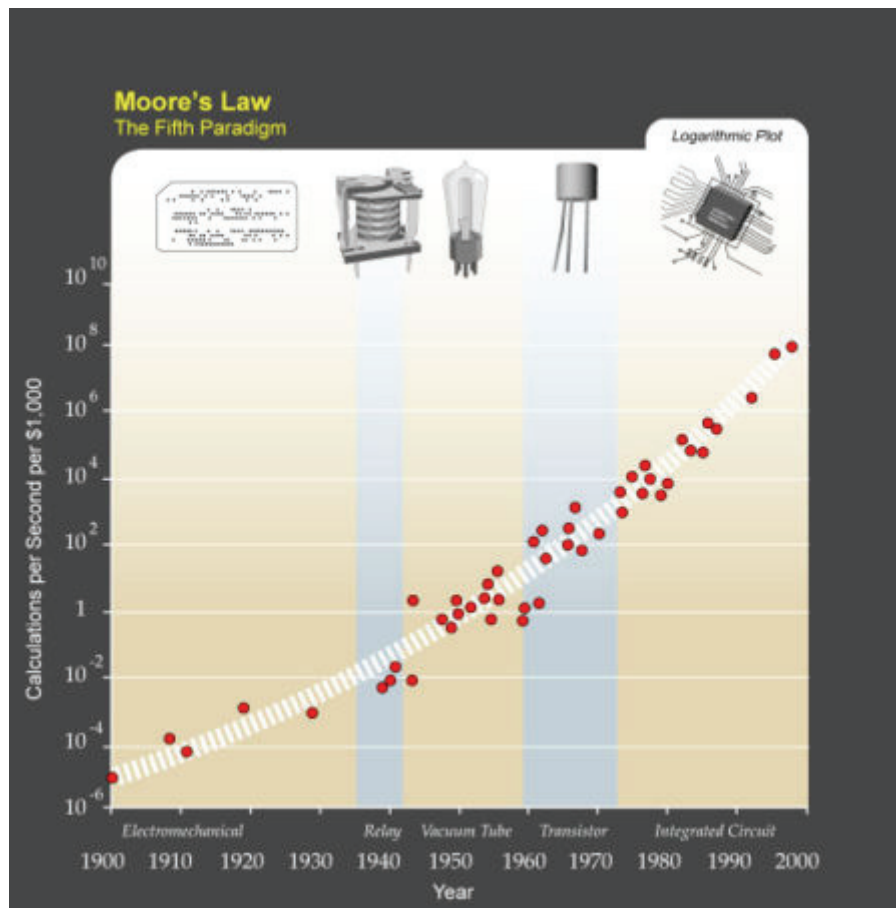


Fig. 3.1 Kurzweil's expansion of Moore's Law shows that due to paradigm shifts the underlying trends holds true from integrated circuits to earlier transistors, vacuum tubes, relays and electromechanical computers (source: Ray Kurzweil in Wikipedia³)

In addition to the integration level –which is the number of components in a circuit- memory circuits are identified by the number of memory bits contained in the circuit. A four-meg memory chip e.g. can store four million bits of memory. Logic circuits are often rated by their number of “gates”. A gate is the basic operational component of a logic circuit (van Zant (2004)).

The reduction in the feature size of the individual components of a chip has been brought about by dramatic increases in the imaging process, known as lithography, and the trend to multiple layers of conductors. The Semiconductor Industry Association (SIA) has projected feature sizes decreasing to 14 nm (0.0014 µm) DRAM pitch by the year 2020 (see Fig. 3.2). Though, as the International Technology Roadmap for Semiconductors mentions in their 2005 Executive Summary Edition, ‘we are still showing DRAM half-pitch at the top of many tables, but it is no longer described as the measure of “technology node”; it is just one among several historical indicators of IC scaling’ (SIA (2005)).

³ Wikipedia Moore's Law (http://en.wikipedia.org/w/index.php?title=Moore%27s_Law), accessed April 10, 2007.

Year	DRAM half-pitch (nm)
2001	130
2004	90
2007	65
2010	45
2013	32
2016	22
2020	14

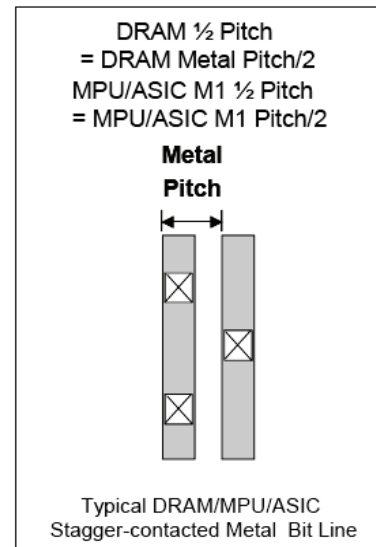


Fig. 3.2 Definition of DRAM Pitches (left) and Lithography DRAM half –pitch size (Source: SIA (2005) and van Zant (2004))

Major IC parameters are interrelated. Moore’s law predicts the future of component density (and size), which triggers the calculation of the integration level, chip size, defect density and the number of interconnection levels required. Tab. 3.2 shows the projection of wafer and chip parameters by the SIA.

Tab. 3.2 Projection of wafer and chip parameters by SIA (Source: van Zant (2004))

Year of Production	2001	2006	2012
Line width (nm)	150	100	50
Logic Bits/cm ²	380M	2.2B	17 B
Chip Size-DRAM (mm ²)	445	790	1580
Mask layers	23	24/26	28
Chip connections-I/O	1195	1970	3585
Wafer diameter (mm)	300	300	450

The semiconductor segment is composed of two major subsegments. One is the firms that actually make the semiconductor solid-state devices and circuits. Within this segment there are three types of chip suppliers:

- *Integrated device manufacturers (IDMs)*: they design, manufacture, package, and market chips;
- *Foundry companies*: they build circuit chips for other chip suppliers;
- *Waferless (or fabless) companies*: they design and market chips, by buying finished chips from chip foundries.

Some firms produce chips for in-house use and also sell on the open market, and others produce specialty chips in house and buy others on the open market.

Captive suppliers on the other hand are firms whose final product is a computer, communications system, or other product, and they produce chips in-house for their own products. Since the 1980s, the trend has been to a greater percentage of chips being fabricated in captive fab areas (van Zant (2004)).

The transition from 200-mm wafers to 300-mm wafers has bumped up equipment costs. The larger wafers generally require more handling time and more process time, resulting in productivity losses (van Zant (2004)). Fig. 3.3 shows the history and projections of wafer size life cycles.

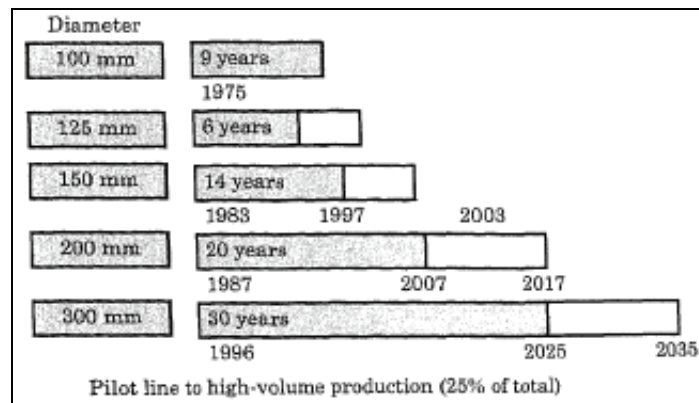


Fig. 3.3 Wafer size life cycles (source: Future Fab International in van Zant (2004))

3.2.2 Semiconductor Materials

Germanium and silicon are the two elemental semiconductors. The first transistor was made with germanium, as were the initial devices of the solid-state era. However, germanium presents problems in processing and in device performance. Its 937°C melting point limits high-temperature processing. More importantly, its lack of a natural occurring oxide leaves the surface prone to electrical leakage (van Zant (2004)).

The development of silicon/silicon dioxide planar processing solved the leakage problem of integrated circuits, flattened the surface profile of circuits and allowed higher temperature processing due to its 1415°C melting point. Consequently, silicon represents over 90 percent of the wafers processed worldwide (van Zant (2004)).

There are many semiconducting compounds formed from elements listed in columns III and IV and II to VI of the periodic table. Of these compounds, the ones most used in commercial semiconductor devices are gallium arsenide (GaAs) and gallium arsenide-phosphide (GaAsP), indium phosphide (InP), gallium aluminium arsenic (GaAlAs), and indium gallium phosphide (InGaP). These compounds have special properties (van Zant (2004)).

Devices of GaAs operate two to three times faster than comparable silicon devices and find applications in super-fast computers and real-time control circuits such as airplane controls. GaAs has a natural resistance to radiation-caused leakage and is also semi-insulating. In an integrated circuit, this property minimises leakage between adjacent devices, allowing a higher packaging density (van Zant (2004)). But despite all of the advantages, GaAs is not expected to replace silicon as the mainstream semiconducting material. The reasons reside in the trade-offs between performance and processing difficulty. While GaAs circuits are very fast, the majority of electronic products do not require their level of speed. Also, the production of large-diameter GaAs wafers has lagged behind that of silicon (van Zant (2004)).

Despite the problems, gallium arsenide is an important semiconducting material that will continue to increase in use and will probably have a major influence on computer performance in the future (van Zant (2004)). The section on future perspectives in the semiconductor industry (Chapter 3.2.6) provides more information on this issue.

3.2.3 Semiconductor production steps

Solid-state devices are manufactured in the following five distinct stages (compare Fig. 3.4): **a.** Material preparation; **b.** Crystal growth and wafer preparation; **c.** Wafer fabrication and sort; **d.** Packaging; **e.** Final and electrical test.

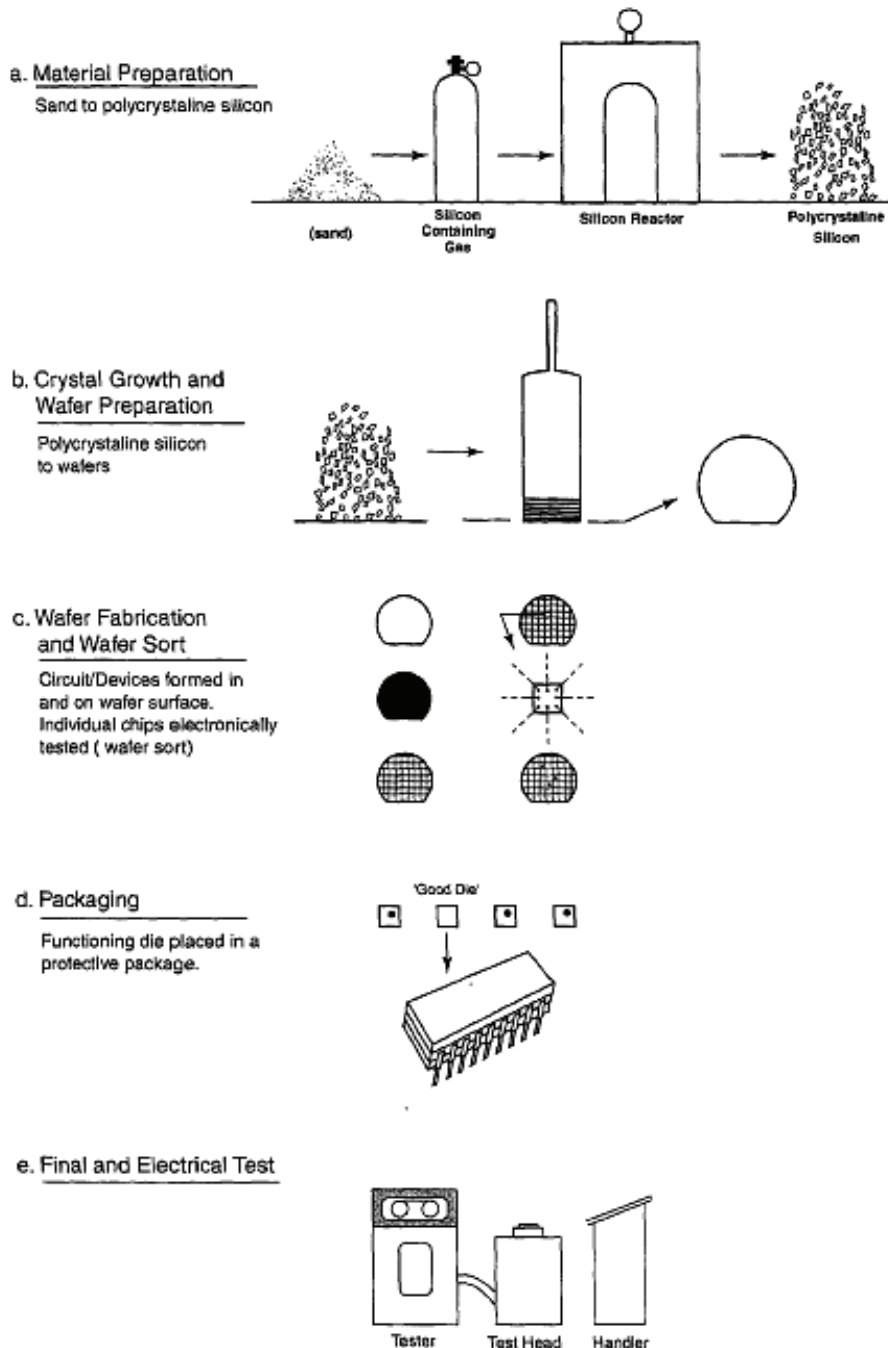


Fig. 3.4 Stages of semiconductor production (Source: van Zant (2004))

The stages **c.** to **e.** are thereby represented with datasets established within this Microchip chapter here (see below), whereas the stages **a.** and **b.** are covered by the ecoinvent report on Photovoltaics (Jungbluth & Tuchschnid (2007)).

Stage a: Material Preparation

In this first step, the raw semiconducting materials are mined and purified to meet semiconductor standards. For silicon, the starting material is sand, which is converted to pure silicon with a polysilicon structure (further details in Jungbluth & Tuchschnid (2007)).

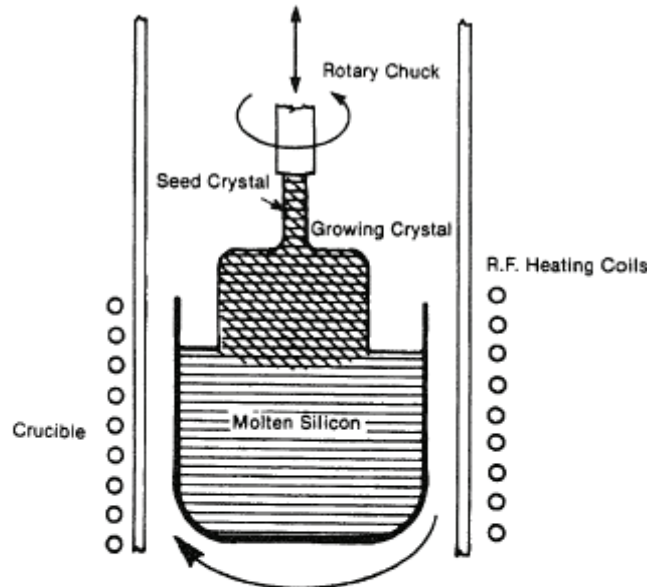


Figure 3.8 Czochralski crystal-growing system.

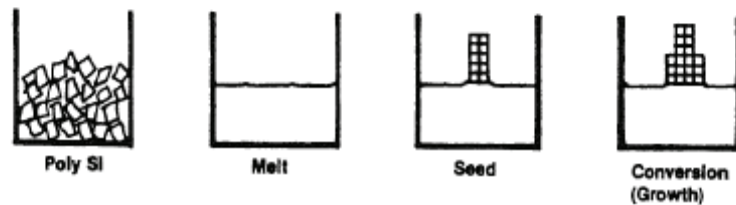


Fig. 3.5 Czochralski crystal-growing system (source: van Zant (2004))

Stage b: Crystal Growth

Here, the material is formed into a crystal with specific electrical and structural parameters. The majority of silicon crystals are grown by the Czochralski (CZ) method (see Fig. 3.5). The equipment consists of a quartz (silica) crucible that is heated by surrounding coils that carry radio frequency (RF) waves or by electric heaters. The crucible is loaded with chunks of polycrystalline of the semiconductor material and small amounts of dopant. The dopant material is selected to create either an N-type or P-type crystal. First, the poly and dopants are heated to the liquid state at 1415 °C. Next, a seed crystal is positioned to just touch the surface of the liquid material (called the melt). Crystal growth starts as the seed is slowly raised above the melt. The surface tension between the seed and the melt causes a thin film of the melt to adhere to the seed and then cool. During the cooling, the atoms in the melted semiconductor material orient themselves to the crystal structure of the seed. The CZ method is capable of producing crystals several feet in length and with diameters up to 12 or more inches. A crystal for 200-mm wafers will weigh some 168 kg and take three days to grow (van Zant (2004)).

Next, thin disks called wafers are cut from the crystal and surface treated in a process called crystal growth and wafer preparation. Further information on the crystal growth stage can again be retrieved from Jungbluth & Tuchschnid (2007).

Stage c: Wafer fabrication

Integrated circuits, like described in Tab. 3.3 are actually formed in and on the wafer surface. The area on the wafer occupied by each discrete device or IC is called a chip, die, microchip, or bar (van Zant (2004)).

Tab. 3.3 Typical 200-mm wafer specification (source: van Zant (2004))

Diameter	200 mm	Resistivity	2.7-4 Ω -cm
Diameter Tolerance	0.25 mm	Oxygen	25-29 ppma
Thickness	725 +/- μ	Carbon	0.3 ppma

Four broad areas of consumption during wafer fabrication are considered – energy, water, elemental gases, and specialty chemicals (see Chapter 3.2.4). Specialty chemicals represent by far the smallest fraction of mass utilised. Although overall usage at a particular site is generally known, it is often difficult to associate flows with specific operations. This makes any activity-based analysis (whether for the purpose of cost or environmental improvement) quite difficult. As a consequence, there are several barriers to direct measurement of water, elemental gases, and energy consumption during wafer fabrication (Murphy et al. (2003)). While a wafer fabrication operation may take several thousand individual steps⁴, there are two major activities (van Zant (2004)):

- FEOL (Front End Of the Line)
- BEOL (Back End Of the Line)

In the *front end of the line* (FEOL), the transistors and other devices are formed in the wafer surface. Companies use four basic operations in a large number of sequences and variations to produce specific microchips. They are layering, patterning, doping, and heat treatments (see Fig. 3.6).

- *Layering* is the operation used to add thin layers to the wafer surface. These layers could be insulators, semiconductors, or conductors. They are of different materials (e.g. silicon dioxide or polycrystalline silicon) and are grown or deposited by a variety of processes (e.g. thermal oxidation, chemical vapor deposition, evaporation, sputtering, electroplating) (van Zant (2004)).
- *Patterning* is the series of steps that results in the removal of selected portions of the added surface layers. The patterning process is known by the names photomasking, masking, photolithography, oxide removing, metal removal, and microlithography. The various physical parts of the transistors, diodes, capacitors or resistors are formed in and on the wafer surface, one layer at the time. The goal of the patterning operation is to create the desired shapes in the exact dimensions required by the circuit design, and to locate them in their proper location on the wafer surface and in relation to the other parts. A typical IC requires 20 to 40 individual patterning steps. Patterning is the most critical of the four basic operations. This operation sets the critical dimension of the devices. As van Zant writes, “patterning is a high-tech version of photography but performed at incredibly small dimensions” van Zant (2004)). Fig. 3.7 shows the ten-step photomasking (patterning) process.

⁴ See <http://www.siliconfareast.com/manufacturing.HTM> to get an idea how wafer fabrication is carried out.

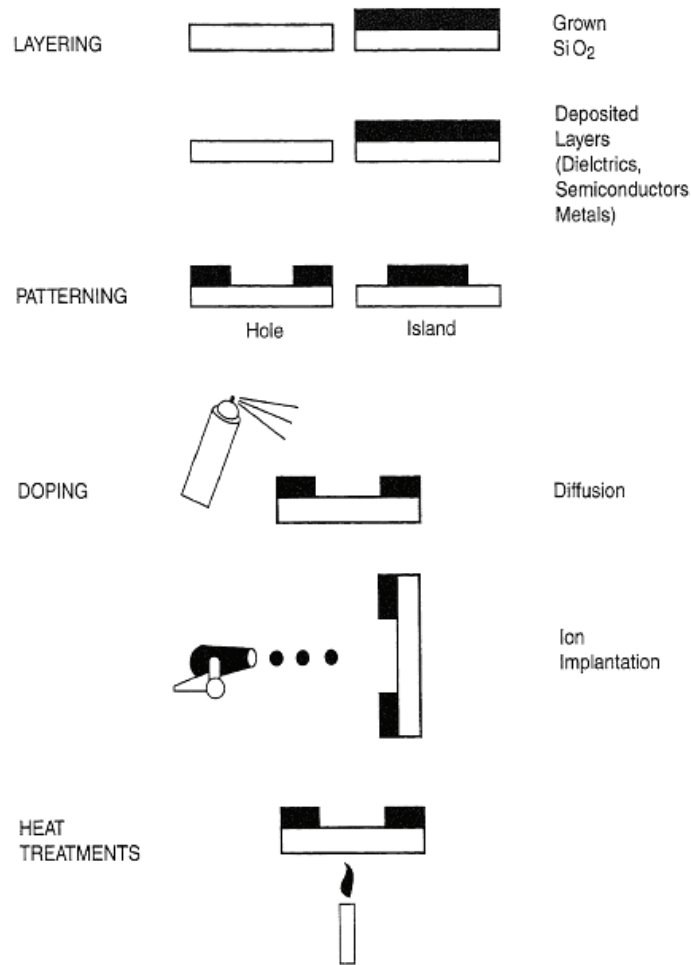


Fig. 3.6 Basic wafer fabrication operations (source: van Zant (2004))

- *Doping* is the process that puts specific amounts of electrically active dopants in the wafer surface through openings in the surface layers. The two techniques are thermal diffusion and ion implantation (see Fig. 3.6). Thermal diffusion is a chemical process that takes place when the wafer is heated to the vicinity of $1000\text{ }^{\circ}\text{C}$ and is exposed to vapors of the proper dopant. Ion implantation on the other hand is a physical process. Wafers are loaded in one end of an implanter and dopant sources (usually in gas form) in the other end. The purpose of the doping operation is to create pockets in the wafer surface that are either rich in electrons (N-type) or rich in electrical holes (P-type). These pockets form the electrically active regions and N-P junctions required for operation of the transistors, diodes, capacitors, and resistors in the circuit (van Zant (2004)).
- *Heat treatments* are the operations in which the wafer is simply heated and cooled to achieve specific results. In the heat treatment operations, no additional material is added or removed from the wafer, although contaminants and vapors may evaporate from the wafer surface. An important heat treatment takes place after ion implantation (disruption of the wafer crystal structure is repaired). Another heat treatment ($450\text{ }^{\circ}\text{C}$) takes place after the conducting stripes of metal are formed on the wafer. This "alloying" of the metal ensures good electrical conduction (van Zant (2004)).

These four basic fabrication operations during FEOL are used to build a particular transistor structure. The other components (diodes, resistors, and capacitors) required for the circuit are formed in other areas of the circuit as the transistors are being formed.

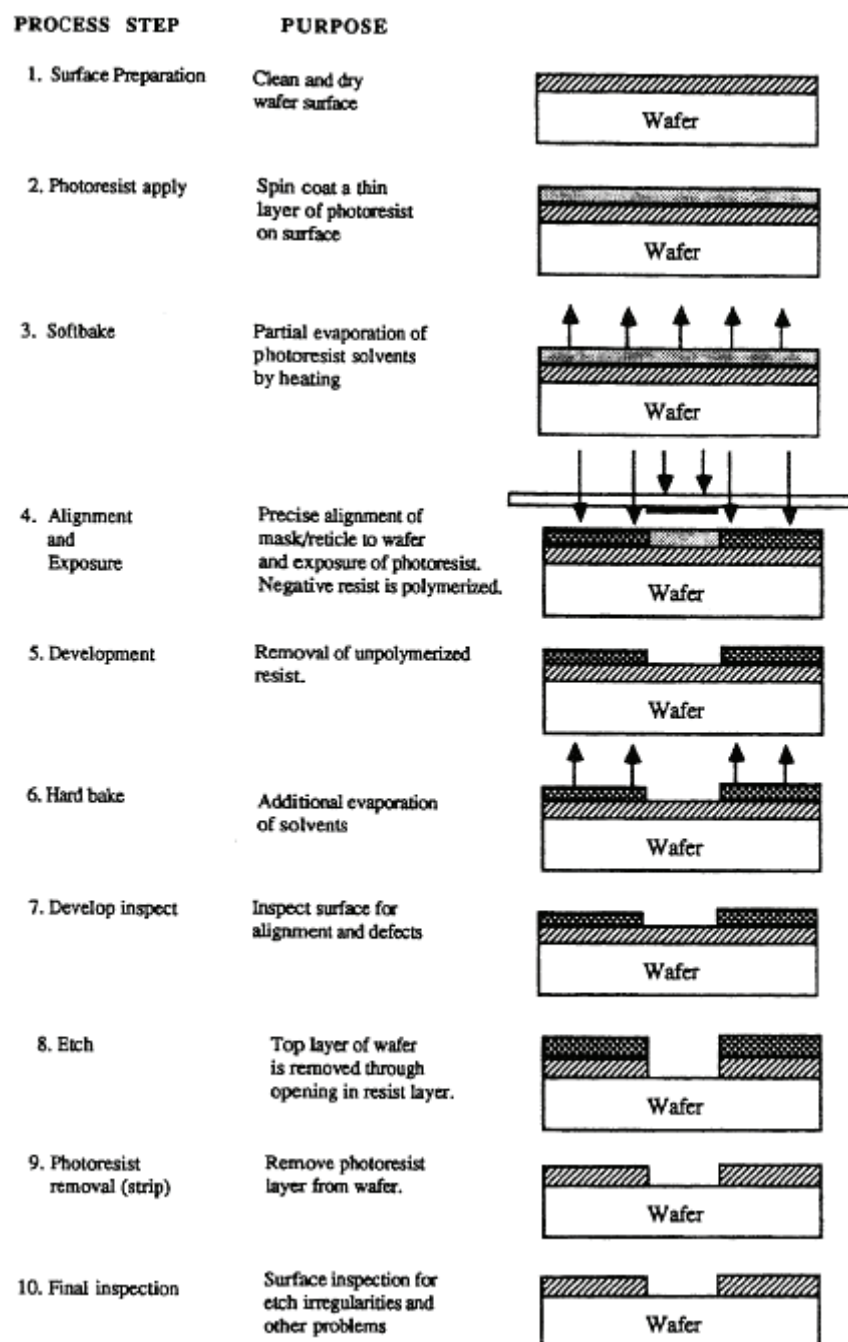


Fig. 3.7 The ten-step photomasking process (source: van Zant (2004))

Modern chip structures (see Fig. 3.8) are many times more complicated than the simple process just described. They have multiple layers and pockets of dopants, and numbers of layers added to the surface (van Zant (2004)).

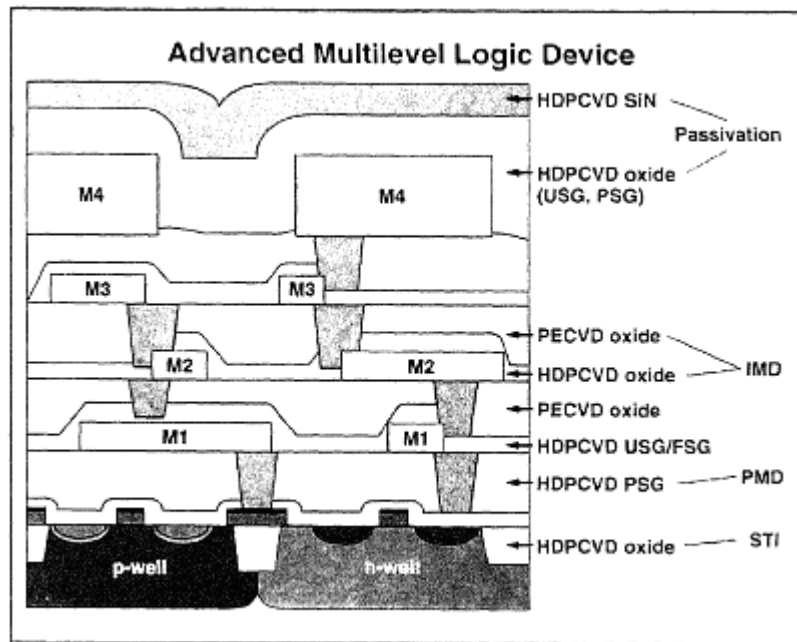


Fig. 3.8 Modern chip structure (source: van Zant (2004))

The components of higher-density circuits are so small that they cannot be distinguished on a photo-micrograph of the entire chip. Generally speaking, the chip features are (van Zant (2004)):

1. A bipolar transistor
2. The circuit designation number
3. Bonding pads for connecting the chip into a package
4. A piece of contamination on a bonding pad
5. Metallization lines
6. Scribe (separation) lines
7. Unconnected component
8. Mask alignment marks
9. Resistor

In the *back end of the line* (BEOL), the devices are wired together with metallization processes, and the circuit is protected with a final sealing layer. Generally, a heat-treatment step, called alloying, is performed after metal patterning to ensure good electrical contact between the metal and the wafer surface. Next comes an electrical test (called “wafer sort”) of every chip to identify those that meet customer specifications. For the test, the wafer is mounted on a vacuum chuck and aligned to thin electrical probes that contact each of the bonding pads on the chip. The number, sequence, and type of tests are directed by a computer program. Wafer sort may be the last step in the wafer fabrication or the first step in the packaging process (van Zant (2004)).

Successful wafer fabrication operations must achieve accumulative fabrication yields over 90 percent to stay profitable and competitive (van Zant (2004)).

Stage d and e: Packaging and electrical test

This step is the series of processes that separate the wafer into individual die and place them into protective packages to protect the chip from contamination and abuse. The industry also refers to this stage as “assembly and test” (A/T; (e)). Packaging takes place in a different department of the semiconductor producer and quite often in a foreign plant.

There are several chip characterisations that affect the packaging process of an integrated circuit (van Zant (2004)):

- integration level
- wafer thickness
- dimensions
- environmental sensitivity
- physical vulnerability
- heat generation
- heat sensitivity

Furthermore, a number of basic packaging operations can be distinguished (van Zant (2004)):

- backside preparation
- die separation (scribing, sawing)
- die pick
- die attach
- inspection
- bonding
- preseal inspection
- packaging sealing
- plating
- lead trim
- marking
- final test

For the datasets of the logic and memory type IC in Chapter 3.6 and 3.7, the operations applied are at least die separation, inspection, die attach, lead bonding, plating, possibly marking and final test. Out of all the operations, the die attach process and the lead bonding is described here in further details:

- *Die attaching*: In the center of every package is an area where the chip is securely attached into the package (Fig. 3.9). Die attachment has several goals: to create a strong physical bond between the chip and the package, to provide either an electrical conducting or insulating contact between the die and the package, and to serve as a medium to transfer heat from the chip to the package.

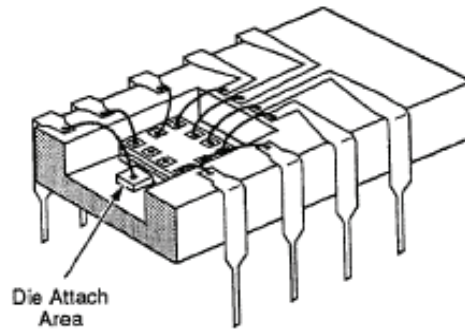


Fig. 3.9 Die attach area of package (source: van Zant 2004)

This die attach material can be a gold-silicon eutectic or an epoxy adhesive material. A major requirement for the die-attachement area is absolute flatness so as to intimately support the chip in the package. Another requirement is the permanency of the die-attachment bond. The bond should not loosen or deteriorate during subsequent processing steps or when the package is in use in an electronic product (van Zant (2004)).

- *Lead bonding*: Thin wires are bonded between the chip bonding pads and the inner leads of the package. There are three primary methods of connecting the chip to the package: wire bonding, bump (ball) technology, and tape automated bonding (TAB) (van Zant (2004)). Wire bonding, which is applied to the logic type microchip described in this report, is simple in concept and procedure. But it is also critical because of the precise wire placement and electrical contact requirements. In addition to accurate placement, each and every wire must make good electrical contact at both ends, and be at a safe distance from neighbouring wires. Distances between adjacent wires are referred to as the *pitch* of the bonding.

3.2.4 Contamination Control Factors

Process water

During the course of fabrication processing, a wafer will be chemically etched and cleaned many times. Each of the etching or cleaning steps is followed by a water rinse. Throughout the entire process, the wafer may spend a total of several hours in water rinse systems. A modern wafer fabrication facility may use up to 10'000 m³ of water per day, representing a substantial investment in water processing, delivery to the process areas, and treatment and discharge of waste water. Given the vulnerability of semiconductor devices to contamination, it is imperative that all process water be treated to meet very specific purity requirements. Normal city system water contains unacceptable amounts of contaminants like dissolved minerals, particulates, bacteria, organics, dissolved oxygen and silica (van Zant (2004)).

The process of removing the electrically active ions changes the water from a conductive medium to a resistive one. This fact is used to improve the quality of deionized (DI) water (van Zant (2004)).

Process Chemicals: Purity and Cleanliness

Extensive processing is required to change the raw semiconducting materials into useful devices. The majority of these processes use chemicals. In fact, microchip fabrication is primarily a chemical process or, more correctly, a series of chemical processes. Up to 20 percent of all process steps are cleaning or wafer surface preparation (van Zant (2004)).

Great quantities of acids, bases, solvents, and water are consumed by a semiconductor plant. Part of this cost is due to the extremely high purities and special formulations required of the chemicals to allow precise and clean processing. When costs of producing a chip are added up, process chemicals can be up to 40 percent of all manufacturing costs (van Zant (2004)).

There is an entire supply industry dedicated to producing the highest-quality chemicals to meet semiconductor processing demands. Chemicals must meet very high purity requirements. In general the target is “six nines” purity. This translates to 99.99999 percent pure. The contaminants of concern are metallics, particulates, and other chemicals. Industrial chemicals are rated by grade: a) commercial; b) reagent; c) electronic; d) semiconductor grade. The first two are generally too dirty for semiconductor use. The delivery of a clean chemical to the processing area involves more than just making a clean chemical. Care must be taken to clean the inside of the containers, using containers that do not dissolve, using particulate-free labels, and placing the clean bottles in bags before shipping (van Zant (2004)). The rapid introduction of new chemicals, materials, and processes requires new rapid assessment methodologies to ensure that new chemicals and materials can be utilised in manufacturing without inducing new hazardous impacts on human health, safety, and the environment (SIA (2005)).

Process gases

In addition to the many wet (liquid) chemical processes, a semiconductor wafer is processed with many gases. The gases are both the air-separation gases (oxygen, nitrogen, and hydrogen) and specialty gases such as arsine and carbon tetrafluoride. Gas quality is measured in four categories: 1) Percentage of purity; 2) Water vapour content; 3) Particulates; 4) Metallic ions. The gases used in oxidation, sputtering, plasma etch, chemical vapor deposition (CVD), reactive ion etch, ion implantation, and diffusion are of particular concern (van Zant (2004)).

Gas purity is specified by the assay number, with typical values ranging from 99.99 to 99.999999 percent (called “six 9s pure” in this industry branch), depending on the gas and the use in the process. The challenge in fab areas is maintaining the gas purity from the manufacturer to the process station. The control of water vapor is also critical. Water vapor is a gas and can enter into unwanted reactions just like other contaminating gases. Silicon for instance oxidises easily wherever free oxygen or water is available (van Zant (2004)).

More about process chemicals, process gases and caused emissions to air and water can be found in the specific dataset description in Chapter 3.5 on wafer fabrication.

Wafer surface cleaning

Clean wafers are essential at all stages of the fabrication process, but are especially necessary before any of the operations are performed at high temperature. Wafer surfaces have four general types of contamination: a) Particulates; b) Organic residues; c) Inorganic residues; d) Unwanted oxide layers. In general, a wafer cleaning process, or series of processes, must a) remove all surface contaminants,

b) not etch or damage the wafer surface, c) be safe and economical in a production setting, and d) be ecologically acceptable (van Zant (2004)).

3.2.5 Types of Integrated Circuits

A solid-state integrated circuit is composed of a number of separate functional areas. Each chip, regardless of the circuit function, has an input and encode section where the incoming signals are “coded” into a form that the circuit can understand. The majority of the circuit area contains the circuitry required to perform the circuit function, either memory or logic. After the data is manipulated by the circuit, it goes to a decode section where it is changed back into a form that is usable by the machine’s output mechanism. Circuit types fall into three broad categories: logic, memory, and logic and memory (microprocessors) (van Zant (2004)).

Logic circuits

Logic circuits perform a specified logical operation on the incoming data. As an example, an onboard automobile computer goes through a logical operation to direct the signal from a sensor indicating an open door to light up the correct warning light on the dashboard.

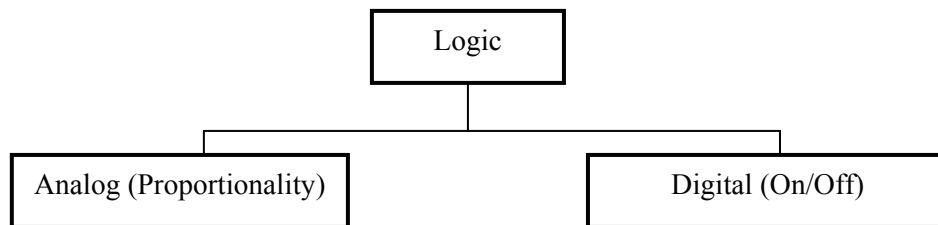


Fig. 3.10 Logic circuit types (source: van Zant (2004))

Logic circuits fall into two main categories: analog and digital (Fig. 3.10). An analog circuit has an output that is proportional to the input. Digital circuits, on the other hand, feature a predetermined output in response to a variety of inputs. A wall light dimmer is an analog device. A standard on-off light switch is a digital device (van Zant (2004)).

Logic circuitry is built around the logic gate. A gate controls and directs passage through a barrier. For example, a room with many “in” doors and only one “out” door is a gate. Many people can enter the room, but their exit is restricted, because only one door is provided. Electronic logic gates perform similar functions but with electrical signals. Analog and digital gates are formed in a logic circuit by wiring together various components.

Memory circuits

Memory circuits are designed to store and give back data in the same form in which it is entered. Pushing the pi (π) key on a calculator activates the memory part of the circuit where the value of π (3.14) is stored.

Metal-oxide semiconductor (MOS) is the favored transistor structure for memory circuits. With process improvements and improved silicon gate structures, complementary metal-oxide semiconductor (CMOS) is the memory technology of choice. Some bipolar memories are favored for their fast speed and switching capabilities. While logic circuits can be (and are) made in MOS technology, most MOS circuits produced are memories, with the majority incorporated into computers (van Zant (2004)).

There are two principal types of memory circuits: volatile and non-volatile (Fig. 3.11).

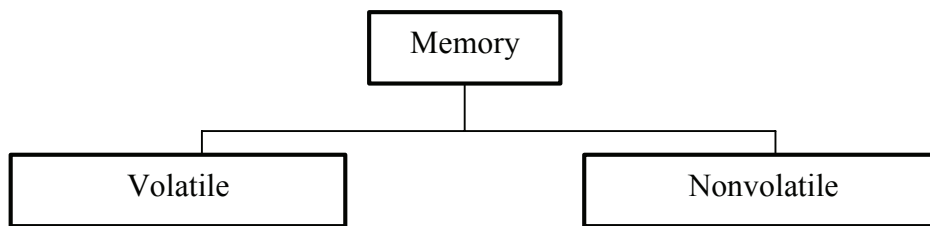


Fig. 3.11 Memory circuit types (source: van Zant (2004))

Nonvolatile memories. A non-volatile memory device is one that does not lose its stored information when it loses its power. An example of this is a compact disk (CD). Nonvolatile memory circuits are (van Zant (2004)):

- ROM (read-only memory); the sole function of this type of circuit is to give back precoded information
- PROM (programmable read-only memory); here, each memory cell is connected into the circuit through a fuse. Users program the PROM to their own memory circuit requirements by blowing fuses at unwanted memory cell locations. After programming, the PROM is changed to a ROM.
- EPROM (erasable programmable ROM); the information stored in the ROM can be changed without having to replace the whole chip. A typical EPROM can be reprogrammed up to ten times.
- EEPROM (electronically erasable PROM); the chip can be programmed and reprogrammed while it is in a socket in the machine. Programming and erasing take place by pulses from the outside that place charges in selected memory cells or drain the charges away. A flash memory is a form of EEPROM.

In case of memory, non-volatile memory provides protection against power loss, but these memories are frequently slow and not very dense.

More important, none of the circuits for non-volatile memories has a write capability, an essential feature in operating a computer. Memory must also be easily erasable so the computer can quickly process new information or accept a completely new program.

Volatile memories. Volatile memory circuits are (van Zant (2004)):

- RAM (random-access memory); unlike a serial memory, the RAM design allows the chip to find the exact information asked for, wherever it is located in the computer memory. This feature allows faster information retrieval and makes the RAM the principal memory circuit in computers.
- DRAM (dynamic random-access memory); DRAMs come in two principle designs: dynamic and static. The memory cell design is based on only one transistor and a small capacitor. The information is stored in it by a charge built up in the capacitor. As the charge drains away very rapidly, the memory information must be re-inputted to the circuit (refreshed) at constant basis. The refreshing of the circuit occurs many thousands of times per second. The goal of the DRAM design is small-cell design for high-density and closely spaced components with small and thin parts for speed. These requirements have driven DRAM design and processing to the highest levels of the technology. This fact makes them the industry's leading-edge circuit. The memory chip described in Chapter 3.7 is a DRAM chip.
- SRAM (static random-access memory); SRAMs are based on a cell design that does not need a refresh function. Once the information is put into the chip, it will stay as long as the power remains

on. This is accomplished with a cell containing several transistors and capacitors. Information can be read and written with a SRAM cell much faster than with a RAM design, since transistors can be switched faster than capacitors can be charged and drained.

- FFERAM (ferroelectric random-access memory); ferroelectric materials used in capacitors create memories that operate faster than conventional SiCMOS technology capacitors.

Microprocessors

Microprocessors are a third circuit type that combines both logic arithmetic functions, and memory. The microprocessor has been dubbed “a computer on a chip”. While it contains all the functional areas of a computer, it is not truly a complete computer. Even simple computers require vast amounts of memory capacity that microprocessors do not have. Within many personal computers, microprocessors function as the central processing unit (CPU). Additional memory chips have to be included to make the computer of practical use (van Zant (2004)).

3.2.6 Semiconductor industry – Past, Present and Future

Chip production saw five different decades of industry development. In the 1950s, basic and crude products and processes launched the semiconductor industry. The 1960s was the decade the industry started growing into a sophisticated industry, driven by new products that demanded new fabrication processes. At the start of the 1970s, the industry was manufacturing ICs primarily at the MSI (Medium Scale Integration, meaning 50-5000 components per chip). The decade also saw the improvement of cleanroom construction and operation. Automation of processes started with spin/bake and develop/bake systems (van Zant (2004)).

The focus in the 1980s was automation of all phases of wafer fabrication and packaging and elimination of operators from the fab areas. The 1980s also started with American and European dominance and ended as a worldwide industry. The one-micron barrier was crossed in the early 1990s when 50 percent of microchip fabrication lines were working at the micron or submicron level. The spread of the technology (competition) and improvements in process control, however, moved the industry to greater emphasis on the production process. The primary productivity factors are automation, cost control, process characterization and control, and worker efficiency. Perhaps the major technological change of the decade was copper wiring. Aluminium wiring ran into limitations in several areas, notably in contact resistance with silicon. Copper has always been a better conductor but was difficult to deposit and pattern. It was also a killer of circuit operation if it got into the silicon (van Zant (2004)).

During recent years, the semiconductor industry has put a lot of hope in nanotechnology. The way to the nano future is among others sketched out in the Semiconductor Industry Association’s International Technology Roadmap for Semiconductors (SIA (2005)). Gate widths of 18 nm are predicted in the year 2018, and DRAM memory size shall go up to 32 gigabits. A gate area is the critical working part of an MOS transistor. Smaller gates are more vulnerable to contamination, which drives the development of cleaner chemicals and processes. As the devices get closer together, they drive the need for a superstructure of metallization layers stacked on top of the surface. As the number of electrical functions on a chip increases, so does the internal temperature, driving the need for heat dissipation techniques (van Zant (2004)). Increasing power densities worsen thermal impact on reliability and performance, while decreasing supply voltages worsen switching currents and noise. These trends stress on-chip interconnect resources, automatic test equipment limits, and burn-in paradigms (SIA (2005)).

Most important and challenging in the area of front-end process modelling is the modelling of ultra-shallow junction formation, which starts from very low energy implant and especially focuses on the thermal annealing, diffusion and activation of dopants under implantation damage and stress (SIA (2005)). It is obvious, that design productivity must be improved to the same degree as design complexity is scaled.

The reduction of the equivalent gate oxide thickness (EOT) has emerged as the most difficult challenge associated with the future device scaling. Apart from that, implementation into manufacturing of new structures such as ultra-thin body, fully depleted silicon-on-insulator (SOI), and multiple-gate MOSFETs (metal-oxide-semiconductor field-effect transistor) is expected. A particularly challenging issue is the control of the thickness, including its variability, of these ultra-thin MOSFETs (SIA (2005)).

The difficulty of integrating analog and high-performance digital functions on a chip increases with scaling in both device geometry and supply voltage. The steady reduction in analog voltage poses a significant circuit design challenge (SIA (2005)).

Furthermore, the gate electrode material and process should be optimized so that the depletion width in the gate electrode may be minimized and the boron-diffusion prevented. These material changes pose a great challenge in MOSFET technology, where silicon dioxide/polysilicon has long played a central role (SIA (2005)).

By 2016, the industry will be near the end of the basic physics of silicon transistors. However, it is difficult for most people in the semiconductor industry to imagine how it could continue to afford the historic trends of increase in process equipment and factory costs for another 15 years. Post-CMOS (Complementary Metal-Oxide-Silicon) must be addressed. Compound semiconductors, such as gallium arsenide (GaAs) are candidates (SIA (2005)).

The ever-improving economies of scale predicted by Moore's Law do not translate to test. Some contributors to the total cost of test, such as test tooling (e.g. probe cards), are not scaling and threaten to dominate the total test cost if present trends continue (SIA (2005)).

Long term future (2014-2020)

To continue MOSFET scaling to less than 15 nm (DRAM half-pitch; see Fig. 3.2), it is quite likely that the device structure will change to advanced non-classical CMOS such as multiple-gate, ultrathin body (UTB) MOSFETs. In these devices, various "technology boosters", such as mobility enhancement by strained Si, elevated source and source/drain spacing, high- K gate dielectric, and metal gate electrode, will likely be simultaneously implemented with the new device structure. Closer to 2020, devices will increasingly be operated in the quasi-ballistic mode, where the current gain will be enhanced by parameters different from those currently known. Eventually, carbon nanotubes, nanowires, and other high transport channel materials (e.g. germanium or III-V thin channels on silicon) may be needed (SIA (2005)).

3.3 System Characterisation

Study object

In the framework of newecoinvent datasets in the field of electronics, the following datasets for integrated circuits are established:

Name of dataset	Unit	Region	Chapter
wafer, fabricated, for integrated circuit, at plant	m ²	GLO	3.5
integrated circuit, IC, logic type, at plant	kg	GLO	3.6
integrated circuit, IC, memory type, at plant	kg	GLO	3.7

Functional unit(s)

The dataset of the wafer fabrication is expressed in 1 square meter (m²), whereas both the IC types (logic and memory) are calculated per kg in analogy to all the other electronic components described in the subsequent chapters of this part of the report.

Sources & Representativeness

Scanning through the web sites of leading semiconductor companies, it appears that there is an enormous variety of integrated circuits available. However, information and data on process technology and input measures is hardly available. Most semiconductor companies either consider their assessed data as confidential, or then no single process line data has ever been collected.

The combination of the datasets “wafer, fabricated, for integrated circuit, at plant” and “IC, logic/memory type” represents the whole “gate-to-gate” phase of a microchip (Fig. 3.12). The dataset “wafer, fabricated, for integrated circuit, at plant” uses as an input the dataset “single-Si wafer, electronics, at plant”. This dataset links to the whole product chain from raw material extraction to the electronic-grade silicon wafer. The electronic-grade silicon wafer is then processed further, expressed by the dataset “wafer, fabricated, for IC”.

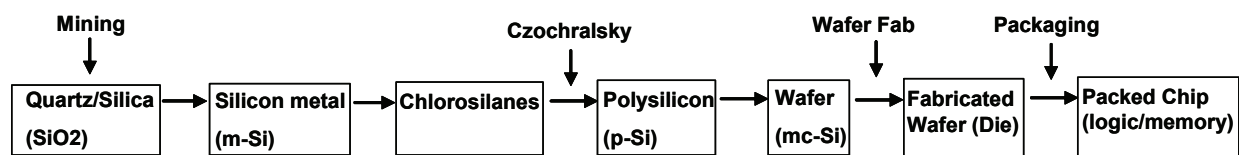


Fig. 3.12 Manufacturing stages from raw material to packaged chip (source: Plepys (2004))

Following these fabrication steps, the wafer (or die) is packaged up with substrate and encapsulation etc. to become a packaged chip. For this report, typical memory and logic type chip configurations are chosen. Sources for wafer fabrication data were mainly literature (Williams et al. (2002)) or Environmental Declarations (ams (2005), Intel (2005c), AMD (2006), ST Microelectronics (2005)), whereas reliable data for packaged chips was found from material declaration reports (ST Microelectronics (2004), Philips Semiconductors (2002)) or literature (Schischke et al. (2002)).

The semiconductor industry has been a global business for decades, and printed wiring boards as carriers for ICs are found in electronic devices throughout all regions of the world. For this reason, the datasets of electronic devices are valid globally (GLO). However, as there are dozens if not hundreds of producer companies in the world, and hundreds of different brands and types of packed integrated circuits, it is impossible to describe an “average” microchip. The information from leading microchip producers and expert literature emphasises the fact that wafer and chip manufacturing is practised with similar processes and technology. Similar amounts of water consumption and emission values for the same production volume supports this statement. Hence, it can be concluded that the data quality represented in this wafer fabrication and IC datasets is rather robust. The authors also tried to picture typical microchips that are a) widely spread, b) world-wide available and c) still applied in electronic devices currently in use.

Configuration of datasets

“wafer, fabricated, for integrated circuit, at plant

The silicon wafer going into wafer fabrication processes is represented by the dataset “single-Si wafer, electronics, at plant”. Wafer fabrication processes include the state-of-the-art etching, deposition or photolithographic processes, applying chemicals, elemental gases and purified water. Wastewater and solid waste flows are also accounted for in the wafer fabrication dataset, as well as all the required process energy input, the water consumption (process and cooling water), air emissions, the rail and road transportation and the infrastructure.

“integrated circuit, IC, logic type, at plant” and “integrated circuit, IC, memory type, at plant”

The two datasets for the logic and memory type microchip are set up in similar ways. Both use as basic input the ecoinvent dataset “wafer, fabricated, for integrated circuit, at plant”(Chapter 3.5), which describes the prepared wafer required for semiconductor fabrication. Further included in the logic and memory type microchip datasets are the leadframe materials (copper etc.), the encapsulation materials (epoxy etc.), the bonding materials, gold wire and leads finishing. These inputs are expressed in material weights; no further processing is inventoried due to unavailable data. The energy used for the assembly process as well as necessary transport for all the materials are included in the logic and memory type IC datasets.

Detailed flow information of the three datasets is provided in Chapters 3.5 to 3.7.

Time period

The data used to represent the microchip fabrication originate mainly from the years 2000 to 2006. All in all, it can be said that the data used to describe the microchip datasets are reasonably recent, maybe not latest available technology (which would be a 300 mm-wafer instead of a 200 mm), but representative for the chips and processors built in computers and other electronic products being currently in use.

Sampling procedure

Parallel to a literature survey with rather poor results, web pages of some leading semiconductor companies (Intel, STMicroelectronics, Philips Semiconductor, AMD etc.) were consulted. For the packaging components of different types of microchips, this collected information was rather good; in case of the wafer fabrication processes, the results of the investigations are sufficient. There, the input flows are established based on two references only (Williams et al. (2002) and ams (2005)). The emission and waste data are derived from AMD (2006), Intel Corporation (2005), ST Microelectronics (2005) and Williams et al. (2002), calculated via the ratio of the respective water consumption data (Chapter 3.5; Tab. 3.5).

3.4 Use & Application

The two IC-datasets listed in Chapter 3.3 can be applied to describe the production of a logic and memory type microchip to be mounted on printed wiring boards of computers and other electronic devices.

Logic type IC

The family of logic type integrated circuits (e.g. microprocessors) is represented by a PBGA, which stands for Plastic Ball Grid Array package.

Memory type IC

Out of dozens of different memory type integrated circuits, one typical kind was chosen, namely the PLCC (Plastic Leaded Chip Carrier). The integrated microchip is a 200-mm wafer 32 MB DRAM.

The same wafer fabrication dataset has been used as basis for the memory and the logic type IC. The two microchip datasets therefore only differ in the constitution of the package components.

Further attention should be put on the fact that cutting-edge ICs are built on 300 mm-wafers and soon on 450 mm-wafers, which is accompanied by even higher purity requirements during wafer fabrica-

tion, shorter gaps between transistors, advanced cooling functions etc. These facts have to be taken into account when applying these datasets.

The microchip datasets are applicable for any region in the world, as integrated circuits are produced in different continents by world-wide operating companies.

3.5 Dataset “wafer, fabricated, for integrated circuit”

3.5.1 Dataset description and meta information

The dataset “Wafer, fabricated, for integrated circuit” represents the number of processes required to prepare an electronic-grade silicon wafer to become an integrated circuit or chip. The functional unit for this dataset is 1 m² of wafer.

General description

Fig. 3.13 shows the schematic setup of input and output flows regarding wafer fabrication processes. Major wafer fabrication steps are layering, patterning, doping and heat treatments (see Chapter 3.2.3). These processes are reflected in the established dataset through input of chemicals and elemental gases on the one hand, and electricity/fossil energy and water input on the other hand. 50 % of the total energy consumption for wafer fabrication is for clean room heating, ventilation and air conditioning, whereas the wafer processing tools account for 30-40 % of the total energy use. A typical purification system will generally take municipal (tap) water with impurity levels in the parts per hundred or parts per thousand to purify it to the few parts per billion level (Williams et al. (2002)). The output side of the dataset is represented by wastewater, solid waste and different types of air emissions (as shown in Tab. 3.9).

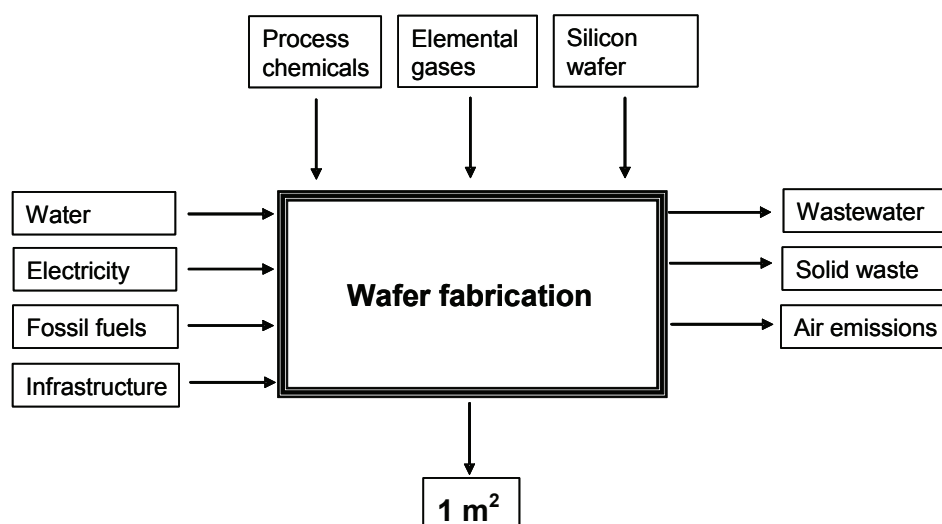


Fig. 3.13 Input/output scheme of the wafer fabrication process (based on Williams et al. (2002) and Schischke et al. (2002))

Input flows

Tab. 3.4 shows producer information for input process data and the chosen values applied to the ecoinvent dataset. The overall process yield of wafer fabrication varies between 16 and 94 %, depending on the complexity and maturity of the technology (Williams et al. (2002)). For the calculations of this dataset, the average of this range has been taken, 55%.

The two sources Williams et al. (2002) and ams (2005) show remarkable differences in values for some elemental gases and especially fossil fuel inputs. For the elemental gases N₂, O₂ and H₂, the values mentioned in ams (2005) are applied. Concerning the energy input to wafer fabrication, modern chip factories mainly use electricity and natural gas as energy carriers (ams (2005); Intel (2005b)). For instance diesel or gasoline only makes up 0.4% of the total energy consumption in 2004 for the entire Intel semiconductor fabrication sites (Intel (2005a)). For this reason, the energy input for this dataset is only represented by electricity (63%) and natural gas (37%) according to the numbers published by ams (2005).

Tab. 3.4 Specific input data according to different sources and values taken for ecoinvent

Substance/chemical:	Williams (2002)	austriamicro-systems (2004)	Value taken for ecoinvent dataset
N ₂	8000 kg/m ²	5060 kg/m ²	5060 kg/m ²
O ₂	54.45 kg/m ²	214 kg/m ²	214 kg/m ²
H ₂	0.836 kg/m ²	4.2 kg/m ²	4.2 kg/m ²
Natural gas (and heavy fuel oil etc.)	18'187 MJ/m ²	58'959 MJ/m ²	58'959 MJ/m ²
Electricity	98'179 MJ/m ²	99'360 MJ/m ²	99'360 MJ/m ²
Water high purity	200 m ³ /m ²	96.5 m ³ /m ²	96 m ³ /m ²
Water for cooling		84.6 m ³ /m ²	85 m ³ /m ²

The calculation of the infrastructure for the wafer fabrication is based on the following information:

- Assumed life-time of the factory: 25 years
 - Production volume: 40'000 wafers per month (200-mm wafer or 0.031415 m²)
 - Process yield: 55%
- > Calculation: = $1/(25 \cdot 40000 \cdot 12 \cdot 0.031415 \cdot (1/0.55))$ --> dataset is assessed per 1 m².

Typical ecoinvent standard transport distances for road (50 km/100 km/200 km; depending on the material) and rail (100 km/200 km/600 km; depending on the material) were applied for the supply of chemicals and raw materials.

Process emissions to air and soil (output flows)

The air emissions caused by wafer fabrication are extracted from the Sustainability Report 2005 by AMD (2006), from the Corporate Responsibility Report of ST Microelectronics (ST Microelectronics (2005)), and from the Project XL Report by Intel (Intel Corporation (2005)). In case of AMD, the values of perfluorocompound gas, corrosives and VOCs stem from the wafer fabrication site in Austin, USA (Fig. 3.14), whereas the various consumption and emission data of Intel stand for the Ocotillo site in Arizona, USA.

Most of the output numbers for the inventory have been calculated indirectly via the water usage. This is due to the fact that often the overall emission values in environmental reports are not related to a production volume. As an example for the calculation setup, the amount for the total water consumption of different companies has been compared to the firm data published in Williams et al. (2002), see Tab. 3.5.

SUSTAINABILITY PERFORMANCE METRICS TABLES, 1 OF 2

Basis of Measurement	AMD recordable injury and illness case rate		Total water use		Normalized water use		Total water conservation ²		Electricity consumed	
	Per 100 workers		Liters		per MP ³		Liters		KWH ⁴	
WAFA FABRICATION										
AUSTIN '05	0.90		2,206,000,489		5.58		55,657,143		284,315,193	
'04	0.94		2,439,321,191		4.51		258,705,896		286,446,993	
'03	1.22		2,885,100,438		8.86		816,441,690		285,594,267	
'02	1.66		3,291,987,495		13.75		1,626,227,778		311,537,186	
'01	1.53		3,511,058,056		9.79		1,650,583,030		334,360,408	

⁴ MMTCE = Million Metric Tons
of Carbon Equivalent

SUSTAINABILITY PERFORMANCE METRICS TABLES, 2 OF 2

Basis of Measurement	Perfluorocompound gas emissions		Air emissions: Corrosives		Air emissions: VOCs	
	MMTCE ⁵		Kilograms		Kilograms	
WAFA FABRICATION						
AUSTIN '05	0.0213		2,809		24,473	
'04	0.0348		3,298		27,947	
'03	0.0260		2,126		33,938	
'02	0.0326		4,384		51,212	
'01	0.0582		8,359		53,195	

Fig. 3.14 a+b Sustainability Performance Metrics Tables for AMD's wafer fabrication site at Austin (USA) for the period 2001-2005 (source: AMD (2006))

To figure out whether the ratio between the water consumption data of different companies would hold also for other measurements, the VOC and PFC emission data are compared (Tab. 3.5). In the case of Intel versus AMD, the ratio factor varies between 8.3 (for VOC) and 9.5 (water consumption). This is an acceptable range that allows using the ratio for further calculations of emission data. Also the water consumption and VOC emission ratio between Intel and STMicroelectronics is within an acceptable range (1.1 and 0.8).

Tab. 3.5 Calculations and ratios to derive values for emissions and other indicators in wafer fabrication

Material type	Williams (for 2000)	AMD Austin (for 2004)	ST Microelec- tronics (for 2004)	Intel (for 2004)	Factor (ratio)
Water consumption (in litres)	364'000 (total for 1m ²)	2'439'321'191 (total, absolute value)	20'550'000'000 (total for entire company, absolute value)	23'144'940'000 (total for entire company, absolute value)	♣ 6701 (AMD : Williams) ♣ 9.5 (Intel : AMD) ♥ 1.1 (Intel : STMicroel.) ♦ 8.4 (STMicroel. : AMD) 63'585 (Intel : Williams) 56'456 (STMicroel. : Wil- liams)
VOC emissions (to air; in kg)	-	27'947 (total, absolute value)	294'000 (total for entire company, absolute value)	232'000 (total for entire company, absolute value)	♣ 8.3 (Intel : AMD) ♥ 0.8 (Intel : STMicroel.) ♦ 10.5 (STMicroel. : AMD)
PFC emissions (to air; in kg Car- bon-equivalents)	-	34'800'000 (total)		300'000'000 (total)	♣ 8.6 (Intel : AMD)

The established emission output data including the calculation setup is displayed in Tab. 3.6. According to ecoinvent standards, water treatment is looked at as a service and is therefore entered as an input flow in the dataset (see under subtitle “Process emissions to water: Waste water effluent” and Tab. 3.7). Inventoried air emissions released during wafer fabrication are Perfluorocompound gas (taken as C_2F_6), corrosives in form of hydrogen fluoride (HF), organic and inorganic HAPs (Hazardous Air Pollutants; summarised as Methanol and Hydrogen fluoride), Phosphine (PH_3), Arsine (AsH_3), VOCs and ozone depleting substances (as R11(CFC-11) equivalents, taken as CFC-113).

Although AMD writes in their Environmental, Health and Safety FAQs Report (AMD (2004)) that “AMD completed the use of ozone depleting substances (ODSs) in manufacturing processes in 1993”, companies like STMicroelectronics still report emissions of ozone depleting substances (in R11-equivalents) until year 2005 (ST Microelectronics (2005)). Therefore, even though numbers are drastically declining, these emissions are inventoried.

Tab. 3.6 Sources, values and calculations of emission data for the wafer fabrication dataset

Emission type	Original data	Source	Calculation ⁵ (see ratios in Tab. 3.5)	ecoinvent value (per m ² of wafer)
Wastewater	188'000 m ³ per 1948 m ²	ams (2005)	= 188'000/1948	9.65E+01 m ³
Water treatment	See Tab. 3.7 for inputs	ST Microelectronics (2005)	Calculation according to waste treatment ecoinvent Excel tool by Doka (2003)	---
Solid waste	7.8 kg per 1 cm ²	Williams et al. (2002)	= 7.8*10'000*(1/0.55)	1.42E+05 kg
Perfluorocompound gas (PFC)	0.0348 MMTCE (2004)	AMD (2006)	= (0.0348*1'000'000'000*12/44/11900)/6701 ⁶	1.19E-01 kg
Corrosives (HF)	3298 kg (2004)	AMD (2006)	= 3298/6701	4.92E-01 kg
Organic HAPs	1060 kg (2004)	Intel Corporation (2005)	= 1060*0.1/63585	1.67E-03 kg
Inorganic HAPs	779.73 kg ⁷ (2004)	Intel Corporation (2005)	= (800-20.27)*0.1/63585	1.23E-03 kg
Phosphine	20 kg (2004)	Intel Corporation (2005)	= 20*0.1/63585	3.15E-05 kg
Arsine	0.62 pounds (2004)	Intel Corporation (2005)	= 0.62*0.435*0.1/63585	4.24E-07 kg
VOCs	27947 kg (2004)	AMD (2006)	= 27947/6701	4.17E+00 kg
Sulfuric acid (H ₂ SO ₄)	30 kg (2004)	Intel Corporation (2005)	= 30*0.1/63585	4.72E-05 kg
Ozone depl. substances (mainly Freon 113)	78 kg CFC-11 equals 97.5 kg CFC-113 ⁸	ST Microelectronics (2005)	= (1/0.8*78)/56456	1.73E-03 kg

Sulfuric acid (H₂SO₄) emissions to soil are also inventoried (Tab. 3.6; compare also flow information in Tab. 3.9). The data retrieved from Intel Corporation (2005) represent numbers for the Ocotillo site in Chandler, Arizona, and make up about 10% of the total consumption and emissions of the entire In-

⁵ Process yield derived from Williams 2002 (average yield 0.55 of mentioned range 0.16 – 0.94)

⁶ 1 kg C_2F_6 is 11900 kg CO₂-eq. (www.epa.gov/climatechange/glossary.html)

⁷ 800 kg for all inorganic HAPs minus 20 kg of Phosphine emissions and minus 0.62*0.435 kg (1 pound) = 0.27 kg of Arsine

⁸ The ODP (ozone depletion potential) of CFC-113 is 0.8 of the potential of CFC-11; <http://www.epa.gov/ozone/defs.html>

tel Company (total of 15 wafer fabs at 9 locations ⁹). Therefore the calculation of those numbers includes a multiplication factor 0.1.

Process emissions to water: Waste water effluent

The emissions to water data is taken from ST Microelectronics (2005). A separate dataset “treatment, wafer fabrication effluent, to wastewater treatment, class 2” is created, based on the Excel tool developed in the framework of the waste section in ecoinvent (see Doka (2007)). Tab. 3.7 shows the major wastewater emission substances and their amount (in kg) per kg wastewater. It is assumed that the eutrophication is split into 50% share of P and 50% share of N. The aquatic ecotoxicity again is 33% Cu, 33% Ni and 33% Si. The heavy metals are not specified either in the CR Report of ST Microelectronics, therefore the composition is assumed to be 20% Cd, 20% As, 20% Cr, 20% Mg and 20% Pb.

Tab. 3.7 Composition of the waste water effluent for the process “Wafer fabrication” (Source: ST Microelectronics (2005))

Impact Category / Substance	Unit	Amount	Remarks
Eutrophication [in P; 50% of total]	kg/m ³ wastewater	0.016 ¹⁰	177'122 kg (total for P and N in 2004)
Eutrophication [in N; 50% of total]	kg/m ³ wastewater	0.016	177'122 kg (total for P and N in 2004)
Heavy metals [in Cd; 20%]	kg/m ³ wastewater	0.0007	19'520 kg (total for heavy metals)
Heavy metals [in As; 20%]	kg/m ³ wastewater	0.0007	19'520 kg (total for heavy metals)
Heavy metals [in Cr; 20%]	kg/m ³ wastewater	0.0007	19'520 kg (total for heavy metals)
Heavy metals [in Mg; 20%]	kg/m ³ wastewater	0.0007	19'520 kg (total for heavy metals)
Heavy metals [in Pb; 20%]	kg/m ³ wastewater	0.0007	19'520 kg (total for heavy metals)
Chemical Oxygen Demand [COD]	kg/m ³ wastewater	0.095	518'935 kg (total for 2004)
Aquatic ecotoxicity [in Cu; 33%]	kg/m ³ wastewater	0.0066	10'772 kg (total for aquatic ecotoxicity in 2004)
Aquatic ecotoxicity [in Ni; 33%]	kg/m ³ wastewater	0.0066	10'772 kg (total for aquatic ecotoxicity in 2004)
Aquatic ecotoxicity [in Si; 33%]	kg/m ³ wastewater	0.0066	10'772 kg (total for aquatic ecotoxicity in 2004)

Further output flows

Solid waste, as well as the required transport by rail and road according to ecoinvent standard distances for chemicals is also inventoried.

More information on the wafer fabrication dataset can be retrieved from the general flow information sheets in Tab. 3.9, as well as from the meta information presented in Tab. 3.8.

⁹ “FAQs about Intel Manufacturing and Production-Related Services”; http://www.intel.com/pressroom/kits/manufacturing/manufacturing_ga.htm#1

¹⁰ Calculation setup: $177122 \text{ kg} / 56456 / 96.5 / 2 = 0.016 \text{ kg [P] per kg wastewater}$; factor 56456 is taken from Tab. 3.5, and factor 96.5 stands for the amount of wastewater (96.5 m³) discharged for 1 m² of wafer (austriamicrosystems 2005).

Tab. 3.8 ecoinvent meta information of the wafer fabrication dataset

Name	wafer, fabricated, for integrated circuit, at plant
Location	GLO
Infrastructure Process	0
Unit	m ²
Dataset Version	2.0
Included Processes	Describes the process of wafer fabrication. Calculated per m ² of wafer. Consumes in the process elemental gases like helium nitrogen or argon; deposition or dopant gases like silane, phosphine or arsine; etchants like ammonia, nitrous oxide, chlorine, hydrogen chloride or hydrofluoric acid; acids like nitric acid or sulfuric acid; photolithographic chemicals like acetone. Further inventoried are electricity input for the whole wafer fabrication process (layering, patterning, doping, heat treatment), consumption of direct fossil fuels (natural gas), water consumption for the purification processes and the cooling, emissions to air (e.g. VOCs, PFC, HAPs, etc.) and water, infrastructure as well as road and train transport for chemicals.
Amount	1
Local Name	Wafer, fabriziert, für integrierte Schaltung, ab Werk
Synonyms	Layering//Patterning//Doping//Heat Treatment
General Comment	This dataset can be applied to describe the processing of a wafer used for integrated circuit fabrication. The original input data for chemicals and elemental gases represents measures in grams per cm ² of input wafer. The information stems partly from an anonymous firm describing material input/output for a mid-sized, state-of-the art integrated circuit fabrication facility in Western Europe in 2000. The infrastructure is calculated for a company producing 40'000 200-mm wafers per month, each weighing 168 kg. The life-span of the facility is assumed to be 25 years, a photovoltaic cell factory is taken as proxy. Transportation input data is based on ecoinvent standard distances for chemicals in Europe.
Start Date	2000
End Date	2006
Data valid for entire period	1
Geography Text	The input data is based on a chip factory in Western Europe representing global average. Emissions, water and energy inputs are retrieved from three different wafer fabrication sites in the USA and one company in Central Europe.
Technology Text	The wafer fabrication can include several thousand individual steps, which are grouped into the major operation classes called layering, patterning, doping and heat treatments. The technology applied during these operations is represented by the material input of chemicals and the energy used in these operations.
Representativeness	unknown
Production Volume	unknown
Sampling Procedure	Manufacturer information from literature and sustainability reports.
Extrapolations	see Geography and Technology
Uncertainty Adjustments	see Geography and Technology

3.5.2 Life Cycle Inventory and General Flow information

The layout of the dataset “wafer, fabricated, for integrated circuit, at plant” with general flow information, remarks, sources, values and uncertainty information is shown in Tab. 3.9 below.

Tab. 3.9 Life cycle inventory for “wafer, fabricated, for integrated circuit” and their representation in the ecoinvent database (PART 1)

General Flow information					Representation in ecoinvent					Uncertainty				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Silicon wafer for IC	Fabricated wafer for integrated circuit PART 1	↑	Electronic grade silicon wafer to be fabricated; process yield 55%	photovoltaic	production of components	No	RER	single-Si wafer, electronics, at plant	182	m2	STM microelectronics (2004)	1	125	(2,4,2,3,153)
He		↑	Helium; elemental gas; input 0.17 g/cm2 of wafer; process yield for wafer fabrication taken as average (55%) of the range (16 - 94 %) mentioned in Williams, E. (2002)	chemicals	inorganics	No	GLO	helium, at plant	3.09	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
N2		↑	Nitrogen; elemental gas	chemicals	inorganics	No	RER	nitrogen, liquid, at plant	5060.0	kg	austriamicrosystems (2005)	1	125	(2,4,2,3,153)
O2		↑	Oxygen; elemental gas; worst case	chemicals	inorganics	No	RER	oxygen, liquid, at plant	214.0	kg	austriamicrosystems (2005)	1	125	(2,4,2,3,153)
Ar		↑	Argon; elemental gas; input 1.7 g/cm2; process yield 55%	chemicals	inorganics	No	RER	argon, liquid, at plant	30.91	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
H2		↑	Hydrogen; elemental gas; worst case	chemicals	inorganics	No	RER	hydrogen, liquid, at plant	4.20	kg	austriamicrosystems (2005)	1	125	(2,4,2,3,153)
Silane SiH4		↑	Deposition/dopant gas; input 0.0078 g/cm2; process yield 55%	chemicals	inorganics	No	RER	silicone product, at plant	0.14	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Phosphine PH3		↑	Deposition/dopant gas; inorganic chemical as proxy; input 0.000017 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	phosphane, at plant	0.0003	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Arsine (AsH3)		↑	Deposition/dopant gas; inorganic chemical as proxy; input 0.0000046 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	arsine, at plant	0.0001	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Diborane (B2H6)		↑	Deposition/dopant gas; input 0.0000046 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	diborane, at plant	0.0001	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Dichlorosilane (SiH2Cl2)		↑	Deposition/dopant gas; input 0.0014 g/cm2; process yield 55%	chemicals	inorganics	No	RER	silicone product, at plant	0.03	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Ammonia (NH3)		↑	Etchant; input 0.012 g/cm2; process yield 55%	chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	0.22	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
N2O		↑	Nitrous oxide; etchant; input 0.072 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	1.31	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)

Tab. 3.9 Life cycle inventory for “wafer, fabricated, for integrated circuit” and their representation in the ecoinvent database (PART 2)

General Flow information				Representation in ecoinvent						Uncertainty				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Cl2	Fabricated wafer for integrated circuit PART 2	↑	Chlorine; etchant; input 0.0048 g/cm2; process yield 55%	chemicals	inorganics	No	RER	chlorine, liquid, production mix, at plant	0.09	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
BCl3		↑	Trichloroborane; etchant; Trichloromethane as proxy; input 0.0087 g/cm2; process yield 55%	chemicals	organics	No	RER	trichloromethane, at plant	0.16	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
BF3		↑	Trifluoroborane; etchant; Trichloromethane as proxy; input 0.000035 g/cm2; process yield 55%	chemicals	organics	No	RER	trichloromethane, at plant	0.0006	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
HBr		↑	Hydrogen bromide; etchant; proxy HCl; input 0.0022 g/cm2; process yield 55%	chemicals	inorganics	No	RER	hydrochloric acid, from the reaction of hydrogen with chlorine, at plant	0.04	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
HCl		↑	Hydrogen chloride; etchant; input 0.005 g/cm2; process yield 55%	chemicals	inorganics	No	RER	hydrochloric acid, from the reaction of hydrogen with chlorine, at plant	0.09	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
HF		↑	Hydrofluoric acid; etchant; input 0.005 g/cm2 plus 1/3 12.84 g/cm2 (of acids/bases; see NH4 below); process yield 55%	chemicals	inorganics	No	GLO	hydrogen fluoride, at plant	168	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
NF3		↑	Nitrogen trifluoride; etchant; proxy "chemicals, inorganic"; input 0.0023 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	0.04	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
WF6		↑	Tungsten fluoride; etchant; input 0.00043 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	0.01	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
SF6		↑	Sulfur hexafluoride; etchant; input 0.0065 g/cm2; process yield 55%	chemicals	inorganics	No	RER	sulphur hexafluoride, liquid, at plant	0.12	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
C2F6		↑	Hexafluoroethane; etchant; input 0.05 g/cm2; process yield 55%	chemicals	organics	No	GLO	hexafluoroethane, at plant	0.91	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
CHF3		↑	Fluoroform; etchant; input 0.031 g/cm2; process yield 55%	chemicals	organics	No	GLO	trifluoromethane, at plant	0.56	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
CF4		↑	Carbon tetrafluoride; etchant; input 0.03 g/cm2; process yield 55%	chemicals	inorganics	No	RER	tetrafluoroethylene, at plant	0.55	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)

Tab. 3.9 Life cycle inventory for “wafer, fabricated, for integrated circuit” and their representation in the ecoinvent database (PART 3)

General Flow information				Representation in ecoinvent						Uncertainty				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
HF and NH4 mix	Fabricated wafer for integrated circuit PART 3	↑	Acid/base: HF 1 vol; NH4 30 vol; hydrofluoric acid and ammonium; the share of HF-acid is included in the HF-at-chant input (above); input of NH4 2.84 g/cm2 (30/31)	chemicals	organics	No	RER	ammonium carbonate, at plant	49.97	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
H3PO4		↑	Acid/base: Phosphoric acid 86% input 0.86*2.41g/cm2; process yield taken as average (55% of the range (16 - 94 % mentioned in Williams, E. (2002)	chemicals	inorganics	No	RER	phosphoric acid, industrial grade, 85%in H2O, at plant	37.68	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Hydrofluoric acid		↑	Acid/base: Hydrofluoric acid 0.5% input 0.005*3.42 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	hydrogen fluoride, at plant	0.31	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Hydrofluoric acid		↑	Acid/base: Hydrofluoric acid 5% input 0.05*0.455 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	hydrogen fluoride, at plant	0.41	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Hydrofluoric acid		↑	Acid/base: Hydrofluoric acid 50% input 0.5*0.252 g/cm2; process yield 55%	chemicals	inorganics	No	GLO	hydrogen fluoride, at plant	2.29	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Nitric acid		↑	Acid/base: Nitric acid (HNO3) 70% input 0.7*1.19 g/cm2; process yield 55% adjusted to applied ecoinvent dat aset	chemicals	inorganics	No	RER	nitric acid, 50%in H2O, at plant	15.15	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Sulfuric acid		↑	Acid/base: Sulfuric acid (H2SO4) 96% input 0.96*7.85 g/cm2; process yield 55%	chemicals	inorganics	No	RER	sulphuric acid, liquid, at plant	137.02	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Hydrochloric acid		↑	Acid/base: Hydrochloric acid 30% input 2.52 g/cm2; process yield 55%	chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	45.82	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Ammonia (NH3)		↑	Acid/base: Ammonia (NH3) 28% input 0.28*0.776 g/cm2; process yield 55%	chemicals	inorganics	No	RER	ammonia, liquid, at regional storehouse	3.95	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Slurry		↑	Slurry for CMP, chemical mechanical polishing; input 0.286 g/cm2	washing agents	builders	No	RER	zeolite, slurry, 50%in H2O, at plant	5.20	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
HCl		↑	Acid/base: Hydrogen chloride 30% input 0.3*0.506 g/cm2; process yield 55%	chemicals	inorganics	No	RER	hydrochloric acid, from the reaction of hydrogen with chlorine, at plant	2.76	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)

Tab. 3.9 Life cycle inventory for “wafer, fabricated, for integrated circuit” and their representation in the ecoinvent database (PART 4)

General Flow information				Representation in ecoinvent					Uncertainty					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
NaOH	Fabricated wafer for integrated circuit PART 4	↑	Acid/base: Sodium hydroxide 50% input 0.570,657 g/cm2; process yield 55%	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	1184	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Hydrogen peroxide		↑	Photolithographic chemical: H2O2 30% input 0.3*4,43 g/cm2; process yield 55%	chemicals	inorganics	No	RER	hydrogen peroxide, 50% in H2O, at plant	24.16	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Isopropyl alcohol		↑	Photolithographic chemical: isopropyl alcohol C3H8O; input 2.02 g/cm2; process yield 55%	chemicals	organics	No	RER	isopropanol, at plant	36.73	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Tetramethylammonium hydroxide		↑	Photolithographic chemical: C4H12N+; Trimethylamine as proxy; input 4.31 g/cm2; process yield 55%	chemicals	organics	No	RER	trimethylamine, at plant	78.36	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Methyl-3-methoxypropionate		↑	Photolithographic chemical: CH3OCH2CH2COOCH3; "Chemicals, organic, at plant" as proxy; input 1.48 g/cm2; process yield 55%	chemicals	organics	No	GLO	methyl-3-methoxypropionate, at plant	26.91	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Acetone		↑	Photolithographic chemical: C3H8O; input 0.554 g/cm2; process yield 55%	chemicals	organics	No	RER	acetone, liquid, at plant	10.07	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Hexamethyldisilazane		↑	Photolithographic chemical: C6H18NSi2; "Chemicals, organic, at plant" as proxy; input 0.022 g/cm2; process yield 55%	chemicals	organics	No	GLO	hexamethyldisilazane, at plant	0.40	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Hydroxyl monoethanolamine		↑	Photolithographic chemical: C2H7NO; input 1.42 g/cm2; process yield 55%	chemicals	organics	No	RER	monoethanolamine, at plant	25.82	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
NaOH		↑	Sodium hydroxide for neutralising wastewater; concentration unclear; input 7.6 g/cm2; process yield 55%	chemicals	inorganics	No	RER	sodium hydroxide, 50% in H2O, production mix, at plant	138.18	kg	Williams, E. (2002)	1	125	(2,4,2,3,153)
Water treatment		↑	Wastewater treatment; included are Eutrophication substances (P+N), COD, heavy metals (Cd, As, Cr, M.g, Pb) and aquatic ecotox/cants (Cu, Ni, Si); output calculated via Excel Tool (ecoinvent)	waste management	wastewater treatment	No	CH	treatment, wafer fabrication effluent, to wastewater treatment, class 2	96.51	m3	austriamicrosystems (2004)	1	124	(2,4,13,156)

Tab. 3.9 Life cycle inventory for “wafer, fabricated, for integrated circuit” and their representation in the ecoinvent database (PART 5)

General Flow information				Representation in ecoinvent					Uncertainty						
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Fabricated wafer for integrated circuit	PART 5	↑	Solid waste	Solid waste 7.8 kg per 1cm2 of wafer input; 10000*7.8 kg for 1m2 of wafer input; process yield 55%	waste management	residual material landfill	No	CH	disposal, waste, Si waferprod., inorg. 9.4%wafer, to residual material landfill	141818.18	kg	Williams, E. (2002)	1	129	(3.4,3.3,156)
		↑	Perfluoro-compound gas	Taken as "Ethane, hexafluoro-, HFC-116" C2F6; input value 0.0348 MM TCE (Million Metric Tons of Carbon Equivalent emissions); 1kg C2F6 equals 1800 kg CO2.	air	high population density			Ethane, hexafluoro-, HFC-116	0.12	kg	AMD (2006)	1	2.07	(2.4,2.3,15.23)
		↑	Corrosives	Taken as Hydrogen fluoride	air	high population density			Hydrogen fluoride	0.49	kg	AMD (2006)	1	1.58	(2.4,2.3,15.31)
		↑	Organic HAPs	Organic HAP emissions to air (Methanol, Xylene, Ethylene glycol, Acetonitrile); represented by Methanol	air	high population density			Methanol	0.0017	kg	Intel Corporation (2005)	1	1.58	(2.4,2.3,15.31)
		↑	Inorganic HAPs	Inorganic HAP emissions to air (Hydrofluoric acid; Chlorine; Hydrochloric acid); represented by Hydrofluoric acid; calculated via water consumption ratio Intel vs. Williams; Phosphine and Arsine (also inorganic HAPs) are inventoried separately and are deducted from this HAP total	air	high population density			Hydrogen fluoride	0.0012	kg	Intel Corporation (2005)	1	1.58	(2.4,2.3,15.31)
		↑	Phosphine	Phosphine (PH3) emissions to air (20 kg Intel Ocotillo site); evaluated separately, although part of inorganic HAPs	air	high population density			Phosphine	0.00003	kg	Intel Corporation (2005)	1	1.58	(2.4,2.3,15.31)
		↑	Arsine	Arsine (AsH3) emissions to air (0.62 pounds for Intel Ocotillo site); evaluated separately, although part of inorganic HAPs	air	high population density			Arsine	0.0000004	kg	Intel Corporation (2005)	1	1.58	(2.4,2.3,15.31)
		↑	VOC	VOC emissions to air; taken as NM VOC	air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	4.17	kg	AMD (2006)	1	2.07	(2.4,2.3,15.23)
		↑	H2SO4	Sulfuric acid emissions (0.03 tons for Intel Ocotillo site)	soil	agricultural			Sulfuric acid	0.00005	kg	Intel Corporation (2005)	1	2.07	(2.4,2.3,15.23)
		↑	Ozone depleting substances	Ozone depleting substances in R11-equivalents (CFC-11); 78 kg absolute terms; taken as CFC-113	air	high population density			Ethane, 1,1,2-trichloro-1,2,2-trifluoro-, CFC-113	0.0017	kg	ST Microelectronics (2005)	1	1.58	(2.4,2.3,15.31)

Tab. 3.9 Life cycle inventory for “wafer, fabricated, for integrated circuit” and their representation in the ecoinvent database (PART 6)

General Flow information				Representation in ecoinvent					Uncertainty					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
Electricity	Fabricated wafer for integrated circuit PART 6		UCTE electricity mix taken as proxy for global mix; electricity input for 1m2 wafer fabrication is approx. 15 kWh	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	27600.0	kWh	austriamicrosystems (2004)	1	120	(2,4,3,3,14,2)
Direct fossil fuels (natural gas)			Natural gas 100% of total fossil energy consumption	natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	58959.0	MJ	austriamicrosystems (2004)	1	127	(3,3,3,3,15,1)
Process water			Water for the purification process; 188'000 m3 for 62000 wafers (equals 1948 m2)	water supply	production	No	GLO	water, ultrapure, at plant	96509.2	kg	austriamicrosystems (2004)	1	125	(2,3,3,3,15,12)
Cooling water			Cooling water including evaporation and input for osmosis process; 165000 m3 for 62000 wafers (equals 1948 m2)	water supply	production	No	RER	tap water, at user	84702.3	kg	austriamicrosystems (2004)	1	125	(2,3,3,3,15,12)
Infrastructure			Infrastructure similar to photovoltaic cell factory; life-time 25 years; production volume 40'000 200-mm wafers per month; process yield 55%	photovoltaic	production of components	Yes	DE	photovoltaic cell factory	0.00000015	unit	ecoinvent	1	3.11	(2,4,13,3,5,9)
Transport road			Transportation of the chemicals to the wafer processing plant by road	transport systems	road	No	RER	transport, lorry >16t, fleet average	357.3	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.08	(3,4,2,3,15,5)
Transport train			Transportation of the chemicals to the wafer processing plant by train	transport systems	train	No	RER	transport, freight, rail	10'17.3	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.08	(3,4,2,3,15,5)
				Input and output data calculated for 1m2 of fabricated wafer	electronics	component	No	GLO	wafer, fabricated, for integrated circuit, at plant	1	m2			

Data Uncertainty and Input Data for ecoinvent Database

The input values of the dataset „wafer, fabricated, for integrated circuit, at plant“ are based on data by an anonymous wafer fabrication company in Western Europe in the year 2000 and partly on ams (2005). The values for process emissions to air are derived from the 2005 Sustainability Report by AMD (2006), from the Corporate Responsibility Report of ST Microelectronics (2005), and from the Project XL Report by Intel Corporation (2005)). It has been considered in the uncertainties of the dataset that the air emission values had to be calculated via the ratio of the water consumption of different companies.

There are no natural resources inventoried for this wafer fabrication dataset. The water input is reported in two different forms, as cooling and process (“high-purity”) water, which is different to groundwater or river water. The infrastructure is approximated by the existing ecoinvent dataset for a photovoltaic cell factory with an assumed life-time of 25 years and monthly production volume of 40'000 200mm-wafers. Nine types of direct air emissions (e.g. PFC gas, corrosives, HAPs, VOCs) are balanced for this fabrication dataset according to the three sources mentioned above. A major part of the emissions goes also into the wastewater (17 kg per cm² of wafer input). The wastewater treatment is inventoried by an auxiliary dataset called “treatment, wafer fabrication effluent, to wastewater treatment, class 2”. Solid waste (7.8 kg per cm² of wafer input) is referred to as “disposal, waste, Si wafer production, inorganic, 9.4% water, to residual material landfill”. This reflects the process rather well.

3.6 Dataset “integrated circuit, IC, logic type”

3.6.1 Dataset description and metainformation

The dataset “integrated circuit, IC, logic type, at plant” represents the processes and materials required to package up a fabricated wafer into a logic type microchip. The functional unit of this dataset is 1 kg of microchip in accordance with all the other printed wiring board components (transistors, resistors, etc.) presented in this report.

Major inputs to this dataset are the fabricated wafer in form of the dataset “wafer, fabricated, for integrated circuit, at plant” (Chapter 3.5), the materials for the substrate (epoxy resin, copper, nickel, gold, resist and TBBA), the materials for the encapsulation (e.g. SiO₂), the die attach material (epoxy and silver glue), the gold wires, and the solderballs (Sn and Pb). Further inventoried are the energy input for the assembly process as well as the required transport by ship, rail and road.

For the transportation data, investigations were made to find out where in the world most of the 200-mm wafer fabrication actually takes place (front-end manufacturing), and where the chips are assembled (back-end manufacturing). This is to assure the quality standards of a globally valid dataset (GLO). Sources of the major chip producers Intel, AMD and STMicroelectronics were checked. As Tab. 3.10 shows, front-end and back-end chip manufacturing happens very often far away from each other. The transport by ship is calculated via an average return travel distance (21'300 km) from Asia (Japan) to Central Europe (Zurich). The distance from the front-end site to the harbour in Asia and from the harbour to the back-end factory in Europe (assumed example) is approximated by the ecoinvent standard distances (2*100 km for road and 2*600 km for rail transport, for chemicals).

Tab. 3.10 Front-end and back-end wafer fabrication sites of three major manufacturers in the world

Company	Front-end wafer fabrication sites: Countries	Back-end wafer fabrication sites: Countries
Intel (information as of 2007) ¹¹	USA: 6 sites Ireland: 1 site Israel: 2 sites	China: 2 sites Costa Rica: 1 site Malaysia: 2 sites Philippines: 1 site
AMD (information as of 2007) ¹²	Austin/USA: 1 site Germany: 1 site Japan: 1 site	Thailand: 1 site Malaysia: 2 sites Singapore: 1 site China: 1 site
STMicroelectronics (information as of 2007) ¹³	Italy: 2 sites France: 2 sites USA: 1 site Singapore: 1 site China: 1 site	Malta: 1 site Morocco: 2 sites Malaysia: 1 site Singapore: 1 site China: 1 site

The technology applied in chip assembly includes wafer backgrind, die preparation (wafer mounting and wafer sawing), die attach, wirebonding, die overcoat, molding, sealing, marking, DTFS (de-flash/trim/form/singulation), leadfinish (see SiliconFarEast for further information on these individual steps¹⁴).

The data for this dataset is based mainly on information from a publication by ST Microelectronics in 2004 named “Chemical content of semiconductor packaging” (ST Microelectronics (2004)), cross-checked with data presented by Philips Semiconductors (2002). The process data for total energy consumption (5.8 MJ per 2 g chip) is retrieved from the paper by Williams et al. (2002). The total energy for assembly is allocated with 17% to “heat from heavy fuel oil” and with 83% to “electricity, medium voltage, UCTE”.

The percentages for the IC package components were chosen based on the two references mentioned above (for details see Fig. 3.15 and Fig. 3.16). The chosen values to calculate this ecoinvent dataset are presented in Tab. 3.11, last column. Out of a large number of different logic chip types, the decision fell on the SOT514 Plastic Ball Grid Array package by Philips and a comparable PBGA256 by ST Microelectronics. The total weight of the two logic type ICs differ, as the SOT514 weighs about 3.6 g and the PBGA256 microchip 2.6 g. Still, the percentage shares of the materials going into the packaged IC are comparable (major differences only in weight shares for substrate and solderballs). The body size of the selected packaged chip is 27mm*27mm*2.36mm. The share of the processed wafer is 2.0% of the total packaged IC weight. The required fabricated wafer surface for a 1 kg packaged logic type chip is therefore calculated as follows:

$$= (0.027*0.027)*0.02/0.00262 = 0.0056 \text{ m}^2$$

with the numbers in brackets representing the surface of the total packaged chip, and the value 0.00262 representing the weight (in kg) of the chosen chip. Factor 0.02 stands for the 2% weight share of the fabricated wafer calculated for a 1kg packaged chip (which is the process unit for this dataset).

¹¹ Intel (2007) http://www.intel.com/pressroom/kits/manufacturing/manufacturing_qa.htm#1; accessed July 2007

¹² AMD (2006) http://www.amd.com/us-en/assets/content_type/DownloadableAssets/2005_Corporate_Responsibility_Report_Final.pdf; accessed July 2007

¹³ STMicroelectronics (2007) <http://www.st.com/stonline/company/identity/slides/euro.pdf>; accessed July 2007

¹⁴ Siliconfareast (2000-2005): Semiconductor Manufacturing; Online at <http://www.siliconfareast.com/manufacturing2.htm>; accessed July 2007

Chemical content SOT514

Device part	substance	mass (mg)
substrate	glass-filled epoxy with Cu, Ni, Au plated pattern	2059
solderballs	SnPb37	0.945
encapsulation	partially brominated epoxy $\text{SiO}_2 = 80\%$ $\text{Sb}_2\text{O}_3 < 4\%$ as Sb TBBA < 1% as Br	1449
die-attach	Ag-filled epoxy	72
active device	doped silicon	15

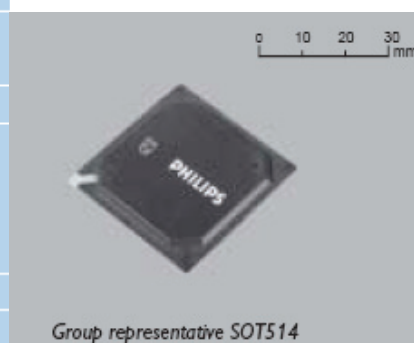


Fig. 3.15 Chemical content of a Philips SOT514 Plastic Ball Grid Array Package - Microchip (source: Philips Semiconductors (2002))

PRODUCT MATERIAL SHEET

Composition part	Material name	Material mass (%)	Element name composition	CAS No	Element name (average by weight-%)
Substrate	Glass epoxy	29	Bismaleimide Triazine (Br) TBBA	--	69
			Cu	79-94-7	1
			Resist	7440-50-8	20
			Ni	---	9
			Au	7440-02-0	0.8
				7440-57-5	0.2
Encapsulation	Epoxy resin	51.53	SiO ₂	60676-86-0	70
			Epoxy	29690-82-2	27
			Sb ₂ O ₃	1309-64-4	2
			Br (TBBA)	79-94-7	1
Chip	Doped Silicon	2.68	Si	7440-21-3	99.4
			Al	7429-90-5	0.6
Die bonding mater.	Glue	0.80	Ag	7440-22-4	75
			Epoxy resin	29690-82-2	25
Wires	Gold	0.99	Au	7440-57-5	100
Balls	Tin/Lead	15	Sn	7440-31-5	63
			Pb	7439-92-1	37
Balls	SnAgCu	15	Sn	7440-31-5	95.5
			Ag	7440-22-4	4
			Cu	7440-50-8	0.5

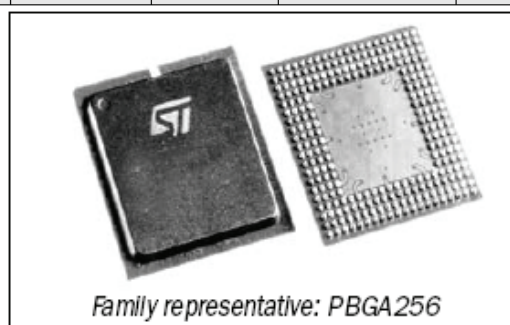


Fig. 3.16 Plastic Ball Grid Array Package PBGA256 of ST Microelectronics as another example for a logic type microchip (source: ST Microelectronics (2004))

According to HD Microsystems (2006), the resist fraction of the substrate very often is polyimide.

Tab. 3.11 Package component percentage shares of a logic type Integrated Circuit

Components (Device parts)	SOT514 (Philips)		PBGA256 (ST Microelectronics)		Chosen for ecoinvent dataset
	mg	Percent	mg	Percent	
<i>Substrate:</i>	2059	57.3% (=100%)	759.8	29.0% (=100%)	41% (=100%)
Glass-filled epoxy			524.3	(69.0%)	(69.0%)
Cu			151.8	(20.0%)	(20.0%)
Ni			6.1	(0.8%)	(0.8%)
Au			1.5	(0.2%)	(0.2%)
TBBA (Br)				(1.0%)	(1.0%)
Resist				(9.0%)	(9.0%)
<i>Encapsulation:</i>	1449	40.3 % (=100%)	1349.3	51.5 % (=100%)	45% (=100%)
SiO2		(80.0%)	944.5	(70.0%)	(70.0%)
Sb2O3 (Sb)		(<4.0%)	27.0	(2.0%)	(2.0%)
TBBA (Br)		(<1.0%)	13.5	(1.0%)	(1.0%)
Epoxy			364.3	(27%)	(27%)
Rest		(ca. 15%)			
<i>Solderballs:</i>	0.945	0.02%	393.0	15.0 % (=100%)	10% (=100%)
Sn			247.6	(63.0%)	(63.0%)
Pb37			145.4	(37.0%)	(37.0%)
<i>Die-attach:</i>	72	2.0%	21.0	0.8 % (=100%)	1% (=100%)
Ag			15.75	(75.0%)	(75.0%)
Epoxy			5.25	(25.0%)	(25.0%)
<i>Wires:</i>			25.9	0.99% (=100%)	1% (=100%)
Au					
<i>Active device (Wafer):</i>	15	0.4%	70.7	2.7% (=100%)	2% (=100%)
Si			70.3	(99.4%)	(99.4%)
Al			0.4	(0.6%)	(0.6%)
Total:	3596	100.0%	2620	100.0%	100.0%

More information on the “logic type integrated circuit” dataset can be retrieved from the general flow information sheet in Tab. 3.13, as well as the meta information presented in Tab. 3.12.

Tab. 3.12 ecoinvent meta information for the “integrated circuit, logic type” dataset

Name	integrated circuit, IC, logic type, at plant
Location	GLO
Infrastructure Process	0
Unit	kg
Dataset Version	2.0
Included Processes	Describes all the processes required to produce a logic type microchip. Included are 'die separation', 'encapsulation', 'die attachment', 'lead bonding', 'plating', and 'marking'. These operations are represented by the material input of glass epoxy, metals, epoxy resin, doped silicon, glue, gold (wires) etc. and the assembly process energy used. The required infrastructure, and the ship, train and road transport are also inventoried. Calculated for 1 kg of packaged logic IC.
Amount	1
Local Name	Integrierte Schaltung, IC, Logik-, ab Werk
Synonyms	Microchip//Microcircuit//Computer chip//Silicon chip//Chip
General Comment	This dataset can be applied to describe the entire production of a logic type integrated circuit. The included wafer fabrication data refers to a 200-mm wafer for a 32 MB DRAM chip, whereas the chip packaging data describes the material contents (in percent) of a plastic ball grid array package (PBGA) microchip (years 2002 to 2004).
Start Date	2000
End Date	2006
Data valid for entire period	1
Geography Text	The data is based on two semiconductor companies in the USA and Western Europe representing international standard.
Technology Text	This dataset describes the chip packaging technology with the typical operation classes die separation, encapsulation, die attachment, lead bonding, plating and marking (inventoried only in form of consumed energy) for a logic-type microchip.
Representativeness	unknown
Production Volume	unknown
Sampling Procedure	Manufacturer information and "chemical content" publications by companies.
Extrapolations	see Geography and Technology
Uncertainty Adjustments	see Geography and Technology

3.6.2 Life Cycle Inventory and General Flow information

The layout of the dataset “integrated circuit, IC, logic type, at plant” with general flow information, remarks, sources, values and uncertainty information are shown in Tab. 3.13 below.

Tab. 3.13 Flows for “integrated circuit, IC, logic type, at plant” and their representation in the ecoinvent database (PART 1)

General Flow information					Representation in ecoinvent					Uncertainty				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Fabricated wafer for ICs	Integrated circuit, IC, logic type, at plant (PART 1)	↑	Fabricated wafer to be packaged; 2% of total packaged chip (IC); chosen example for original microchip with body size 27*27mm (equals 7.29 cm2) and weight 2.620 g.	electronics	component	No	GLO	wafer, fabricated, for integrated circuit, at plant	0.0056	m2	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
Epoxy resin in substrate		↑	Glass-filled epoxy moulding compound for final IC production; 28 % of total IC-package (69% of substrate)	plastics	others	No	RER	glass fibre reinforced plastic, polyamide, injection moulding, at plant	0.28	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
Copper in substrate		↑	Copper in substrate final IC production; 8.2% of total IC-package (20% of substrate)	metals	extraction	No	GLO	copper, primary, at refinery	0.082	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
Nickel in substrate		↑	Nickel in substrate final IC production; 0.33% of total IC-package (0.8% of substrate)	metals	extraction	No	GLO	nickel, 99.5% at plant	0.003	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
Gold in substrate		↑	Gold in substrate final IC production; 0.08% of total IC-package (0.2% of substrate)	metals	extraction	No	GLO	gold, primary, at refinery	0.001	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
Resist in substrate		↑	Polyimide in substrate final IC production; 3.7% of total IC-package (9% of substrate); polyamide taken as proxy.	plastics	others	No	RER	glass fibre reinforced plastic, polyamide, injection moulding, at plant	0.04	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
TBBA in substrate		↑	TBBA (Br) in substrate final IC production; 0.4 % of total IC-package (% of substrate). "Chemicals organic" as proxy.	chemicals	organics	No	GLO	chemicals organic, at plant	0.004	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
SiO2 in encapsulation		↑	SiO2 makes up 3.15% of total IC-package (70% of encapsulation)	plastics	monomers	No	RER	epoxy resin/insulator (SiO2), at plant	0.32	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
Sb2O3 in encapsulation		↑	Sb2O3 (Antimony oxide) makes up 0.9% of total IC-package (2% of encapsulation). Zinc taken as proxy. M of weight ratio Sb2:Sb2O3 ca. 243:297=0.84; this share of the Sb2O3 input is replaced by Zn.	metals	extraction	No	RER	zinc, primary, at regional storage	0.008	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
TBBA in encapsulation		↑	TBBA (Br) makes up 0.45% of total IC-package (% of encapsulation); Tetra bromide phenol (TBBA) is a flame retardant. "Chemicals organic" as proxy.	chemicals	organics	No	GLO	chemicals organic, at plant	0.0045	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)
Epoxy in encapsulation		↑	Epoxy makes up 12.15% of total IC-package (27% of encapsulation)	plastics	monomers	No	RER	epoxy resin/insulator (SiO2), at plant	0.12	kg	STM Icroelectronics (2004)	1	125	(2,4,2,3,153)

3. Microchips

Tab. 3.13 Flows for “integrated circuit, IC, logic type, at plant” and their representation in the ecoinvent database (PART 2)

General Flow information				Representation in ecoinvent					Uncertainty					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Die attach material Epoxy	Integrated circuit, IC, logic type, at plant (PART 2)	↑	Glue: share of epoxy resin (25% of total glue)	plastics	monomers	No	RER	epoxy resin, liquid, at plant	0.0025	kg	STMicroelectronics (2004)	1	125	(2.4,2.3,153)
Die attach material Ag		↑	Glue: share of silver (75% of total glue)	metals	extraction	No	RER	silver, at regional storage	0.008	kg	STMicroelectronics (2004)	1	125	(2.4,2.3,153)
Wires		↑	Gold wires: 1% of total packaged chip (IC)	metals	extraction	No	GLO	gold, primary, at refinery	0.010	kg	STMicroelectronics (2004)	1	125	(2.4,2.3,153)
Solderballs Sn		↑	Solderballs make up around 10% of packaged microchip; tin 63% of total solderballs.	metals	extraction	No	RER	tin, at regional storage	0.063	kg	STMicroelectronics (2004)	1	125	(2.4,2.3,153)
Solderballs Pb		↑	Solderballs make up around 10% of packaged microchip; lead 37% of total solderballs.	metals	extraction	No	RER	lead, at regional storage	0.037	kg	STMicroelectronics (2004)	1	125	(2.4,2.3,153)
Assembly process energy heat		↑	5.8 MJ per 32 MB DRAM packaged chip (weights 2 g in total, here calculated for 1kg IC); 17% is heat from heavy fuel oil	oil	heating systems	No	RER	heat, heavy fuel oil, at industrial furnace 11MW	493.0	MJ	Williams, E. (2002)	1	126	(3.4,2.3,152)
Assembly energy electricity		↑	5.8 MJ per 32 MB DRAM packaged chip (weights 2 g in total, here calculated for 1kg IC); 83% is electricity	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	668.6	kWh	Williams, E. (2002)	1	126	(3.4,2.3,152)
Infrastructure		↑	Infrastructure; electronic component production plant as proxy; annual production volume 2 mio. pieces, life-span 25 years	electronics	component	Yes	GLO	electronic component production plant	0.00000002	unit	According to electronic component production plant (ecoinvent)	1	3.11	(2.4,13.3,59)
Transport ship		↑	Transportation of the package components to the chip assembly plant by freight ship. Assumed to be 21300 km on average as many Back-end chip factories are located in Asia. Electronic grade wafer input (0.00556 m2) is 0.02 kg.	transport systems	ship	No	OCE	transport, transoceanic freight ship	20.5	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.11	(4.4,13,155)
Transport road		↑	Transportation of the package components to the chip assembly plant by road. Electronic grade wafer input (0.00556 m2) is 0.02 kg.	transport systems	road	No	RER	transport, lorry >16t, fleet average	0.2	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.08	(3.4,2.3,155)
Transport train		↑	Transportation of the package components to the chip assembly plant by train. Electronic grade wafer input (0.00556 m2) is 0.02 kg.	transport systems	train	No	RER	transport, freight, rail	12	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.08	(3.4,2.3,155)
				electronics	component	No	GLO	integrated circuit, IC, logic type, at plant	1	kg				

1kg integrated circuit, IC, logic type

Data Uncertainty and Input Data for ecoinvent Database

The dataset „integrated circuit, IC, logic type, at plant“ is mainly based on data by ST Microelectronics (2004) and Philips Semiconductors (2002), some data is supported by information of Schischke et al. (2002). The energy input for the chip packaging is based on published data by Williams et al. (2002) on a DRAM chip. The fact that no further data was available to improve the quality of the dataset has been taken into account in the uncertainties. There are no natural resources inventoried for this logic IC dataset. No water is required for the assembly of the integrated circuit. Neither direct air emissions nor any waste are balanced for this packaging dataset.

3.7 Dataset “integrated circuit, IC, memory type”

3.7.1 Dataset description and metainformation

The dataset “integrated circuit, IC, memory type, at plant” represents the processes and materials required to package up a fabricated wafer into a memory type microchip. The functional unit of this dataset is 1 kg of microchip in accordance with all the other printed wiring board components (transistors, resistors, etc.) presented in this ecoinvent report.

Major inputs to this dataset are the fabricated wafer in form of the dataset “wafer, fabricated, for integrated circuit, at plant” (Chapter 3.5), the materials for the leadframe (copper, iron, phosphor), the materials for the encapsulation or moulding compound (e.g. SiO₂), the die bonding material (epoxy and silver glue), the gold wires, and the leads finishing (Sn and Pb). A cross section of the here described memory chip is shown in Fig. 3.17. Further inventoried are the energy input for the assembly process as well as the required transport by ship, rail and road.

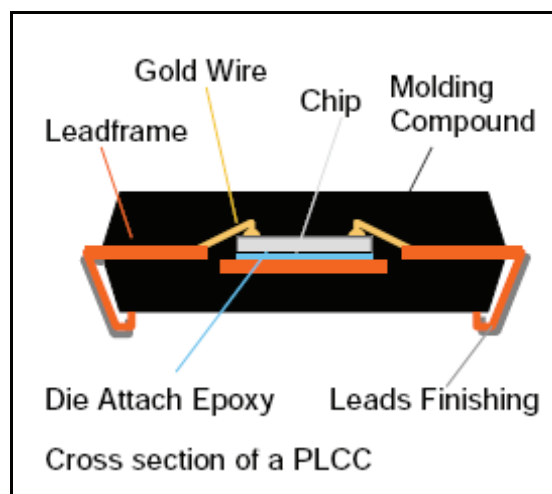


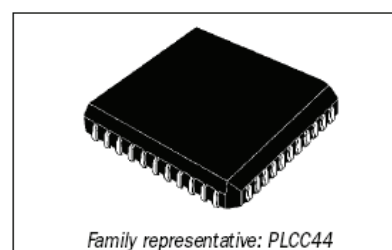
Fig. 3.17 Cross section of a Plastic Leaded Chip Carrier (source: Schischke et al. (2002))

For the transportation data, investigations were made (Tab. 3.10) to find out where in the world most of the 200-mm wafer fabrication actually takes place (front-end manufacturing), and where the chips are assembled (back-end manufacturing). This is to assure the quality standards of a globally valid dataset (GLO). Sources of the major chip producers Intel, AMD and STMicroelectronics were checked. As Tab. 3.10 in Chapter 3.6.1 shows, front-end and back-end chip manufacturing happens very often far away from each other. The transport by ship is calculated via an average return travel distance (21'300 km) from Asia (Japan) to Central Europe (Zurich). The distance from the front-end site to the harbour in Asia and from the harbour to the back-end factory in Europe is approximated by the ecoinvent standard distances (2*100km for road and 2*600km for rail transport, for chemicals).

The technology applied in chip assembly includes wafer backgrind, die preparation (wafer mounting and wafer sawing), die attach, wirebonding, die overcoat, molding, sealing, marking, DTFS (de-flash/trim/form/singulation), leadfinish (see SiliconFarEast for further information on these individual steps¹⁵).

The data for this dataset is based mainly on information from a publication by ST Microelectronics in 2004 named “Chemical content of semiconductor packaging” (ST Microelectronics (2004)), respectively by Schischke et al. (2002). The process data for total energy consumption (5.8 MJ per 2 g chip) was retrieved from the paper by Williams et al. (2002). The total energy for assembly is allocated with 17% to “heat from heavy fuel oil” and with 83% to “electricity, medium voltage, UCTE”.

The percentages for the IC package components (Fig. 3.18 and Tab. 3.12) were chosen based on the references mentioned above. Out of a large number of different memory chip types, the decision fell on the PLCC, as it offered the best information available.



PRODUCT MATERIAL SHEET

Composition part	Material name	Material mass (%)	Element name composition	CAS No	Element name (average by weight-%)	RoHS/Lead Status
Leadframe	Cu alloy	32.1	Cu	7440-50-8	97.5	
			Fe	7439-89-6	2.4	
			P	7723-14-0	0.1	
Encapsulation	Epoxy resin	66.1	SiO ₂	60676-86-0	70	
			Epoxy	29690-82-2	27	
			Sb ₂ O ₃	1309-64-4	2	
			Br (TBBA)	79-94-7	1	
Chip	Doped Silicon	0.9	Si	7440-21-3	99.4	
			Al	7429-90-5	0.6	
Die bonding mater.	Glue	0.28	Ag	7440-22-4	75	
			Epoxy resin	29690-82-2	25	
Wires	Gold	0.1	Au	7440-57-5	100	
Leadfinishing	Tin/Lead	0.52	Sn	7440-31-5	85	
			Pb	7439-92-1	15	

Fig. 3.18 Plastic Leaded Chip Carrier PLCC44 of ST Microelectronics as example for a memory type microchip (source: ST Microelectronics (2004))

The total weight of this PLCC44 type ICs is 2.3 g, its body size is 16.6mm*16.6mm. The share of the processed wafer is 0.9% of the total packaged IC weight (ST Microelectronics (2004)). The required fabricated wafer surface for a 1 kg packaged memory type chip is therefore calculated as follows:

¹⁵ Siliconfareast (2000-2005): Semiconductor Manufacturing; Online under <http://www.siliconfareast.com/manufacturing2.htm>

$$= (0.0166 \times 0.0166) \times 0.009 / 0.0023 = 0.0011 \text{ m}^2$$

with the numbers in brackets representing the surface of the total packaged chip, and the value 0.0023 representing the weight (in kg) of the chosen chip. Factor 0.009 stands for the 0.9% weight share of the fabricated wafer calculated for a 1kg packaged chip (which is the process unit for this dataset).

More information on the “memory type integrated circuit” dataset can be retrieved from the general flow information sheet in Tab. 3.15, as well as the meta information presented in Tab. 3.14.

Tab. 3.14 ecoinvent meta information for the “integrated circuit, memory type,” dataset

Name	integrated circuit, IC, memory type, at plant
Location	GLO
Infrastructure Process	0
Unit	kg
Dataset Version	2.0
Included Processes	Describes all the processes required to produce a memory type microchip. Included are 'die separation', 'encapsulation', 'die attachment', 'lead bonding', 'plating', and 'marking'. These operations are represented by the material input of copper alloy, epoxy resin, doped silicon, glue, gold (wires) etc. and the assembly process energy used. The required infrastructure, and the ship, train and road transport are also inventoried. Calculated for 1 kg of packaged memory IC.
Amount	1
Local Name	Integrierte Schaltung, IC, Speicher-, ab Werk
Synonyms	Microchip//Microcircuit//Computer chip//Silicon chip//Chip
General Comment	This dataset can be applied to describe the entire production of a memory type integrated circuit. The included wafer fabrication data refers to a 200-mm wafer for a 32 MB DRAM chip, whereas the chip packaging data describes the material contents (in percent) of a plastic leaded chip carrier (PLCC) microchip (years 2002 to 2004).
Start Date	2000
End Date	2006
Data valid for entire period	1
Geography Text	The data is based on a semiconductor company in the USA representing international standard.
Technology Text	This dataset describes the chip packaging technology with the typical operation classes die separation, encapsulation, die attachment, lead bonding, plating and marking (inventoried only in form of consumed energy) for a memory-type microchip.
Representativeness	unknown
Production Volume	unknown
Sampling Procedure	Manufacturer information and "chemical content" publications by companies.
Extrapolations	see Geography and Technology
Uncertainty Adjustments	see Geography and Technology

3.7.2 Life Cycle Inventory and General Flow information

The layout of the dataset “integrated circuit, IC, memory type, at plant” with general flow information, remarks, sources, values and uncertainty information are shown in Tab. 3.15 below.

Tab. 3.15 Flows for “integrated circuit, IC, memory type, at plant” and their representation in the ecoinvent database (PART 1)

General Flow information				Representation in ecoinvent						Uncertainty				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Fabricated wafer for ICs	Integrated circuit, IC, memory type, at plant (PART 1)	↑	Fabricated wafer to be packaged; 0.9%(or 0.09 kg) of total packaged chip (IC) weight; chosen example for body size of packed chip 16.6*16.6 mm (equals 2.76 cm2) with a weight of 2.3 g.	electronics	component	No	GLO	wafer, fabricated, for integrated circuit, at plant	0.0011	m2	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Leadframe Copper		↑	Copper makes up 31.3%of total IC package; share of copper in leadframe 97.5%	metals	extraction	No	GLO	copper, primary, at refinery	0.31	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Leadframe Iron		↑	Iron makes up 0.7%of total IC package; share of Fe in leadframe 2.4 %	metals	extraction	No	GLO	pig iron, at plant	0.008	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Leadframe Phosphor		↑	Phosphorus makes up 0.032%of total IC package; share of P in leadframe 0.1%	chemicals	inorganics	No	RER	phosphorus, white, liquid, at plant	0.0003	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Epoxy in encapsulation		↑	Epoxy makes up 17.8%of total IC-package (27%of encapsulation)	plastics	monomers	No	RER	epoxy resin/insulator (SiO2), at plant	0.48	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
SiO2 in encapsulation		↑	SiO2 makes up 46.3%of total IC-package (70%of encapsulation).	plastics	monomers	No	RER	epoxy resin/insulator (SiO2), at plant	0.46	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Sb2O3 in encapsulation		↑	Sb2O3 (Antimony oxide) makes up 13%of total IC-package (2%of encapsulation). Zinc taken as proxy. Mol weight ratio Sb2 : Sb2O3 ca. 243:291 = 0.84; this share of the Sb2O3 input is replaced by Zn.	metals	extraction	No	RER	zinc, primary, at regional storage	0.011	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
TBBA in encapsulation		↑	TBBA (Br) makes up 0.66%of total IC-package (1%of encapsulation); Tetra bromide/bisphenol (TBBA) is a flame retardant. "Chemical, organic" as proxy.	chemicals	organics	No	GLO	chemicals organic, at plant	0.007	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Diebonding material Epoxy		↑	Glue: share of epoxy resin (25%of total glue; 0.07%of total packed IC)	plastics	monomers	No	RER	epoxy resin, liquid, at plant	0.0007	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Diebonding material Ag		↑	Glue: share of silver (75%of total glue; 0.2 % of total packed IC)	metals	extraction	No	RER	silver, at regional storage	0.0021	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)
Wires		↑	Gold wires; 0.1%of total packaged chip (IC)	metals	extraction	No	GLO	gold, primary, at refinery	0.0010	kg	STM microelectronics (2004)	1	125	(2,4,2,3,153)

Tab. 3.15 Flows for “integrated circuit, IC, memory type, at plant” and their representation in the ecoinvent database (PART 2)

General Flow information				Representation in ecoinvent					Uncertainty					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
Leads finishing Sn	Integrated circuit, IC, memory type, at plant (PART 2)	↑	Leads finishing (tin/lead alloy) equals 0.52% of total packed IC; share of tin makes up 85% of total leads finishing	metals	extraction	No	RER	tin, at regional storage	0.0044	kg	STM icroelectronics (2004)	1	125	(2,4,2,3,15,3)
Leads finishing Pb		↑	Leads finishing (tin/lead alloy) equals 0.52% of total packed IC; share of lead makes up 15% of total leads finishing	metals	extraction	No	RER	lead, at regional storage	0.0008	kg	STM icroelectronics (2004)	1	125	(2,4,2,3,15,3)
Assembly energy process heat		↑	5.8 MJ per 32 MB DRAM packaged chip (weights 2 g in total, here calculated for 1kg IC); 77% is heat from heavy fuel oil	oil	heating systems	No	RER	heat, heavy fuel oil, at industrial furnace 1MW	493.0	MJ	Williams, E. (2002)	1	126	(3,4,2,3,15,2)
Assembly energy electricity		↑	5.8 MJ per 32 MB DRAM packaged chip (weights 2 g in total, here calculated for 1kg IC); 83% is electricity	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	668.6	kWh	Williams, E. (2002)	1	126	(3,4,2,3,15,2)
Infrastructure		↑	Infrastructure; electronic component production plant as proxy; annual production volume 2 mio. pieces, life-span 25 years	electronics	component	Yes	GLO	electronic component production plant	0.00000002	unit	According to electronic component production plant (ecoinvent)	1	3.11	(2,4,13,3,5,9)
Transport ship		↑	Transportation of the package components to the chip assembly plant by freight ship. Assumed to be 21300 km on average as many Back-end chip factories are located in Asia. Electronic grade wafer input (0.001 m2) is 0.009 kg.	transport systems	ship	No	OCE	transport, transoceanic freight ship	213	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.11	(4,4,13,15,5)
Transport road		↑	Transportation of the package components to the chip assembly plant by road. Assumed to be 200 km on average according to ecoinvent standard distances. Electronic grade wafer input (0.001 m2) is 0.009 kg.	transport systems	road	No	RER	transport, lorry >16t, fleet average	0.2	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.08	(3,4,2,3,15,5)
Transport train		↑	Transportation of the package components to the chip assembly plant by train. Assumed to be 200 km on average according to ecoinvent standard distances. Electronic grade wafer input (0.001 m2) is 0.009 kg.	transport systems	train	No	RER	transport, freight, rail	12	tkm	Calculated according to ecoinvent standard distances for chemicals	1	2.08	(3,4,2,3,15,5)
			1kg Integrated circuit, IC, memory type	electronics	component	No	GLO	integrated circuit, IC, memory type, at plant	1	kg				

Data Uncertainty and Input Data for ecoinvent Database

The dataset „integrated circuit, IC, memory type, at plant“ is mainly based on data mainly by ST Microelectronics (ST Microelectronics (2004)), some data is supported by information of Schischke et al. (2002). The energy input for the chip packaging is based on published data by Williams et al. (2002) on a DRAM chip. The fact that no further data was available to improve the quality of the dataset has been taken into account in the uncertainties.

There are no natural resources inventoried for this memory IC dataset. No water is required for the assembly of the integrated circuit. Neither direct air emissions nor any waste are balanced for this packaging dataset.

4 Transistors

A **transistor** “is a semiconductor device, the most common type of which acts as a current amplifier. There do however exist less common types of transistor - FETs (field effect transistors) - which act as voltage controlled resistors.”¹⁶

Hence a transistor is a fundamental building block of all the circuitries that govern the operation of computers, cellular phones, and all other modern electronics. Due to their fast response and accuracy, a transistor may be used in a wide variety of digital and analogue functions, including amplification, switching, voltage regulation, signal modulation, and oscillators. Transistors may be packaged individually or as part of an integrated circuit chip, which may hold thousands of transistors in a very small area.

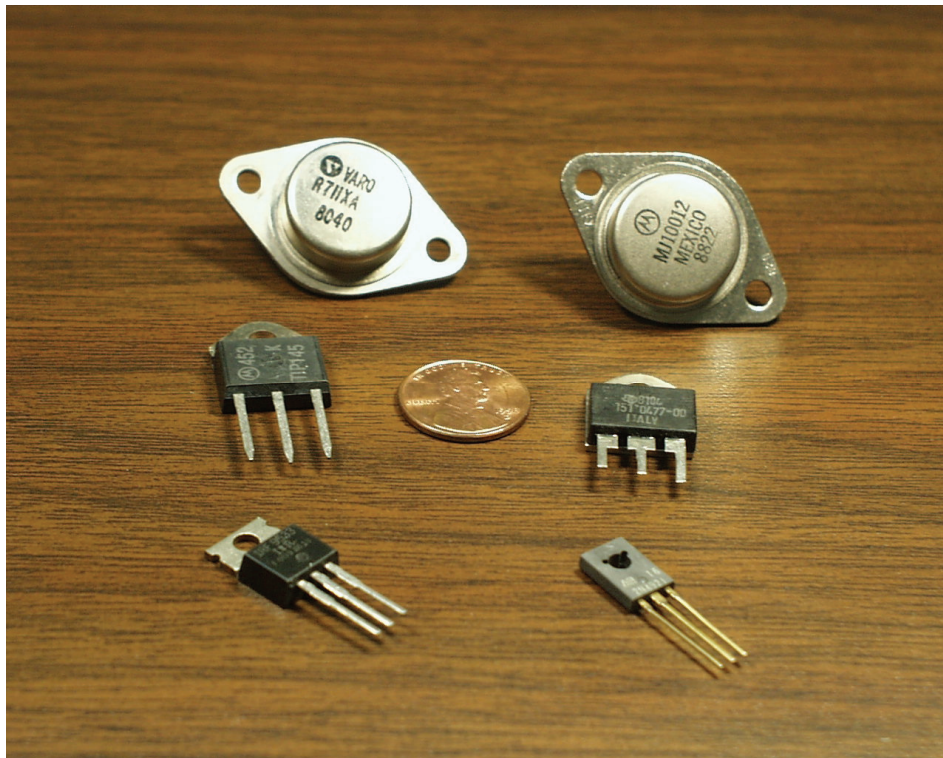


Fig. 4.1 Different types of transistors (picture from <http://www.xcvcorp.com/>).

4.1 Characterisation, Use & Market

Transistors are considered by many to be one of the greatest inventions in modern history, ranking in importance with the printing press, car and Electric/Electronic communication. It is the key active component in practically all modern electronics. Its importance in today's society rests on its ability to be mass produced using a highly automated process (fabrication) that achieves vanishingly low per-transistor costs.

Besides the use of millions of individual or discrete transistors, the vast majority of transistors are fabricated into integrated circuits today (see preceeding chapter 3 of this report here) together with diodes, resistors, capacitors and other electronic components in order to establish a complete electronic circuit.

¹⁶ Wikipedia: Transistor (<http://en.wikipedia.org/wiki/Transistor>), accessed March 16, 2007.

Low cost, flexibility and reliability have made out of transistors a universal device for non-mechanical tasks, such as digital computing. Transistorized circuits have replaced electromechanical devices for the control of appliances and machinery as well. It is often less expensive and more effective to use a standard microcontroller and write a computer program to carry out a control function than to design an equivalent mechanical control function¹⁷.

Currently, there is a huge variety of different types of transistors on the market – a variety that can not be covered 100% by the here described ecoinvent datasets. Main characteristics are differences in the semiconductor material used, the structure, the polarity, the maximum power rating, the maximum operating frequency or the physical packaging. Hence, for this project here a choice had to be made. Based on the available data sources, three different types of transistors have been chosen as being represented within this project here: SMD type transistors, small and big wired type transistors.

4.2 System Characterisation

Similar like for capacitors, a variety of data sources can be found about the composition of various types of transistors, but it is very hard to get any information concerning the actual production efforts for the production of such an electronic component. Hence, for transistors, again, the composition is much more accurate and in details shown in this study than the production itself. Thus – for the production efforts of transistors, no distinction between the different here examined types of transistors is made – instead, one average dataset representing the “production efforts” is established; and all types of transistors will then have this dataset as one of their input values. All in all this ends up with the following scheme for the representation of the electronic components “Transistors” in the framework of the ecoinvent database.

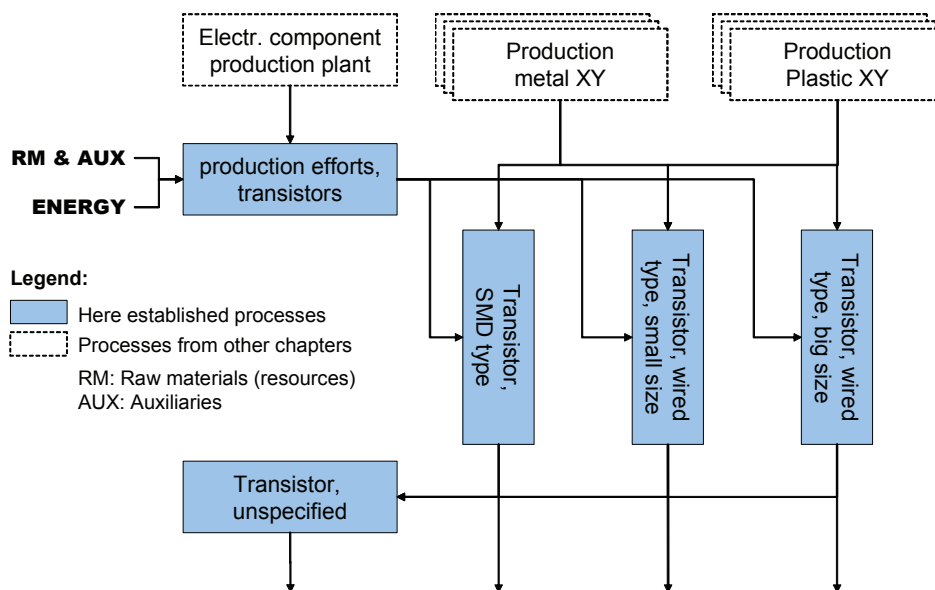


Fig. 4.2 Scheme of the representation of electronic components „Transistor” in the ecoinvent database

All these datasets are based on information representing current models of transistors. The used data sources are mainly material specification sheets from industry reported in Behrendt et al. (1998), CPM (2006), Philips Semiconductors (2002) and ST Microelectronics (2004).

¹⁷ Wikipedia: Transistor (<http://en.wikipedia.org/wiki/Transistor>), accessed March 16, 2007.

They represent transistors that have been in use in the last about 10 years – as well as current models. The companies behind this information are producing on a global scale – hence, the here developed data can be considered as “global” valid data.

The respective meta information of the various datasets of transistors can be found in Tab. 4.1.

Tab. 4.1 ecoinvent meta information of the transistor's datasets.

Name	production efforts, transistors	transistor, unspecified, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	Kg	kg
Dataset Version	2.0	2.0
Included Processes	This module includes auxillary and energy consumption, emissions (air, water) and waste amounts, as well as an estimation of the efforts for transportation and of the infrastructure for the production of transistors.	This dataset covers raw material input and production efforts for the production of current transistors.
Amount	1	1
Local Name	Transistoren, Herstellaufwände	Transistoren, unspezifisch, ab Werk
Synonyms	-	-
General Comment	Data are a rough estimation of the average production efforts over all types of transistors. Data for auxillaries, energy, emissions and waste are taken from a dataset out of an international LCI database. Infrastructure and transportation efforts are based on own assumptions.	This module represents an unweighted average of all specific transistor datasets represented in the ecoinvent database.
Start Date	1998	1994
End Date	2005	2007
Data valid for entire period	1	1
Geography Text	Data from literature – here representing the global situation.	Data represent global valuable composition data for transistors.
Technology Text	average production technology comprising Si-crystal die bonding; wire bonding; encapsulation; plating; trimming; forming and marking.	average production technology comprising Si-crystal die bonding; wire bonding; encapsulation; plating; trimming; forming and marking.
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	estimation – based on literature	Literature & own estimation
Extrapolations	see Geography & Technology	see Geography and General Comments
Uncertainty Adjustments	None	None

Tab. 4.1 (Cont.) ecoinvent meta information of the transistor's datasets.

Name	transistor, SMD type, for surface-mounting, at plant	transistor, wired type, small size, for through-hole mounting, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used SMD transistors for surface mounting technology.	This dataset covers raw material input and production efforts for the production of currently used small wired transistors for through-hole mounting technology.
Amount	1	1
Local Name	Transistor, SMD-Typ, surface mounting, ab Werk	Kondensator, Film-, through-hole mounting, ab Werk
Synonyms	-	film condenser
General Comment	The data represent a typical SMD transistor for surface mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such transistors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of transistors with an average weight of 0.593 gram/unit.	The data represent a typical wired, small-size, transistor for through-hole mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such transistors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of transistors with an average weight of 0.818 gram/unit.
Start Date	1994	1994
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of SMD type transistors.	Data represent a global valuable composition of small wired type transistors.
Technology Text	average production technology comprising Si-crystal die bonding; wire bonding; encapsulation; plating; trimming; forming and marking.	average production technology comprising Si-crystal die bonding; wire bonding; encapsulation; plating; trimming; forming and marking
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature	literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 4.1 (Cont.) ecoinvent meta information of the transistor's datasets.

Name	transistor, wired type, big size, for through-hole mounting, at plant
Location	GLO
Infrastructure Process	0
Unit	kg
Dataset Version	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used biz-sized wired transistors for through-hole mounting technology.
Amount	1
Local Name	Transistor, Draht-, gross, through-hole mounting, ab Werk
Synonyms	film condenser
General Comment	The data represent a typical wired, big-sized, transistor for through-hole mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such transistors. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of transistors with an average weight of 6.34 gram/unit.
Start Date	1994
End Date	2007
Data valid for entire period	1
Geography Text	Data represent a global valuable composition of big wired transistors.
Technology Text	average production technology comprising Si-crystal die bonding; wire bonding; encapsulation; plating; trimming; forming and marking.
Representativeness	see General Comments
Production Volume	-
Sampling Procedure	Literature
Extrapolations	see Geography & Technology
Uncertainty Adjustments	None

4.3 Dataset “production efforts, transistors”

This dataset describes the production efforts for the production of the various types of transistors examined within ecoinvent. The dataset has the functional unit “efforts for the production of 1 kg of transistors”.

4.3.1 System Characterization

The dataset established here comprises the input of auxiliaries and energy as well as the output of emissions and waste due to these auxiliaries in the framework of the transistor production. The input of the used infrastructure, as well as the transport efforts for the auxiliaries and the raw materials of the

various transistors are included as well. Main data sources for this information are the data about transistors in CPM (2006) and exact (1998).

4.3.2 Life Cycle inventory

As the raw material input is taken into account in the respective transistor datasets – hence this dataset has no input of raw materials. Instead, this dataset here contains all other input and output flows – i.e. auxiliaries, energy, transport amount (for auxiliaries and raw materials!), infrastructure as well as the emissions and waste amounts produced. In details these various aspects are taken into account based on the following information:

- **Auxiliaries:** In CPM (2006) and exact (1998) an average input of about 8.3 kg of nitrogen (process gas) and of further 0.38 kg of various chemicals – reported here as “chemicals organics, at plant” – per kg of produced transistor is shown and used for this dataset.
- **Energy:** From CPM (2006) and exact (1998) an average input of electricity of 200 kWh per kg of produced transistors can be calculated.
- **Emissions to air:** According to CPM (2006) all reported chemicals are emitted a 100% into air – here reported as “NMVOC, unspecified, to air”. Furthermore, it was assumed, that 100% of the electricity consumed, i.e. 200 kWh per kg transistor produced, is converted to waste heat. In accordance with the ecoinvent guidelines, it is assumed that 100% of the waste heat is released to the air.
- **Emissions to water & Water input:** According to CPM (2006) and exact (1998), an average water consumption of 7.3 kg per kg of transistors takes place. It is assumed due to lack of more precise information, that 20% of this is used as cooling water – and 80% as process water. Due to a lack of respective emission information, the outflow of the latter amount is approximated by the use of the dataset “treatment, sewage, to waste water treatment, class 2”.
- **Waste:** According to CPM (2006) and exact (1998), an average of 4.8 kg of non-used raw materials can be found in the production process. Based on the available information in these sources, this amount of waste is taken into account as following:

Type of waste	Amount	Used dataset
capsules	0.3%	disposal, waste, Si waferprod., inorg, 9.4% water, to residual material landfill
solder	14.1%	disposal, tin sheet, 0% water, to sanitary landfill
lead frame	36.3%	- (assumed to be 100% recycled)
epoxy	49.4%	disposal, plastics, mixture, 15.3% water, to municipal incineration

- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here. The amount takes into account the consumed auxiliaries as well as the raw materials for the production of any type of transistor. The amount of raw materials used equals the material in the transistor (i.e. 1 kg) as well as the waste out of raw materials (i.e. 4.9 kg).
- **Infrastructure and land-use:** Due to a lack of more specific information, the infrastructure module established in chapter 2, representing an electronic component production plant, is used also for the production of transistors.

4.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is a rather rough estimation of the actual transistor production efforts, based mainly on information from one dataset out of the Swedish LCI database (CPM (2006)). Hence, the respective uncertainty values are accordingly high.

The resulting unit process “production efforts, transistors” is shown in Tab. 4.5 (on page 73).

4.4 Dataset “transistor, SMD type, for surface mounting, at plant”

This dataset here describes the production efforts of the production of SMD type transistors for the surface mounting technology. The dataset has a functional unit of “1 kg of transistors”, based on the assumption that a typical SMD transistor in electronic modules has an average weight of **0.593 g**.

4.4.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 4.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Philips Semiconductors (2002) and ST Microelectronics (2004).

4.4.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, transistors”, the production of 1 kg of transistors lead to a waste amount of 4.9 kg unused raw materials. Therof about 49.4% are unused amount of the encapsulation (i.e. epoxy), another 36.3% consists of the lead frame (here represented by 100% Cu), 14.1% are solder waste (i.e. 37% lead, 63% tin – as used for the solder material reported in chapter 2 of part II of this report) and the remaining 0.3% are unused. In other words, this dataset here has a total input of 5.9 kg of raw materials per kg of transistor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Philips Semiconductors (2002) and ST Microelectronics (2004), as well as the above information about waste composition. All this is summarized in Tab. 4.2 together with the respective ecoinvent datasets used.

Tab. 4.2 Material specifications of SMD type transistors (data sources: see text)

		SMD	Used ecoinvent dataset
		transistor	
[i] Metals			
aluminium	%	0.21	aluminium, primary, at plant (RER)
iron	%	1.04	pig iron, at plant (GLO)
copper	%	36.30	copper, primary, at plant (RER)
nickel	%	0.80	nickel, 99.5%, at plant (GLO)
lead	%	4.41	lead, at regional storage (RER)
tin	%	7.61	tin, at regional storage (RER)
[ii] Others			
encapsulation (SiO ₂ ,)	%	49.13	epoxy resin insulator (SiO ₂), at plant (RER)
doped silicium	%	0.51	silicon, electronic grade, at plant (RER)

4.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical SMD transistor for surface mounting, based on composition information from various worldwide transistor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific transistor type and have thus, accordingly higher uncertainties.

The resulting unit process “transistor, SMD type, for surface mounting, at plant” is shown in Tab. 4.5 (on page 74).

4.5 Dataset “transistor, wired type, small size, for through-hole mounting, at plant”

This dataset here describes the production efforts of the production of small size wired type transistors for the through-hole mounting technology. The dataset here has a functional unit of “1 kg of transistors”, based on the assumption that this kind of transistors in electronic modules has an average weight of **0.818 g**.

4.5.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 4.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Philips Semiconductors (2002) and ST Microelectronics (2004).

4.5.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, transistors”, the production of 1 kg of transistors lead to a waste amount of 4.9 kg unused raw materials – for more details see chapter 4.5.1. In other words, this dataset here has a total input of 5.9 kg of raw materials per kg of transistor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Philips Semiconductors (2002) and ST Microelectronics (2004), as well as the above information about waste composition. All this is summarized in Tab. 4.3 together with the respective ecoinvent datasets used.

4.5.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical small size wired transistor for through-hole mounting, based on composition information from various worldwide transistor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific transistor type and have thus, accordingly higher uncertainties.

The resulting unit process “transistor, wired type, small size, for through-hole mounting, at plant” is shown in Tab. 4.7 (on page 76).

Tab. 4.3 Material specifications of small size wired type transistors (data sources: see text)

		small size wired transistor	Used ecoinvent dataset
[i] Metals			
aluminium	%	0.01	aluminium, primary, at plant (RER)
iron	%	1.91	pig iron, at plant (GLO)
copper	%	37.29	copper, primary, at plant (RER)
nickel	%	1.40	nickel, 99.5%, at plant (GLO)
lead	%	4.37	lead, at regional storage (RER)
zinc	%	0.23	zinc, for coating, at plant (RER)
tin	%	7.46	tin, at regional storage (RER)
[ii] Others			
glass	%	0.07	glass tube, borosilicate, at plant (DE)
encapsulation (SiO ₂ ,)	%	46.95	epoxy resin insulator (SiO ₂), at plant (RER)
doped silicium	%	0.32	silicon, electronic grade, at plant (RER)

4.6 Dataset “transistor, wired type, big size, for through-hole mounting, at plant”

This dataset here describes the production efforts of the production of big size wired type transistors for the through-hole mounting technology. The dataset here has a functional unit of “1 kg of transistors”, based on the assumption that this kind of transistors in electronic modules has an average weight of **6.34 gram**.

4.6.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 4.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials related to this dataset here are Behrendt et al. (1998), Philips Semiconductors (2002) and ST Microelectronics (2004).

4.6.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, transistors”, the production of 1 kg of transistors lead to a waste amount of 4.9 kg unused raw materials – for more details see chapter 4.5.1. In other words, this dataset here has a total input of 5.9 kg of raw materials per kg of transistor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998), Philips Semiconductors (2002) and ST Microelectronics (2004) and above information about waste composition. All this is summarized in Tab. 4.4 together with the respective ecoinvent datasets used.

4.6.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical big size wired

transistor for through-hole mounting, based on composition information from various worldwide transistor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific transistor type and have thus, accordingly higher uncertainties.

The resulting unit process “transistor, wired type, big size, for through-hole mounting, at plant” is shown in Tab. 4.8 (on page 77).

Tab. 4.4 Material specifications of big size wired type transistors (data sources: see text)

		big size wired transistor	Used ecoinvent dataset
[i] Metals			
aluminium	%	0.04	aluminium, primary, at plant (RER)
iron	%	2.35	pig iron, at plant (GLO)
copper	%	35.41	copper, primary, at plant (RER)
nickel	%	0.13	nickel, 99.5%, at plant (GLO)
lead	%	3.69	lead, at regional storage (RER)
zinc	%	0.49	zinc, for coating, at plant (RER)
tin	%	8.11	tin, at regional storage (RER)
wolfram	%	0.83	proxy "chemicals, inorganic" used
[ii] Others			
Al ₂ O ₃	%	1.35	aluminium oxide, at plant (RER)
plastics (PA, ...)	%	3.62	nylon 6, at plant (RER)
BeO	%	0.81	proxy "chemicals, inorganic" used
glass	%	1.29	glass tube, borosilicate, at plant (DE)
encapsulation (SiO ₂ ,)	%	41.60	epoxy resin insulator (SiO ₂), at plant (RER)
doped silicium	%	0.28	silicon, electronic grade, at plant (RER)

4.7 Dataset “transistor, unspecified, at plant”

Last but not least, for all those cases where no information is given about the transistor type used, an unspecified dataset is established as well. This latter one is an unweighted mean value of all three above described transistor types. The dataset has as functional unit also “1 kg of transistors”.

4.7.1 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an unweighted average of the various transistor types shown in this database. Hence, the uncertainty values are accordingly high.

The resulting unit process “transistor, unspecified, at plant” is shown in Tab. 4.9 (on page 78).

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Tab. 4.5 Unit process inventory of the dataset “Production efforts, transistors (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment	
Water (process)	production efforts, transistors	↑		resource	in water			Water, unspecified natural origin	584E-03	m3	data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,12); estimation, based on total water amount from 1 literature source	
				resource	in water			Water, cooling, unspecified natural origin	146E-03	m3	data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,12); estimation, based on total water amount from 1 literature source	
				chemicals	inorganics	No	RER	nitrogen, liquid, at plant	8.30E-00	kg	data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,3); value from 1 literature source	
				chemicals	organics	No	GLO	chemicals organic, at plant	3.80E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,3); value from 1 literature source	
various chemicals					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.00E-02	kWh	data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,2); value from 1 literature source
electricity					transport systems	road	No	RER	transport, lorry >8t, fleet average	14,5E-00	tkm	standard distances	1	2.09	(4,5,nanana,5); standard distances
Transport (part by lorry)					transport systems	train	No	RER	transport, freight, rail	8.69E-00	tkm	standard distances	1	2.09	(4,5,nanana,5); standard distances
Transport (part by train)					electronics	component	Yes	GLO	electronic component production plant	2,00E-08	unt	rough estimation	1	3.77	(4,5,13,5,4,9); rough assumption
Infrastructure				waste management	municipal incineration	No	CH	disposal, plastics, mixture, incineration	2,42E-00	kg	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,15,6); calculated from input of auxiliaries	
continued ▶		↑		approximated by the dataset "disposal, Si waferprod., inorganic, to landfill"	waste management	residual material landfill	No	CH	disposal, waste, Si waferprod., inorg, 9.4 % water, to residual material landfill	100E-02	kg	estimation from data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,6); estimation, based on total waste amount from 1 literature source
		↑		approximated by the dataset "disposal, tin sheets, to landfill"	waste management	sanitary landfill	No	CH	disposal, tin sheet, 0% water, to sanitary landfill	6.90E-01	kg	estimation from data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,6); estimation, based on total waste amount from 1 literature source
		↑		due to a lack of knowledge of content, dataset "treatment, sewage, WWTP class 2" is used as proxy	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 2	584E-03	m3	estimation from data reported in a dataset in CPM (2006)	1	127	(2,4,3,3,15,6); estimation, based on total waste amount from 1 literature source
		↑													

Tab. 4.5 (Cont.) Unit process inventory of the dataset “Production efforts, transistors (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
▼ continued	production efforts, transistors	waste heat	↑	air	high population density			Heat, waste	7.20E+02	MJ	calculated from the electricity input	1	1.27	(2,4,3,3,15,13); calculated, from electricity input
		chemicals, to air	↑	air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	3.80E-01	kg	calculated from input of auxiliaries	1	2.08	(2,4,3,3,15,23); calculated from input of auxiliaries
		Production efforts for transistors	↑	electronics	component	No	GLO	production efforts, transistors	1.00E+00	kg		1		

4. Transistors

Tab. 4.6 Unit process inventory of the dataset “transistor, SMD type, surface mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	SDv 95%	General Comment
production efforts	transistor, SMD type, surface mounting, at plant	SMD transistor		electronics	component	No	GLO	production efforts, transistors	1.00E+00	kg	transistor production according to CPM (2006)	1	1.61	(3,4,3,1,4,5,1); rough estimation
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	122E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	6.03E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	2.11E+00	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	4.64E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
lead				metals	extraction	No	RER	lead, at regional storage	2.56E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	4.41E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
encapsulation (SiO2, ...)				plastics	monomers	No	RER	epoxy resin insulator (SiO2), at plant	2.89E+00	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
doped silicon				metals	refinement	No	DE	silicon, electronic grade, at plant	2.96E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
				electronics	component	No	GLO	transistor, SMD type, surface mounting, at plant	1.00E+00	kg		1		

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Tab. 4.7 Unit process inventory of the dataset “transistor, wired, small size, through-hole mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	transistor, wired, small size, through-hole mounting, at plant	small-sized wired transistor		electronics	component	No	GLO	production efforts, transistors	1.00E+00	kg	transistor production according to CPM (2006)	1	1.61	(3,4,3,1,4,5,1); rough estimation
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	5.80E-04	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	1.11E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	2.16E+00	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	8.12E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
lead				metals	extraction	No	RER	lead, at regional storage	2.53E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
zinc				metals	extraction	No	RER	zinc, primary, at regional storage	1.33E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	4.33E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
glass				glass	construction	No	DE	glass tube, borosilicate, at plant	4.06E-03	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
encapsulation (SiO2, ...)				plastics	monomers	No	RER	epoxy resin/insulator (SiO2), at plant	2.72E+00	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
doped silicon				metals	refinement	No	DE	silicon, electronic grade, at plant	1.86E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,1,14,3); average from literature & company information
				electronics	component	No	GLO	transistor, wired, small size, through-hole mounting, at plant	1.00E+00	kg		1		

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Tab. 4.8 Unit process inventory of the dataset “transistor, wired, big size, through-hole mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment	
production efforts	transistor, wired, big size, through-hole mounting, at plant			electronics	component	No	GLO	production efforts, transistors	1.00E+00	kg	transistor production according to CPM (2006)	1	1.61	(3,4,3,14,5,11); rough estimation	
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	2.32E-03	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); company information	
iron				metals	extraction	No	GLO	pig iron, at plant	1.36E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); company information	
copper				metals	extraction	No	RER	copper, primary, at refinery	2.05E+00	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); company information	
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	7.54E-03	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); company information	
lead				metals	extraction	No	RER	lead, at regional storage	2.14E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information	
zinc				metals	extraction	No	RER	zinc, primary, at regional storage	2.84E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information	
tin				metals	extraction	No	RER	tin, at regional storage	4.70E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information	
wolfram				chemicals inorganic used as a proxy	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	4.81E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information
aluminium oxide				chemicals	chemicals	inorganics	No	RER	aluminium oxide, at plant	7.83E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information
nylon				plastics	plastics	polymers	No	RER	nylon 6, at plant	2.10E-01	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information
BeO				chemicals inorganic used as a proxy	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	4.70E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information
glass					glass	construction	No	DE	glass tube, borosilicate, at plant	7.48E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(14,2,114,3); average from literature & company information

continued ▼

Tab. 4.8 (Cont.) Unit process inventory of the dataset “transistor, wired, big size, through-hole mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
encapsulation (SiO2, ...)	transistor, wired, big size, through-hole mounting, at plant	continued		plastics	monomers	No	RER	epoxy resin/insulator (SiO2), at plant	2.4 IE+00	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(1,4,2,1,1,4,3); average from literature & company information
doped silicon				metals	refinement	No	DE	silicon, electronic grade, at plant	162E-02	kg	Average according to Philips (2002) and ST (2004)	1	1.16	(1,4,2,1,1,4,3); average from literature & company information
				electronics	component	No	GLO	transistor, wired, big size, through-hole mounting, at plant	100E+00	kg		1		

Tab. 4.9 Unit process inventory of the dataset “transistor, unspecified, at plant (GLO)”

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
SMD transistors	transistor, unspecified, at plant	unspecified transistors		electronics	component	No	GLO	transistor, SMD type, surface mounting, at plant	3.33E-01	kg	assumption	1	189	(5,4,4,1,4,5,3); assumption, based on specific data
small wired-type transistors				electronics	component	No	GLO	transistor, wired, small size, through-hole mounting, at plant	3.33E-01	kg	assumption	1	189	(5,4,4,1,4,5,3); assumption, based on specific data
big wired-type transistors				electronics	component	No	GLO	transistor, wired, big size, through-hole mounting, at plant	3.33E-01	kg	assumption	1	189	(5,4,4,1,4,5,3); assumption, based on specific data
				electronics	component	No	GLO	transistor, unspecified, at plant	100E+00	kg		1		

5 Diodes

A **diode** is “a component that restricts the direction of movement of charge carriers. Essentially, it allows an electric current to flow in one direction, but blocks it in the opposite direction.”¹⁸

Hence, a diode can be seen as an electronic version of a check valve. A circuit that requires current flow in only one direction will consequently include at least one diode in its circuit design. Currently most of the diodes are made from traditional semiconductor materials like silicium or germanium.



Fig. 5.1 Different types of diodes (taken from Wikipedia)

A **light-emitting diode** (LED) is “a semiconductor device that emits incoherent narrow-spectrum light when electrically biased in the forward direction”¹⁹. This effect is a kind of electroluminescence. Similar to a normal diode, an LED contains a chip of doped semi conducting material. And as in these other diodes, a current can flow easily from the anode to the cathode, but not in the reverse direction. The color of the emitted light by a LED depends on the composition and condition of the semiconducting material used, and can vary from infrared via visible to near-ultraviolet.

5.1 Characterisation, Use & Market

Most modern diodes are based on semiconductor p-n junctions. In a p-n diode, conventional current flows from the p-type side (the anode) to the n-type side (the cathode), but not in the opposite direction.

Currently, there is a huge variety of different types of diodes on the market – a variety that can not be covered 100% by the here described ecoinvent datasets. Based on the available data sources, it has been decided that within this project here, only glass diodes (both mounting technologies) and light-emitting diodes (LEDs) are further investigated.

5.2 System Characterisation

Similar like for capacitors, a variety of data sources can be found about the composition of various types of diodes (including LED as a special type of diodes), but it is very hard to get any information concerning the actual production efforts for the production of such an electronic component. Hence, also for diodes, the composition is much more accurate and in details shown in this study than the production itself. Thus – for the production efforts of diodes, no distinction between the different here ex-

¹⁸ Wikipedia: diode (<http://en.wikipedia.org/wiki/diode>), accessed December 12, 2006.

¹⁹ Wikipedia: Light-emitting diode (<http://en.wikipedia.org/wiki/LED>), accessed March 16, 2007.

amined types of diodes is made – instead, one average dataset representing the “production efforts” is established; and all types of diodes will then have this dataset as one of their input values. All in all this ends up with the following scheme for the representation of the electronic components “diodes”, including “LED” in the framework of the ecoinvent database.

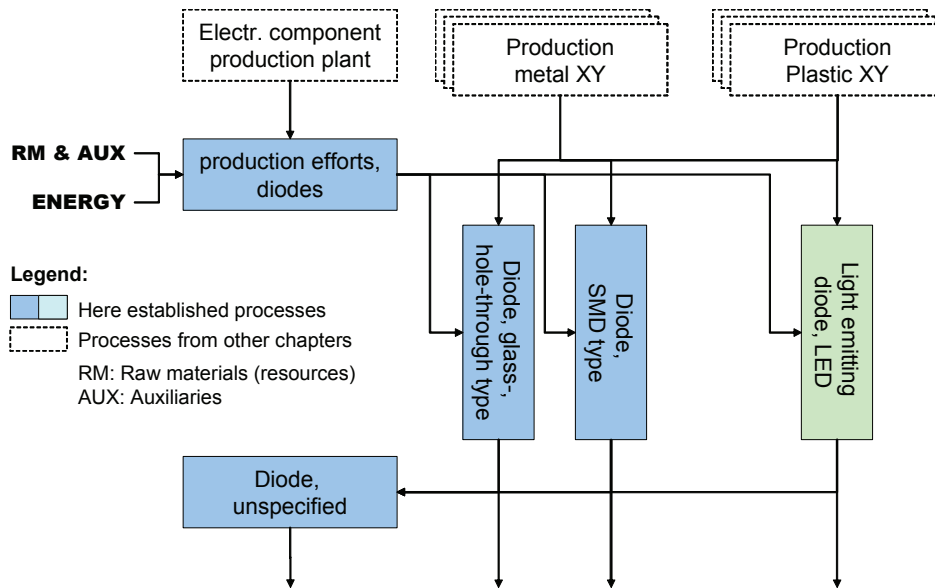


Fig. 5.2 Scheme of the representation of electronic components „diodes” (incl. LED) in the ecoinvent database

All these datasets are based on information representing current models of diodes. The used data sources are mainly Behrendt et al. (1998), CPM (2006), Philips Semiconductors (2002) and ST Microelectronics (2004), representing material specification sheets from industry as well as further data sources.

They represent diodes and LEDs that have been in use in the last about 10 years – as well as current models. The companies behind this information are producing on a global scale – hence, the here developed data can be considered as “global” valid data.

The respective meta information for the various datasets of diodes can be found in Tab. 5.1.

Tab. 5.1 Ecoinvent meta information for the processes for diodes within the database.

Name	production efforts, diodes	diode, unspecified, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This module includes auxiliary and energy consumption, emissions (air, water) and waste amounts, as well as an estimation of the efforts for transportation and of the infrastructure for the production of diodes (incl. LEDs).	This dataset covers raw material input and production efforts for the production of current diodes.
Amount	1	1
Local Name	Dioden, Herstellaufwände	Diode, unspezifisch, ab Werk
Synonyms	-	-
General Comment	Data are a rough estimation of the average production efforts over all types of diodes. Data for auxiliaries, energy, emissions and waste are taken from a dataset out of an international LCI database. Infrastructure and transportation efforts are based on own assumptions.	This module represents an unweighted average of all specific diode datasets (incl. LEDs) represented in the ecoinvent database.
Start Date	1998	1994
End Date	2005	2007
Data valid for entire period	1	1
Geography Text	Data from literature – here representing the global situation.	Data represent global valuable composition data for diodes.
Technology Text	average diode production technology, comprising the diode wafer production (cleaing, masking, etching, doping, oxidising, metal deposition) and the final assembly of the diode (wafer sawing, die bonding, moulding, trimming and forming)	average diode production technology, comprising the diode wafer production (cleaing, masking, etching, doping, oxidising, metal deposition) and the final assembly of the diode (wafer sawing, die bonding, moulding, trimming and forming)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature & own estimations	own estimation
Extrapolations	see Geography and General Comments	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 5.1 (Cont.) Ecoinvent meta information for the processes for diodes within the database.

Name	diode, glass-, through-hole mounting, at plant	diode, glass-, SMD type, for surface mounting, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used glass diodes for through-hole mounting technology.	This dataset covers raw material input and production efforts for the production of currently used SMD diodes for surface mounting technology.
Amount	1	1
Local Name	Diode, Glas-, through-hole mounting, ab Werk	Diode, Glas-, SMD-Typ, surface mounting, ab Werk
Synonyms	-	-
General Comment	The data represent a typical glass diode for hole-through mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such diodes and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of diodes with an average weight of 0.596 gram/unit.	The data represent a typical SMD type diode for surface mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such diodes and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of diodes with an average weight of 32 miligram/unit.
Start Date	1994	1994
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of glass diodes for through-hole mounting technology.	Data represent a global valuable composition of SMD type glass diodes.
Technology Text	average diode production technology, comprising the diode wafer production (cleaing, masking, etching, doping, oxidising, metal deposition) and the final assembly of the diode (wafer sawing, die bonding, moulding, trimming and forming)	average diode production technology, comprising the diode wafer production (cleaing, masking, etching, doping, oxidising, metal deposition) and the final assembly of the diode (wafer sawing, die bonding, moulding, trimming and forming)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	literature	literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 5.1 (Cont.) Ecoinvent meta information for the processes for diodes within the database.

Name	light emitting diode, LED, at plant
Location	GLO
Infrastructure Process	0
Unit	kg
Dataset Version	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used light emitting diodes (LED) for through-hole mounting technology.
Amount	1
Local Name	Leuchtdiode, LED, ab Werk
Synonyms	-
General Comment	The data represent a typical light emitting diode (LED) for hole-through mounting, used in the information and communication technology. Material data are approximated by using respective information from the dataset "diode, glass-, for through-hole mounting". Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of 5mm LEDs with an average weight of 0.35 gram/unit.
Start Date	1994
End Date	2007
Data valid for entire period	1
Geography Text	Data represent a global valuable composition of glass diodes for through-hole mounting technology, used as a first approximation for LEDs.
Technology Text	average diode production technology, comprising the diode wafer production (cleaing, masking, etching, doping, oxidising, metal deposition) and the final assembly of the diode (wafer sawing, die bonding, moulding, trimming and forming)
Representativeness	see General Comments
Production Volume	-
Sampling Procedure	literature
Extrapolations	see Geography & Technology
Uncertainty Adjustments	None

5.3 Dataset "production efforts, diodes"

This dataset describes the production efforts for the production of the various types of diodes examined within ecoinvent. The dataset has the functional unit "efforts for the production of 1 kg of diodes".

5.3.1 System Characterization

The dataset established here comprises the input of auxiliaries and energy as well as the output of emissions and waste due to these auxiliaries in the framework of the diode production. The input of the used infrastructure, as well as the transport efforts for the auxiliaries and the raw materials of the various diodes are included as well. Main data source for this information is the information about the diode production in CPM (2006). The datasets covers the diode production itself as well as the preceding wafer production step.

5.3.2 Life Cycle inventory

As the raw material input is taken into account in the respective diodes datasets – hence this dataset here has no input of raw materials. Instead, this dataset here contains all other input and output flows – i.e. auxiliaries, energy, transport amount (for auxiliaries and raw materials!), infrastructure as well as the emissions and waste amounts produced. In details these various aspects are taken into account based on the following information:

- **Auxiliaries:** CPM (2006) reports a total input of auxiliaries in the order of almost 100 kg per kg of produced diode. However, the major parts of this auxiliary input are process gas (nitrogen, oxygen, hydrogen). The following table (Tab. 5.2) gives an overview of the in CPM (2006) reported amounts and their translation into the ecoinvent dataset in the framework of this project here.

Tab. 5.2 Auxiliary input into the production of diodes (incl. wafer production) per kg of diodes – according to CPM (2006)

Auxiliary		per kg	Used ecoinvent dataset
[i] Process gases			
nitrogen	kg	92.6	nitrogen, liquid, at plant (RER)
oxygen	kg	0.750	oxygen, liquid, at plant (RER)
hydrogen	kg	0.090	hydrogen, liquid, at plant (RER)
[ii] Process chemicals			
HF	kg	0.620	hydrogen fluoride, at plant (RER)
H ₂ SO ₄	kg	0.600	sulphuric acid, at plant (RER)
Methanol	kg	0.550	methanol, at plant (GLO)
xylene	kg	0.296	xylene, at plant (RER)
HNO ₃	kg	0.280	nitric acid, 50% in H ₂ O at plant (RER)
H ₂ O ₂	kg	0.220	hydrogen peroxide, 50% in H ₂ O, at plant (RER)
HCl	kg	0.099	hydrochloric acid, 30% in H ₂ O, at plant (RER)
further chemicals ^{a)}	kg	0.871	chemicals organic, at plant (RER)

a) **comprising:** 2-Propanol, methanesulfonic acid, butyl acetate, acetic acid, ammonia, ethyl alcohol, trisodium citrate, 2-(2-butoxyethoxy)ethanol, ammonium chloride and photoresist

- **Energy:** From CPM (2006) and exact (1998) an average input of electricity of 241 kWh, and a fuel input of 244 MJ of oil (as light fuel oil) and of 280 MJ of natural gas per kg of produced diodes can be calculated.
- **Emissions to air:** CPM (2006) reports a total output in form of emissions to air of about 0.54 kg per kg of produced diode. The following table (Tab. 5.3) gives an overview of the in CPM (2006) reported amounts of emissions into air.

Tab. 5.3 emissions into air from the production of diodes (incl. wafer production) per kg of diodes – according to CPM (2006)

Emissions to air		per kg
xylene, to air	kg	0.101
acetone, to air	kg	0.073
NO ₂ , to air	kg	0.073
CO, to air	kg	0.042
SO ₂ , to air	kg	0.025
HF, to air	kg	0.015
NH ₃ , to air	kg	0.011
Emission to air, various ^{a)}	kg	0.201

a) **comprising:** acetic acid, butyl acetate, trimethyl benzene, hydrocarbons, nitric acid, trichloroethene and fluoride

Furthermore, it was assumed, that 100% of the electricity consumed, i.e. 241 kWh per kg diodes produced, is converted to waste heat. In accordance with the general rules for ecoinvent, it was assumed that 100% of the waste heat is released to the air.

- **Emissions to water & Water input:** In CPM (2006) no quantitative information about the total water consumption is given. It is assumed here, that the amount used is about 50% compared to the amount used in the production of wafers for the microchip production (see chapter 3.5) – i.e. an amount of 0.286 m³ process water and 0.253 m³ cooling water (per kg of diodes). The emissions into water are taken from CPM (2006) and reported in Tab. 5.4.

Tab. 5.4 emissions into water from the production of diodes (incl. wafer production) per kg of diodes – according to CPM (2006)

Emissions to water		per kg
suspended solids, to water	kg	0.110
BOD, to water	kg	0.040
COD, to water	kg	3.6E-03
NO ₂ -, to water	kg	0.064
SO ₄ --, to water	kg	0.013
F-, to water	kg	0.012
Cl-, to water	kg	3.9E-03
PO ₄ ---, to water	kg	3.0E-03
Sn, to water	kg	5.0E-05

- **Waste:** CPM (2006) reports an amount of 0.37 kg of waste per kg of produced diodes. Most of this waste is from the used auxiliaries (solvents, photoresist, ...) – however, a quantification of the amount of raw material in the waste is not possible. Hence, about 90% of the waste are assumed to be from auxiliaries – reported as “disposal, solvents mixture, 16.5% water, to hazardous waste incineration” – and only 0.04 kg are assumed to be waste raw materials – reported as “disposal, waste, Si waferprod., inorg, 9.4% water, to residual material landfill”.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here. The amount takes into account the consumed auxiliaries as well as the raw materials for the production of any type of diodes. The amount of raw materials used equals the material in the diodes (i.e. 1 kg) as well as the waste out of raw materials (i.e. 0.04 kg).

- **Infrastructure and land-use:** Due to a lack of more specific information, the infrastructure module established in chapter 2, representing an electronic component production plant, is used also for the production of diodes.

5.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is only a rough estimation of the actual diodes production efforts, based on information from one single dataset out of the Swedish LCI database (CPM (2006)). Hence, the respective uncertainty values are accordingly high.

The resulting unit process “production efforts, diodes” is shown in Tab. 5.7 (on page 90).

5.4 Dataset “diode, glass-, SMD type, for surface mounting, at plant”

This dataset here describes the production efforts for the production of SMD glass diodes for the surface mounting technology. The dataset has a functional unit of “1 kg of diodes”, based on the assumption that a typical SMD diode in electronic modules has an average weight of **32 mg**.

5.4.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 5.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Behrendt et al. (1998), Philips Semiconductors (2002) and ST Microelectronics (2004).

5.4.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, diodes”, the production of 1 kg of diodes lead to a waste amount of 0.37 kg – whereof 0.04 kg are calculated to be unused raw materials. Hence, this dataset here has an input of 1.04 kg of raw materials per kg of diode. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998), Philips Semiconductors (2002) and ST Microelectronics (2004) and is summarized in Tab. 5.5. In this table the respective ecoinvent datasets used are also shown.

Tab. 5.5 Material specifications of SMD type glass diodes (data sources: see text)

		SMD type	Used ecoinvent dataset
		diode	
Weight	g	0.032	
[i] Metals			
lead	%	0.79	lead, at regional storage (RER)
copper	%	24.03	copper, primary, at plant (RER)
molybdenum	%	15.45	molybdenum, at regional storage (RER)
tin	%	1.47	tin, at regional storage (RER)
[ii] Others			
Al ₂ O ₃	%	30.65	aluminium oxide, at plant (RER)
Glass (Pb containing)	%	15.36	for both inputs the proxy "funnel glass, CRT screen, at plant (RER)" used
Encapsulation (<72% SiO ₂)	%	9.74	
Epoxy resin	%	2.11	epoxy resin, liquid, at plant (RER)
doped silica	%	0.46	silicon, electronic grade, at plant (RER)

5.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical SMD glass diode for surface mounting, based on composition information from various worldwide diodes producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific diode type and have thus, accordingly higher uncertainties.

The resulting unit process “diode, glass-, SMD type, for surface mounting, at plant” is shown in Tab. 5.8 (on page 93).

5.5 Dataset “diode, glass-, for through-hole mounting, at plant”

This dataset here describes the production efforts of the production of diodes based on the through-hole mounting technology. The dataset here has a functional unit of “1 kg of diodes”, based on the assumption that a typical through-hole mounting diode in electronic modules has an average weight of 0.596 g.

5.5.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 5.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Behrendt et al. (1998), Philips Semiconductors (2002) and ST Microelectronics (2004).

5.5.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, diodes”, the production of 1 kg of diodes lead to a waste amount of 0.37 kg –

whereof 0.04 kg are calculated to be unused raw materials. Hence, this dataset here has an input of 1.04 kg of raw materials per kg of diode. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998), Philips Semiconductors (2002) and ST Microelectronics (2004) and is summarized in Tab. 5.6. In this table the respective ecoinvent datasets used are also shown.

Tab. 5.6 Material specifications of through-hole mounted diodes (data sources: see text)

		hole-thr.	Used ecoinvent dataset
		diode	
Weight	g	0.596	
[i] Metals			
lead	%	1.57	lead, at regional storage (RER)
iron	%	43.34	pig iron, at plant (GLO)
copper	%	22.61	copper, primary, at plant (RER)
nickel	%	0.37	nickel, 99.5%, at plant (GLO)
tin	%	4.47	tin, at regional storage (RER)
molybdänum	%	2.02	molybdenum, at regional storage (RER)
[ii] Others			
Glass (Pb containing)	%	15.62	for both inputs the proxy "funnel glass, CRT
Encapsulation (<72% SiO ₂)	%	9.52	screen, at plant (RER)" used
Epoxy resin	%	0.01	epoxy resin, liquid, at plant (RER)
doted silica	%	0.45	silicon, electronic grade, at plant (RER)

5.5.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical diode for through-hole mounting, based on composition information from various worldwide diode producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific capacitor type and have thus, accordingly higher uncertainties.

The resulting unit process “diode, glass-, for through-hole mounting, at plant” is shown in Tab. 5.9 (on page 94).

5.6 Dataset “light emitting diode, LED, at plant”

This dataset here describes the material composition of the production of a light emitting diode (LED). The dataset has a functional unit of “1 kg of LEDs”, based on the assumption that a typical 5mm LED in electronic modules has an average weight of **0.35 g** (Osram (2006)).

5.6.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 5.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data source for the input of raw materials in

this dataset here are – due to a lack of other specific information – the material data of traditional through-hole mounted diodes described in chapter 5.5.

5.6.2 Life Cycle inventory

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 5.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Due to a lack of respective information concerning the actual composition of LEDs – the composition of the dataset “Diode, glass-, for through-hole mounting, at plant” (see chapter 5.5) is used as a proxy.

5.6.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an estimation for a 5mm LED for through-hole mounting, based on the composition information from a traditional through-hole mounted diode. Hence, the uncertainty values are accordingly high. All production efforts (summed up in a separate dataset) represent not a specific capacitor type and have thus, accordingly higher uncertainties.

The resulting unit process “light emitting diode, LED, at plant” is shown in Tab. 5.10 (on page 95).

5.7 Dataset “diode, unspecified, at plant”

Last but not least, for all those cases where no information is given about the diode type used, an unspecified dataset is established as well. This latter one is an unweighted mean value of all three above described diode types (including LEDs). The dataset has as functional unit also “1 kg of diodes”.

5.7.1 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an unweighted average of the various diode types shown in this database. Hence, the uncertainty values are accordingly high.

The resulting unit process “diode, unspecified, at plant” is shown in Tab. 5.11 (on page 96).

Tab. 5.7 Unit process inventory of the dataset “Production efforts, diodes (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment	
process water (pure water)	production efforts, diodes			water supply	production	No	GLO	water, ultrapure, at plant	2.86E+02	kg	estimation as 50% of microchip production data	1	168	(4.52,3.4,5.3); estimation- based on data for microchip production	
nitrogen				chemicals	inorganics	No	RER	nitrogen, liquid, at plant	9.26E+01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
oxygen				chemicals	inorganics	No	RER	oxygen, liquid, at plant	7.50E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
hydrogen				chemicals	inorganics	No	RER	hydrogen, liquid, at plant	9.00E-02	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
hydrogen fluoride				chemicals	inorganics	No	GLO	hydrogen fluoride, at plant	6.20E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
sulfuric acid				chemicals	inorganics	No	RER	sulphuric acid, liquid, at plant	6.00E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
methanol				chemicals	organics	No	GLO	methanol, at plant	5.50E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
xylene				chemicals	organics	No	RER	xylene, at plant	2.96E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
nitric acid				chemicals	inorganics	No	RER	nitric acid, 50% in H2O, at plant	2.80E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
hydrogen peroxide				chemicals	inorganics	No	RER	hydrogen peroxide, 50% in H2O, at plant	2.20E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
hydrogen chloride				chemicals	inorganics	No	RER	hydrochloric acid, 30% in H2O, at plant	9.90E-02	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source	
further chemicals				dataset "chemicals organic" used as a proxy for a list of further chemical inputs	chemicals	organics	No	GLO	chemicals organic, at plant	8.7E-01	kg	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.53); value from 1 literature source
electricity					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	2.41E+02	kWh	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.52); value from 1 literature source
light fuel oil (heat)					oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 1MW	2.44E+02	MJ	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.5); value from 1 literature source
natural gas (heat)					natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	2.80E+02	MJ	data reported in a dataset in CPM (2006)	1	127	(2.4,3.3,1.5); value from 1 literature source

cont inued ▶

continued ▼

Tab. 5.7 (Cont.) Unit process inventory of the dataset “Production efforts, diodes (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
Transport (part by lorry)	production efforts, diodes	continued		transport systems	road	No	RER	transport, lorry >8t, fleet average	9.80E+00	tkm	standard distances	1	2.09	(4.5,na,na,na,5); standard distances	
Transport (part by train)				transport systems	train	No	RER	transport, freight, rail	5.88E+01	tkm	standard distances	1	2.09	(4.5,na,na,na,5); standard distances	
Infrastructure (capacitor plant)				electronics	component	Yes	GLO	electronic component production plant	2.00E+08	unt	rough estimation	1	3.77	(4.5,13.54,9); rough assumption	
		↑ auxiliaries to waste		due to a lack of data, the dataset "disposal, solvent mixture, to HWI" is used as a proxy	waste management	hazardous waste incineration	No	CH	disposal, solvent's mixture, 16.5%water, to hazardous waste incineration	3.33E-01	kg	assumption, based on input and waste amount reported in CPM (2006)	1	1.27	(2.4,3.3,156); assumption, based on total waste information in 1 literature source
		↑ raw materials to waste		due to a lack of data, the dataset "disposal, waste, Si waferprod., inorg., to landfill" is used as a proxy	waste management	residual material landfill	No	CH	disposal, waste, Si waferprod., inorg, 9.4% water, to residual material landfill	3.70E-02	kg	assumption, based on input and waste amount reported in CPM (2006)	1	1.27	(2.4,3.3,156); assumption, based on total waste information in 1 literature source
		↑ waste water to treatment		due to a lack of knowledge of content, dataset "treatment, sewage, WWTP class 2" is used as proxy	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 2	2.86E-01	m3	estimation, based on microchip production data	1	1.27	(2.4,3.3,156); assumption, based on literature
		↑ waste heat			air	high population density			Heat, waste	8.68E+02	MJ	calculated from the electricity input	1	1.27	(2.4,3.3,15,13); calculated, from electricity input
		↑ xylene, to air			air	high population density			Xylene	101E-01	kg	data reported in a dataset in CPM (2006)	1	2.08	(2.4,3.3,15,23); value from 1 literature source
		↑ acetone, to air			air	high population density			Acetone	730E-02	kg	data reported in a dataset in CPM (2006)	1	2.08	(2.4,3.3,15,23); value from 1 literature source
		↑ NO2, to air			air	high population density			Nitrogen oxides	730E-02	kg	data reported in a dataset in CPM (2006)	1	160	(2.4,3.3,15,16); value from 1 literature source
		↑ CO, to air			air	high population density			Carbon monoxide, fossil	4.20E-02	kg	data reported in a dataset in CPM (2006)	1	508	(2.4,3.3,15,17); value from 1 literature source
		↑ SO2, to air			air	high population density			Sulfur dioxide	2.50E-02	kg	data reported in a dataset in CPM (2006)	1	1.27	(2.4,3.3,15,15); value from 1 literature source
		↑ HF, to air			air	high population density			Hydrogen fluoride	150E-02	kg	data reported in a dataset in CPM (2006)	1	160	(2.4,3.3,15,31); value from 1 literature source
continued			↑ NH3, to air		air	high population density			Ammonia	110E-02	kg	data reported in a dataset in CPM (2006)	1	134	(2.4,3.3,15,28); value from 1 literature source

Tab. 5.7 (Cont.) Unit process inventory of the dataset “Production efforts, diodes (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
▼ continued	production efforts, diodes	↑ suspended solids, to water		water	river			Suspended solids, unspecified	1.0E-01	kg	data reported in a dataset in CPM (2006)	1	160	(2,4,3,3,1,5,32); value from 1 literature source
		↑ BOD, to water		water	river			BOD5, Biological Oxygen Demand	4.00E-02	kg	data reported in a dataset in CPM (2006)	1	160	(2,4,3,3,1,5,32); value from 1 literature source
		↑ COD, to water		water	river			COD, Chemical Oxygen Demand	3.60E-03	kg	data reported in a dataset in CPM (2006)	1	160	(2,4,3,3,1,5,32); value from 1 literature source
		↑ NO2-, to water		water	river			Nitrite	6.40E-02	kg	data reported in a dataset in CPM (2006)	1	160	(2,4,3,3,1,5,33); value from 1 literature source
		↑ SO4--, to water		water	river			Sulfate	130E-02	kg	data reported in a dataset in CPM (2006)	1	160	(2,4,3,3,1,5,33); value from 1 literature source
		↑ F-, to water		water	river			Fluoride	120E-02	kg	data reported in a dataset in CPM (2006)	1	160	(2,4,3,3,1,5,33); value from 1 literature source
		↑ Cl-, to water		water	river			Chloride	3.90E-03	kg	data reported in a dataset in CPM (2006)	1	3.07	(2,4,3,3,1,5,34); value from 1 literature source
		↑ PO4---, to water		water	river			Phosphate	3.00E-03	kg	data reported in a dataset in CPM (2006)	1	160	(2,4,3,3,1,5,33); value from 1 literature source
		↑ Sn, to water		water	river			Tin, ion	5.00E-05	kg	data reported in a dataset in CPM (2006)	1	5.08	(2,4,3,3,1,5,35); value from 1 literature source
		↑ Production efforts for capacitors		electronics	component	No	GLO	production efforts, diodes	100E+00	kg		1		

Tab. 5.8 Unit process inventory of the dataset “Diode, glass, SMD type, surface mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
production efforts	diode, glass-, SMD type, surface mounting, at plant	SMD diode		electronics	component	No	GLO	production efforts, diodes	100E+00	kg	average resistor production from CPM (2006)	1	1.61	(3,4,3,1,4,5,1); rough estimation
lead				metals	extraction	No	RER	lead, at regional storage	8.22E-03	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	2.50E-01	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information
molybdenum				metals	extraction	No	RER	molybdenum, at regional storage	1.61E-01	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	1.53E-02	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information
Aluminium oxide				chemicals	inorganics	No	RER	aluminium oxide, at plant	3.19E-01	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information
glass (incl. Encapsulation)				electronics	component	No	GLO	funnel glass, CRT screen, at plant	2.61E-01	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	2.19E-02	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information
doped silica				metals	refinement	No	DE	silicon, electronic grade, at plant	4.78E-03	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14,2,11,4,3); average from literature & company information

Tab. 5.9 Unit process inventory of the dataset “Diode, glass, through-hole mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
production efforts	diode, glass-, through-hole mounting, at plant	through-hole mounting diode		electronics	component	No	GLO	production efforts, diodes	100E+00	kg	average resistor production from CPM (2006)	1	1.61	(3.4, 3.14, 5.11); rough estimation
lead				metals	extraction	No	RER	lead, at regional storage	1.63E-02	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	4.5E-01	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	2.35E-01	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	3.85E-03	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	4.65E-02	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
molybdenum				metals	extraction	No	RER	molybdenum, at regional storage	2.10E-02	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
glass (ind. Encapsulation)			CRT funnel glass used as a proxy due to the Pb content	electronics	component	No	GLO	funnel glass, CRT screen, at plant	2.61E-01	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	1.04E-04	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
doped silica				metals	refinement	No	DE	silicon, electronic grade, at plant	4.68E-03	kg	Average according to Behrendt (1998) and Philips (2002) and ST (2004)	1	1.16	(14.2, 11.4, 3); average from literature & company information
				electronics	component	No	GLO	diode, glass-, through-hole mounting, at plant	100E+00	kg		1		(14.2, 11.4, 3); average from literature & company information

Tab. 5.10 Unit process inventory of the dataset “light emitting diode, LED, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	light emitting diode, LED, at plant	LED		electronics	component	No	GLO	production efforts, diodes	100E+00	kg	average resistor production from CPM (2006)	1	1.61	(3.4.3.14.5.1); rough estimation
lead			metals	extraction	No	RER	lead, at regional storage	163E-02	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy (using through-hole mounted diodes)	
iron			metals	extraction	No	GLO	pig iron, at plant	4.51E-01	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy	
copper			metals	extraction	No	RER	copper, primary, at refinery	2.35E-01	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy	
nickel			metals	extraction	No	GLO	nickel, 99.5% at plant	3.85E-03	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy	
tin			metals	extraction	No	RER	tin, at regional storage	4.65E-02	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy	
molybdenum			metals	extraction	No	RER	molybdenum, at regional storage	2.70E-02	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy	
glass (incl. Encapsulation)			CRT funnel glass used as a proxy due to the Pb content	electronics	component	No	GLO	funnel glass, CRT screen, at plant	2.61E-01	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	104E-04	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy
doped silica				metals	refinement	No	DE	silicon, electronic grade, at plant	4.68E-03	kg	rough estimation- based on data for through-hole mounted diodes	1	1.61	(3.4.3.14.5.3); assumption from conclusion by analogy
				electronics	component	No	GLO	light emitting diode, LED, at plant	100E+00	kg		1		

Tab. 5.11 Unit process inventory of the dataset “Diode, unspecified, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
SMD diodes	diode, unspecified, at plant	↑	↑	electronics	component	No	GLO	diode, glass-, SMD type, surface mounting, at plant	3.33E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
through-hole mounting diode				electronics	component	No	GLO	diode, glass-, through-hole mounting, at plant	3.33E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
LED				electronics	component	No	GLO	light emitting diode, LED, at plant	3.33E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
				electronics	component	No	GLO	diode, unspecified, at plant	100E+00	kg		1		

6 Capacitors

A **capacitor** “is an electrical device that can store energy in the electric field between a pair of closely-spaced conductors (called 'plates').”¹

As soon as voltage is applied to a capacitor, this leads to the building of electric charges of equal magnitude, but opposite polarity, on each plate. Within electrical circuits, hence, capacitors are used as devices for energy-storage. In an electronic filter, a capacitor can differentiate between signals of different frequencies – i.e. between high-frequency and low-frequency signals.

Capacitors are occasionally referred to as *condensers*, however this is considered today an antiquated term.

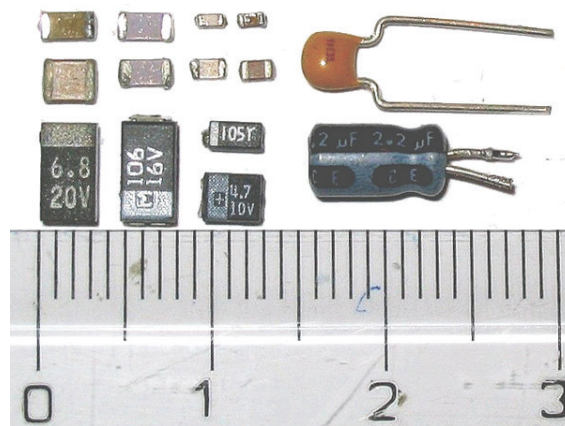


Fig. 6.1 Capacitors: SMD ceramic (top left); SMD tantalum (bottom left); through-hole tantalum (top right); through-hole electrolytic (bottom right); scale in cm (Picture taken from Wikipedia).

6.1 Characterisation, Use & Market

Many types of capacitor are available commercially, with capacitances ranging from the picofarad range to more than a farad, and voltage ratings up to hundreds of kilovolts. In general, the higher the capacitance and voltage rating, the larger the physical size of the capacitor and the higher the cost (e.g. Fig. 6.2). Tolerances in capacitance value for discrete capacitors are usually specified as a percentage of the nominal value. Tolerances ranging from 50% (electrolytic types) to less than 1% are commonly available.

Currently, there is a huge variety of different types of capacitors on the market – a variety that can not be covered 100% by the here described ecoinvent datasets. Main characteristics are different dielectric media used in these different types of capacitors – from paper, over various plastics up to aluminium oxide.¹ Hence, for this project here a choice had to be made, taking into account the following types of capacitors: film type (for through-hole mounting) and SMD type (for surface mounting) as small capacitors, electrolyt capacitors in both forms (i.e. small and big ones) and last but not least tantalum-containing ceramic capacitors.

These most important three types – SMD type, film type and electrolyt- – can, according to EPCOS (2006), be characterized as the following:

- **Film type:** They have almost unlimited selfhealing capability. Short circuits are thus largely nonexistent. They exhibit high pulse handling and ripple current capability. Typical applications are mobile phones, television, automobiles, PCs and lamp ballasts.

¹ Wikipedia: Capacitor (<http://en.wikipedia.org/wiki/Capacitors>), accessed March 12, 2007.

- **SMD type:** They offer maximum capacitance within the smallest space, and hence, can be found in many areas of advanced microelectronics – for instance information, entertainment, automotive, and industrial electronics.
- **Electrolyte:** Mainly out of aluminium, they are notable for their high capacitance per unit volume (CV product) and excellent current handling capability. Therefore they are essential components in frequency converters, as DC link in traction, in electronic lamp ballasts, automotive and studio flash applications.



Fig. 6.2 Various types of capacitors (Picture taken from Wikipedia²).

6.2 System Characterisation

While about the composition of the various types of capacitors, a variety of data sources can be found, it is very hard to get any information concerning the actual production efforts for the production of a capacitor. Hence, the composition is much more accurate and in details shown in this study than the production itself. Thus – for the production efforts of capacitors, no distinction between the different here examined types of capacitors is made – instead, one average dataset representing the “production efforts” is established; and all types of capacitors will then have this dataset as one of their input values. All in all this ends up with the scheme for the representation of the electronic components “Capacitors” in the framework of the ecoinvent database shown in Fig. 6.3.

² Wikipedia: Capacitor (<http://en.wikipedia.org/wiki/Capacitors>), accessed March 12, 2007.

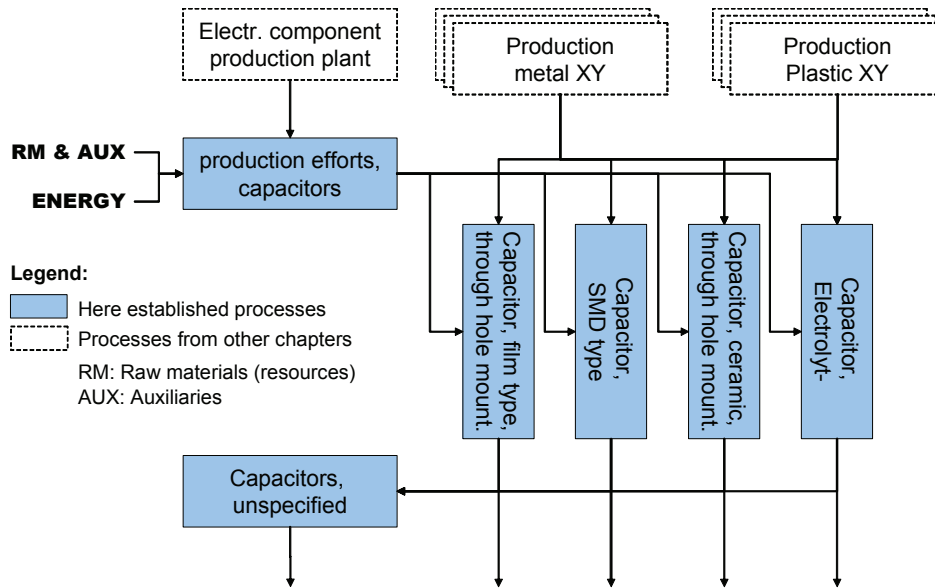


Fig. 6.3 Scheme of the representation of the electronic components „Capacitors“ in the ecoinvent database

All these datasets are based on information representing current models of capacitors. The used data sources are mainly Behrendt et al. (1998), CPM (2006), EPCOS (2006), exact (1998) and ZVEI (2006), representing material specification sheets from industry as well as further data sources.

They represent capacitors that have been in use in the last about 10 years – as well as current models. The companies behind this information are producing on a global scale – hence, the here developed data can be considered as “global” valid data.

The respective meta information for the various datasets of capacitors can be found in Tab. 6.1.

Tab. 6.1 ecoinvent meta information for the processes for capacitors within the database.

Name	production efforts, capacitors	capacitor, unspecified, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This module includes auxillary and energy consumption, emissions (air, water) and waste amounts, as well as an estimation of the efforts for transportation and of the infrastructure for the production of capacitors.	This dataset covers raw material input and production efforts for the production of current capacitors.
Amount	1	1
Local Name	Kondensatoren, Herstellaufwände	Kondensator, unspezifisch, ab Werk
Synonyms	Condenser production	Condenser, unspecified
General Comment	Data are a rough estimation of the average production efforts over all types of capacitors. Data for auxillaries, energy, emissions and waste are taken from the Environmental report of a global producer as well as from two datasets out of an international LCI database. Infrastructure and transportation efforts are based on own assumptions.	This module represents an unweighted average of all specific capacitor datasets represented in the ecoinvent database.
Start Date	1998	1994
End Date	2005	2007
Data valid for entire period	1	1
Geography Text	Data from literature and one global producer – here representing the global situation.	Data represent global valuable composition data for capacitors.
Technology Text	average production of: formulation; build-up process; cutting; bisquing; Pre-Firing; Firing; Tumbling; Termination (SMT) resp. metallization; slitting; winding; metal spray; heat treatment; sawing; lead welding; epoxy filling (THT)	average production of: formulation; build-up process; cutting; bisquing; Pre-Firing; Firing; Tumbling; Termination (SMT) resp. metallization; slitting; winding; metal spray; heat treatment; sawing; lead welding; epoxy filling (THT)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature & own estimations	own estimation
Extrapolations	see Geography and General Comments	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 6.1 (Cont.) ecoinvent meta information for the processes for capacitors within the database.

Name	capacitor, film type, for through-hole mounting, at plant	capacitor, SMD type, for surface-mounting, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used film capacitors for through-hole mounting technology.	This dataset covers raw material input and production efforts for the production of currently used SMD capacitors for surface mounting technology.
Amount	1	1
Local Name	Kondensator, Film-, through-hole mounting, ab Werk	Kondensator, SMD-Typ, surface mounting, ab Werk
Synonyms	film condenser	SMD condenser
General Comment	The data represent a typical film type capacitor for hole-through mounting, used in the information and communication technology. Material data are taken from data-sheets of producers of such capacitors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of capacitors with an average weight of 0.7 gram/unit.	The data represent a typical SMD type capacitor for surface mounting, used in the information and communication technology. Material data are taken from data-sheets of producers of such capacitors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of capacitors with an average weight of 0.086 gram/unit.
Start Date	1994	1994
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of film type capacitors.	Data represent a global valuable composition of SMD type capacitors.
Technology Text	average production of: formulation; build-up process; cutting; bisquing; Pre-Firing; Firing; Tumbling; Termination (SMT) resp. metallization; slitting; winding; metal spray; heat treatment; sawing; lead welding; epoxy filling (THT)	average production of: formulation; build-up process; cutting; bisquing; Pre-Firing; Firing; Tumbling; Termination (SMT) resp. metallization; slitting; winding; metal spray; heat treatment; sawing; lead welding; epoxy filling (THT)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	literature	literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 6.1 (Cont.) ecoinvent meta information for the processes for capacitors within the database.

Name	capacitor, electrolyt-, < 2 cm hight, at plant	capacitor, electrolyt-, > 2 cm hight, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used small (< 2 cm) electrolyt-capacitors.	This dataset covers raw material input and production efforts for the production of currently used big (> 2 cm) electrolyt-capacitors.
Amount	1	1
Local Name	Kondensator, Elektrolyt-, < 2cm Höhe, ab Werk	Kondensator, Elektrolyt-, > 2cm Höhe, ab Werk
Synonyms	electrolyt condenser	electrolyt condenser
General Comment	The data represent a typical small electrolyt-capacitor (hight < 2 cm), used in the information and communication technology. Material data are taken from datasheets of producers of such capacitors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of capacitor with an average weight of 1.29 gram/unit.	The data represent a typical small electrolyt-capacitor (hight between 2 and 8 cm), used in the information and communication technology. Material data are taken from datasheets of producers of such capacitors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of capacitors with an average weight of 50.5 gram/unit.
Start Date	1994	1994
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of small electrolyt-capacitors.	Data represent a global valuable composition of big electrolyt-capacitors.
Technology Text	average production of: formulation; build-up process; cutting; bisquing; Pre-Firing; Firing; Tumbling; Termination (SMT) resp. metallization; slitting; winding; metal spray; heat treatment; sawing; lead welding; epoxy filling (THT)	average production of: formulation; build-up process; cutting; bisquing; Pre-Firing; Firing; Tumbling; Termination (SMT) resp. metallization; slitting; winding; metal spray; heat treatment; sawing; lead welding; epoxy filling (THT)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	literature	literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 6.1 (Cont.) ecoinvent meta information for the processes for capacitors within the database.

Name	capacitor, Tantalum-, for through-hole mounting, at plant
Location	GLO
Infrastructure Process	0
Unit	kg
Dataset Version	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used tantalum-containing capacitors for through-hole mounting.
Amount	1
Local Name	Kondensator, Tantal-, für through-hole mounting, ab Werk
Synonyms	Condenser
General Comment	The data represent a tantalum-capacitor for the hole-through mounting technology, used in the information and communication technology. Material data are taken from datasheets of producers of such capacitors. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of capacitors with an average weight of 0.254 gram/unit.
Start Date	1994
End Date	2007
Data valid for entire period	1
Geography Text	Data represent a global valuable composition of small electrolyt-capacitors.
Technology Text	average production of: formulation; build-up process; cutting; bisquing; Pre-Firing; Firing; Tumbling; Termination (SMT) resp. metallization; slitting; winding; metal spray; heat treatment; sawing; lead welding; epoxy filling (THT)
Representativeness	see General Comments
Production Volume	-
Sampling Procedure	literature
Extrapolations	see Geography & Technology
Uncertainty Adjustments	None

6.3 Dataset “production efforts, capacitors”

This dataset describes the production efforts for the production of the various types of capacitor examined within the ecoinvent database. The dataset has the functional unit “production of 1 kg of capacitors”.

6.3.1 System Characterization

The dataset established here comprises the input of auxiliaries and energy as well as the output of emissions and waste due to these auxiliaries in the framework of the capacitor production. The input of the used infrastructure, as well as the transport efforts for the auxiliaries and the raw materials of the

various capacitors are included as well. Main data sources for this information are CPM (2006), EPCOS (2006) and US DoE (2007).

6.3.2 Life Cycle inventory

As the raw material input is taken into account in the respective capacitor datasets – hence this dataset has no input of raw materials. Instead, this dataset here contains all other input and output flows – i.e. auxiliaries, energy, transport amount (for auxiliaries and raw materials!), infrastructure as well as the emissions and waste amounts produced. In details these various aspects are taken into account based on the following information:

- **Auxiliaries:** From CPM (2006) and EPCOS (2006) an average input of about 0.99 kg of auxiliaries per kg of produced capacitors can be calculated. Based on the information in CPM (2006), the following substances are those with the highest amount – and hence, the overall amount is shown as:

Acetone	10%	-
Ethyl Acetate	40%	-
Methyl ethyl ketone	40%	-
Hardener	10%	As “chemicals organic, at plant”

- **Energy:** From CPM (2006) and EPCOS (2006) an average input of electricity of 48.7 kWh and an input of natural gas (fuel) of 25.3 MJ per kg of produced capacitors can be calculated.
- **Emissions to air:** From CPM (2006) and EPCOS (2006) an average output of 0.34 kg of air emissions per kg of produced capacitors can be calculated. In accordance with the above shown treatment of the auxiliaries input, this output amount is split up in the following way – assuming that the emissions have a similar distribution like the auxiliaries:

Acetone, to air	10%	-
Ethyl Acetate, to air	40%	-
Methyl ethyl ketone, to air	40%	-
Hardener, to air	10%	as “NMVOC, ”

Furthermore, it was assumed, that 100% of the electricity consumed, i.e. 48.7 kWh per kg capacitor produced, is converted to waste heat. In accordance with the general rules for ecoinvent, it was assumed that 100% of the waste heat is released to the air.

- **Emissions to water & Water input:** From EPCOS (2007), an average cooling water amount of 0.13 m³ resp. a process water amount of 0.15 m³ per kg of produced capacitors can be estimated. Due to a lack of respective emission information, the outflow of the latter amount is approximated by the use of the dataset “treatment, sewage, to waste water treatment, class 2”.
- **Waste:** From CPM (2006) and EPCOS (2006) a total average output of 1.13 kg of waste per kg of produced capacitors can be calculated. Assuming that the auxiliaries are emitted either as emissions (air, water, soil) or as waste, it can be concluded that the remaining part of 0.48 kg waste are non-used raw materials. Due to a lack of information about the specific composition of this kind of waste, for 50% the dataset “disposal, waste, Si waferprod, to residual material landfill” and for the remaining 50% the dataset “disposal, plastic mixture, to sanitary landfill” are used as proxies. For the remaining 0.65 kg of auxiliaries, the dataset “disposal, solvent mixture, to hazardous waste incineration” is used.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here. The amount takes

into account the consumed auxiliaries as well as the raw materials for the production of any type of capacitor. The amount of raw materials used equals the material in the capacitor (i.e. 1 kg) as well as the waste out of raw materials (i.e. 0.48 kg).

- **Infrastructure and land-use:** Due to a lack of more specific information, the infrastructure module established in chapter 2, representing an electronic component production plant, is used also for the production of capacitors.

6.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is only a rough estimation of the actual capacitor production efforts, based on information from two datasets out of the Swedish LCI database (CPM (2006)) and from the company wide information of a leading capacitor producer (EPCOS (2007)). Hence, the respective uncertainty values are accordingly high.

The resulting unit process “production efforts, capacitors” is shown in Tab. 6.6 (on page 112).

6.4 Dataset “capacitor, film type, for through-hole mounting, at plant”

This dataset here describes the material composition of the production of film type capacitors for the through-hole mounting technology. As already mentioned above, this type of capacitor has almost unlimited selfhealing capabilities and short circuits are thus largely nonexistent. Typical applications for this type of capacitor are mobile phones, television, automobiles, PCs and lamp ballasts (EPCOS (2006)).

The dataset here has a functional unit of “1 kg of capacitors”, based on the assumption that a typical film capacitor in electronic modules has an average weight of **0.7 g**.

6.4.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 6.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Behrendt et al. (1998) and ZVEI (2006).

6.4.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, capacitors”, the production of 1 kg of capacitors lead to a waste amount of 1.13 kg – whereof about 0.48 kg are unused raw materials. Hence, this dataset here has an input of 1.48 kg of raw materials per kg of capacitor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998) and ZVEI (2006) and is summarized in Tab. 6.2. In this table the respective ecoinvent datasets used are also shown.

Tab. 6.2 Material specifications of film-type capacitors (data sources: see text)

		film type capacitor	Used ecoinvent dataset
Total weight	g	0.7	
[i] Metals			
Aluminium	%	3.15	aluminium, primary, at plant (RER)
Copper	%	24.59	copper, primary, at refinery (RER)
Iron	%	7.90	pig iron, at plant (RER)
Lead	%	0.41	lead, at regional storage (RER)
Nickel	%	0.90	nickel, 99.5%, at plant (RER)
Phosphor	%	1.05	phosphorus, white, liquid, at plant (RER)
Silver	%	2.82	silver, at regional storage (RER)
Tin	%	2.29	tin, at regional storage (RER)
Zinc	%	8.38	zinc, for coating, at regional storage (RER)
[ii] Others			
PPS	%	6.47	polyphenylene sulfide, at plant (GLO)
PEN	%	6.03	as "polyphenylene sulfide, at plant (GLO)"
PP/PBT	%	2.64	100% as "polypropylene, at plant (RER)"
PP/PET	%	4.75	50% as "polypropylene, at plant (RER)" and 50% as "PET, amorphous, at plant (RER)"
Phenol resins	%	0.24	phenolic resin, at plant (RER)
epoxy resins	%	15.50	epoxy resin, liquid, at plant (RER)
Thermosetting resins	%	2.73	as "epoxy resin, liquid, at plant (RER)"
plastics, unspec.	%	10.15	as 50% "polyphenylene sulfide" and 50% "epoxy resin, liquid"

6.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical film capacitor for through-hole mounting, based on composition information from various worldwide capacitor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific capacitor type and have thus, accordingly higher uncertainties.

The resulting unit process “capacitor, film type, for through-hole mounting, at plant” is shown in Tab. 6.7 (on page 114).

6.5 Dataset “capacitor, SMD type, for surface-mounting, at plant”

This dataset here describes the material composition of the production of SMD capacitors for the surface-mounting technology. As already mentioned above, this type of capacitor offers a maximum capacitance within the smallest space, and hence, can be found in many areas of advanced microelectronics – for instance information, entertainment, automotive, and industrial electronics (EPCOS (2006)). The dataset here has a functional unit of “1 kg of capacitors”, based on the assumption that a typical SMD capacitor has an average weight of **0.086 g**.

6.5.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 6.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Behrendt et al. (1998), Venkel (2006), as well as ZVEI (2006).

6.5.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, capacitors”, the production of 1 kg of capacitors lead to a waste amount of 1.13 kg – whereof about 0.48 kg are unused raw materials. Hence, this dataset here has an input of 1.48 kg of raw materials per kg of capacitor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998) and Venkel (2006) and is summarized in Tab. 6.3. In this table the respective ecoinvent datasets used are also shown.

Tab. 6.3 Material specifications of SMD capacitors (data sources: see text)

		SMD capacitor	Used ecoinvent dataset
Weight	g	0.086	
[i] Metals			
Copper	%	0.95	copper, primary, at refinery (GLO)
Nickel	%	13.57	nickel, 99.5%, at plant (RER)
Palladium	%	0.09	palladium, at regional storage (RER)
Silver	%	2.65	silver, primary, at plant
Tin	%	1.65	tin, at regional storage (RER)
[ii] Others			
Glas	%	0.15	flat glass, uncoated, at plant (RER)
Ceramics (BaTiO ₃ / CoO)	%	39.13	34% as TiO ₂ / 66% as BaCO ₃ (with BaSO ₄ as proxy) according to stoichiometric proportions
BaTiO ₃ or CaZrO ₃	%	2.34	
CaMgTiO ₃	%	6.98	
BaTiO ₃	%	27.15	
CaTiO ₃	%	5.34	

6.5.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical SMD capacitor for surface-mounting, based on composition information from various worldwide capacitor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific capacitor type and have thus, accordingly higher uncertainties.

The resulting unit process “capacitor, SMD type, for surface-mounting, at plant” is shown in Tab. 6.8 (on page 116).

6.6 Datasets for Electrolyt Capacitors

The two datasets here describe the material composition of the production of small (i.e. height below 2 cm) and big (i.e. height above 2, up to 8 cm) electrolyt capacitors. As already mentioned above, an electrolyt capacitor is made mainly out of aluminium and such capacitors are notable for their high capacitance per unit volume (CV product) and their excellent current handling capability. Therefore they are essential components in frequency converters, as DC link in traction, in electronic lamp ballasts, automotive and studio flash applications (EPCOS (2006)). The datasets here have as functional unit both “1 kg of capacitors”, based on the assumption that a typical small electrolyt capacitor has an average weight of **1.29 g**, a big electrolyt capacitor an average weight of **50.5 g**.

6.6.1 System Characterization

The datasets established here comprise the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 6.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in these datasets here are Behrendt et al. (1998) and ZVEI (2006).

6.6.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, capacitors”, the production of 1 kg of capacitors lead to a waste amount of 1.13 kg – whereof about 0.48 kg are unused raw materials. Hence, this dataset here has an input of 1.48 kg of raw materials per kg of capacitor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998) and ZVEI (2006) and is summarized in Tab. 6.4. In this table the respective ecoinvent datasets used are also shown.

6.6.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a small electrolyt capacitor, based on composition information from various worldwide capacitor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific capacitor type and have thus, accordingly higher uncertainties.

The two resulting unit processes “capacitor, electrolyt-, < 2 cm height, at plant” and “capacitor, electrolyt-, > 2 cm height, at plant” are shown in Tab. 6.9 (on page 116) and Tab. 6.10 (on page 119).

Tab. 6.4 Material specifications of small and big electrolyt capacitors (data sources: see text)

		small	big	Used ecoinvent dataset
		electrolyt	electrolyt	
		capacitor	capacitor	
Total weight	g	1.29	50.5	
[i] Metals				
Aluminium	%	47.2	50.9	aluminium, primary, at plant (RER)
Copper	%	0.920	0.919	copper, primary, at refinery (RER)
Iron	%	5.7	2.0	pig iron, at plant (RER)
Lead	%	0.003	-	lead, at regional storage (RER)
Silver	%	1.7	-	silver, at regional storage (RER)
Tin	%	0.139	0.073	tin, at regional storage (RER)
[ii] Others				
electrolyte (Et-glycol, ...)	%	18.5	25.0	ethylene glycol, at plant (RER)
paper	%	5.3	8.3	kraft paper, unbleached, at plant (RER)
rubber	%	4.1	1.6	synthetic rubber, at plant (RER)
paper / rubber / teflon	%	0.690	1.150	as 50% paper / 50% rupper
PVC	%	0.963	0.963	polyvinylchloride, at regional storage (RER)
PVC / PET	%	0.8	1.3	50% as "PVC, at regional storage (RER)" and 50% as "PET, amorphous, at plant (RER)"
plastics, unspec.	%	6.6	6.6	50% as "PVC, at regional storage (RER)" and 50% as "PET, amorphous, at plant (RER)"
phenolic resin / nylon	%	-	1.3	50% as "phenolic resin, at plant (RER)" and 50% as "nylon 6, at plant (RER)"
Polyamide-6	%	0.025	-	as nylon
PPS	%	1.4	-	polyphenylene sulfide, at plant (GLO)
Polypyrrole	%	0.213	-	as "polyphenylene, sulfide, at plant (GLO)"
Epoxy resin	%	5.8	-	epoxy resin, liquid, at plant (RER)

6.7 Dataset “capacitor, Tantalum-, for through-hole mounting, at plant”

This dataset here describes the material composition of the production of tantalum capacitors for the through-hole mounting technology. This type of capacitor offers a large capacitance to volume ratio, smaller size, good stability, a wide operating temperature range, and a long reliable operating life.³ Hence, they are extensively used in miniaturized equipment and computers. The dataset here has a functional unit of “1 kg of capacitors”, based on the assumption that a typical tantalum ceramic capacitor has an average weight of **0.254 g**.

6.7.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 6.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as

³ Wikipedia: Capacitor (<http://en.wikipedia.org/wiki/capacitor>), accessed March 12, 2007.

well as the produced emissions and waste amounts. Main data source for the input of raw materials in this dataset here is ZVEI (2006).

6.7.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, capacitors”, the production of 1 kg of capacitors lead to a waste amount of 1.13 kg – whereof about 0.48 kg are unused raw materials. Hence, this dataset here has an input of 1.48 kg of raw materials per kg of capacitor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in ZVEI (2006) and is summarized in Tab. 6.5. In this table the respective ecoinvent datasets used are also shown.

Tab. 6.5 Material specifications of tantalum (ceramic) capacitors (data sources: see text)

		Tantalum capacitor	Used ecoinvent dataset
Weight	g	0.254	
[i] Metals			
Iron	%	5.86	pig iron, at plant (RER)
Nickel	%	4.24	nickel, 99.5%, at plant (RER)
Silver	%	2.60	silver, primary, at plant
Tin	%	2.30	tin, at regional storage (RER)
Tantalum	%	36.70	tantalum, powder, capacitor-grade, at regional storage (GLO)
[ii] Others			
Manganese Dioxide	%	5.70	63% as Manganese (stoichiometric amount) - remaining part omitted
Sb ₂ O ₃	%	1.10	as "chemical inorganic, at plant (GLO)"
Teflon	%	0.50	tetrafluoroethylene, at plant
Epoxid (brominated)	%	1.00	as "epoxy resin, liquid, at plant (RER)"
Epoxy resin	%	40.00	epoxy resin, liquid, at plant (RER)

6.7.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a tantalum capacitor for through-hole mounting, based on composition information from a worldwide capacitor producer. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific capacitor type and have thus, accordingly higher uncertainties.

The resulting unit process “capacitor, tantalum-, for through-hole mounting, at plant” is shown in Tab. 6.11 (on page 121).

6.8 Dataset “capacitor, unspecified, at plant”

Last but not least, for all those cases where no information is given about the capacitor type used, an unspecified dataset is established as well. This latter one is an unweighted mean value of all five above described capacitor types. The dataset has as functional unit also “1 kg of capacitors”.

6.8.1 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an unweighted average of the various capacitor types shown in this database. Hence, the uncertainty values are accordingly high.

The resulting unit process “capacitor, unspecified, at plant” is shown in Tab. 6.12 (on page 122).

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Tab. 6.6 Unit process inventory of the dataset "Production efforts, capacitor (GLO)"

General Flow information					Representation in ecoinvent					Uncertainty information					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment	
Water (process)	production efforts, capacitors			resource	in water			Water, unspecified natural origin	150E-01 m3		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,12); average from literature & CER	
Water (cooling)				resource	in water			Water, cooling, unspecified natural origin	130E-01 m3		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,12); average from literature & CER	
acetone				chemicals	organics	No	RER	acetone, liquid, at plant	9.90E-02 kg		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,3); average from literature & CER	
ethyl acetate				chemicals	organics	No	RER	ethyl acetate from butane, at plant	3.96E-01 kg		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,3); average from literature & CER	
methyl ethyl ketone				chemicals	organics	No	RER	methyl ethyl ketone from butane, at plant	3.96E-01 kg		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,3); average from literature & CER	
hardener				dataset "chemical organic" as proxy used	chemicals	organics	No	GLO	chemicals organic, at plant	9.90E-02 kg		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,3); average from literature & CER
electricity					electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.87E+01 kWh		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,2); average from literature & CER
natural gas (heat)					natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	2.53E+01 MJ		calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,1); average from literature & CER
Transport (part by lorry)					transport systems	road	No	RER	transport, lorry >6t, fleet average	2.47E-01 tkm		standard distances according to Frischknecht et al. (2004)	1	2.09	(4,5,nanana,5); standard distances
Transport (part by train)					transport systems	train	No	RER	transport, freight, rail	1.48E+00 tkm		standard distances according to Frischknecht et al. (2004)	1	2.09	(4,5,nanana,5); standard distances
Infrastructure (capacitor plant)					electronics	component	Yes	GLO	electronic component production plant	2.00E+08 unit		rough estimation	1	3.77	(4,5,13,5,4,9); rough estimation of infrastructure
continued ▼				due to a lack of data, the dataset "disposal, solvent mixture, to HWT" is used as a proxy	waste management	hazardous waste incineration	No	CH	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	6.50E-01 kg	calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,6); average from literature & CER	
				due to a lack of information, 50% are assumed as plastics waste to sanitary landfill.	waste management	sanitary landfill	No	CH	disposal, plastics, mixture, 15.3% water, to sanitary landfill	2.40E-01 kg	amount calculated from CPM (2006) and EPCOS (2007)	1	127	(2,4,3,3,15,6); assumption, based on literature & CER	

Tab. 6.6 (Cont.) Unit process inventory of the dataset “Production efforts, capacitor (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
▼ continued	production efforts, capacitors	raw materials to waste	...and the remaining 50%are assumed to be anorganic waste to landfill (proxy of Si wafer waste used)	waste management	residual material landfill	No	CH	disposal, waste, Si waferprod., inorg, 9.4 % water, to residual material landfill	2.40E-01	kg	amount calculated from CPM (2006) and EPCOS (2007)	1	1.27	(2.4.3.3.156); assumption based on literature & CER
		waste water to treatment	due to a lack of knowledge of content, dataset "treatment, sewage, WWTP class 2" is used as proxy	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 2	150E-01	m3	calculated as 100%of process water input ...	1	1.27	(2.4.3.3.156); average from literature & CER
		waste heat	100%of electricity input (according to quality guidelines of ecoinvent)	air	high population density			Heat, waste	175E+02	MJ	calculated from electricity input	1	2.08	(2.4.3.3.1523); calculated from electricity input
		acetone, to air		air	high population density			Acetone	3.40E-02	kg	calculated frominput of auxiliaries	1	2.08	(2.4.3.3.1523); calculated frominput of auxiliaries
		ethyl acetate, to air		air	high population density			Ethyl acetate	136E-01	kg	calculated frominput of auxiliaries	1	2.08	(2.4.3.3.1523); calculated frominput of auxiliaries
		methyl ethyl ketone, to air		air	high population density			Methyl ethyl ketone	136E-01	kg	calculated frominput of auxiliaries	1	2.08	(2.4.3.3.1523); calculated frominput of auxiliaries
		hardener, to air		air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	3.40E-02	kg	calculated frominput of auxiliaries	1	2.08	(2.4.3.3.1523); calculated frominput of auxiliaries
		Production efforts for capacitors		electronics	component	No	GLO	production efforts, capacitors	100E+00	kg		1		

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Tab. 6.7 Unit process inventory of the dataset “Capacitor, film, through-hole mounting, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
production efforts	capacitor, film, through-hole mounting, at plant			electronics	component	No	GLO	production efforts, capacitors	1.00E+00	kg	average capacitor production from CPM (2006) and EPCOS (2007)	1	165	(3,4,4,14,5,11); rough estimation
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	4.65E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	3.64E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	1.17E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
lead				metals	extraction	No	RER	lead, at regional storage	6.07E-03	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	1,33E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
phosphor				chemicals	inorganics	No	RER	phosphorus, white, liquid, at plant	1,55E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
silver				metals	extraction	No	RER	silver, at regional storage	4,16E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	3,39E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
zinc				metals	extraction	No	RER	zinc, primary, at regional storage	1,24E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information
PPS				plastics	polymers	No	GLO	polyphenylene sulfide, at plant	2,60E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	126	(14,4,11,4,3); average from literature & company information

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Tab. 6.7 (Cont.) Unit process inventory of the dataset “Capacitor, SMD type, surface mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
polypropylene	capacitor, film, through-hole mounting, at plant	continued		plastics	polymers	No	RER	polypropylene, granulate, at plant	7.41E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	126	(14.4, 114.3); average from literature & company information
PET				plastics	polymers	No	RER	polyethylene terephthalate, granulate, amorphous, at plant	3.52E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	126	(14.4, 114.3); average from literature & company information
phenolic resin				paintings	production	No	RER	phenolic resin, at plant	3.55E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	126	(14.4, 114.3); average from literature & company information
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	3.45E-01	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	126	(14.4, 114.3); average from literature & company information
				electronics	component	No	GLO	capacitor, film, through-hole mounting, at plant	100E+00	kg			1	

Tab. 6.8 Unit process inventory of the dataset “Capacitor, SMD type, surface mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	SDv 95%	General Comment
production efforts	capacitor, SMD type, surface-mounting, at plant	SMD capacitor		electronics	component	No	GLO	production efforts, capacitors	1.00E+00	kg	average capacitor production from CPM (2006) and EPCOS (2007)	1	1.61	(3,4,3,1,4,5,1); rough estimation
copper				metals	extraction	No	RER	copper, primary, at refinery	1.41E-02	kg	Average according to Beherndt (1998) and Venkel (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	2.01E-01	kg	Average according to Beherndt (1998) and Venkel (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
palladium				metals	extraction	No	RER	palladium, at regional storage	1.33E-03	kg	Average according to Beherndt (1998) and Venkel (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
silver				metals	extraction	No	RER	silver, at regional storage	3.92E-02	kg	Average according to Beherndt (1998) and Venkel (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	2.44E-02	kg	Average according to Beherndt (1998) and Venkel (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
Glass				glass	construction	No	RER	flat glass, uncoated, at plant	2.22E-03	kg	Average according to Beherndt (1998) and Venkel (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
Ceramics (BaTiO3)				chemicals	inorganics	No	RER	barite, at plant	7.91E-01	kg	Average according to Beherndt (1998) and Venkel (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
				electronics	component	No	GLO	capacitor, SMD type, surface-mounting, at plant	1.00E+00	kg		1		

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Tab. 6.9 Unit process inventory of the dataset “Capacitor, electrolyte type, < 2 cm height, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
production efforts	capacitor, electrolyte type, < 2cm height, at plant			electronics	component	No	GLO	production efforts, capacitors	100E+00	kg	average capacitor production from CPM (2006) and EPCOS (2007)	1	1.61	(3,4,3,14,5,11); rough estimation
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	6.99E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	1.36E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	8.44E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
lead				metals	extraction	No	RER	lead, at regional storage	4.88E-05	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
silver				metals	extraction	No	RER	silver, at regional storage	2.52E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	2.07E-03	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
Electrolyte (ethylene glycol)				chemicals	organics	No	RER	ethylene glycol, at plant	2.74E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
paper				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	8.35E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
rubber				plastics	polymers	No	RER	synthetic rubber, at plant	6.58E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
PVC				plastics	polymers	No	RER	polyvinylchloride, at regional storage	6.96E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information

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Tab. 6.9 (Cont.) Unit process inventory of the dataset “Capacitor, electrolyte type, < 2 cm height, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	SDv 95%	General Comment
PET	capacitor, electrolyte type, < 2cm height, at plant	continued		plastics	polymers	No	RER	polyethylene terephthalate, granulate, amorphous, at plant	5.48E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2,114.3); average from literature & company information
Nylon 6				plastics	polymers	No	RER	nylon 6, at plant	3.70E-04	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2,114.3); average from literature & company information
PPS				plastics	polymers	No	GLO	polyphenylene sulfide, at plant	2.38E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2,114.3); average from literature & company information
Epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	8.58E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2,114.3); average from literature & company information
				electronics	component	No	GLO	capacitor, electrolyte type, < 2cm height, at plant	1.00E+00	kg			1	

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Tab. 6.10 Unit process inventory of the dataset “Capacitor, electrolyte type, > 2 cm height, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	capacitor, electrolyte type, > 2cm height, at plant			electronics	component	No	GLO	production efforts, capacitors	100E+00	kg	average capacitor production from CPM (2006) and EPCOS (2007)	1	1.61	(3,4,3,14,5,1); rough estimation
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	7.53E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	1.36E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	2.96E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	1.08E-03	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
Electrolyte (ethylene glycol)				chemicals	organics	No	RER	ethylene glycol, at plant	3.70E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
paper				paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	1.32E-01	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
rubber				plastics	polymers	No	RER	synthetic rubber, at plant	3.26E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
PVC				plastics	polymers	No	RER	polyvinylchloride, at regional storage	7.33E-02	kg	Average according to Beherndt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
continued ▶														

continued ▼

Tab. 6.10 (Cont.) Unit process inventory of the dataset “Capacitor, electrolyte type, > 2 cm height, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
PET	capacitor, electrolyte type, > 2cm height, at plant	continued		plastics	polymers	No	RER	polyethylene terephthalate, granulate, amorphous, at plant	5.85E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2,114.3); average from literature & company information
Nylon6				plastics	polymers	No	RER	nylon 6, at plant	9.62E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2,114.3); average from literature & company information
Phenolic resin				paintings	production	No	RER	phenolic resin, at plant	9.62E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2,114.3); average from literature & company information
				electronics	component	No	GLO	capacitor, electrolyte type, > 2cm height, at plant	100E+00	kg		1		

6. Capacitors

Tab. 6.11 Unit process inventory of the dataset “Capacitor, Tantalum-, through-hole mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information						
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment		
production efforts	capacitor, Tantalum-, through-hole mounting, at plant	Tantalum capacitor		electronics	component	No	GLO	production efforts, capacitors	100E+00	kg	average capacitor production from CPM (2006) and EPCOS (2007)	1	1.61	(3,4,3,14,5,1); rough estimation		
iron			metals	extraction	No	GLO	pig iron, at plant	8.67E-02	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information			
nickel			metals	extraction	No	GLO	nickel, 99.5% at plant	6.28E-02	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information			
silver			metals	extraction	No	RER	silver, at regional storage	3.85E-02	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information			
tin			metals	extraction	No	RER	tin, at regional storage	3.40E-02	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information			
Tantalum			metals	extraction	No	GLO	tantalum, powder, capacitor-grade, at regional storage	5.43E-01	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information			
Manganese Dioxide			metals	extraction	63% of amount as manganese (stoichiometric amount) - remaining part omitted	metals	extraction	No	RER	manganese, at regional storage	5.31E-02	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information
Sb2O3			chemicals	"chemical inorganics" used as proxy ...	chemicals	inorganics	No	GLO	chemicals inorganic, at plant	1.63E-02	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information	
Teflon			chemicals		chemicals	inorganics	No	RER	tetrafluoroethylene, at plant	7.40E-03	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information	
Epoxy resin			plastics		plastics	monomers	No	RER	epoxy resin, liquid, at plant	6.07E-01	kg	Typical value according to ZVEI (2006)	1	1.16	(14,2,114,3); company information	

Tab. 6.12 Unit process inventory of the dataset “Capacitor, unspecified, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
SMD type capacitors	capacitor, unspecified, at plant	↑	↑	electronics	component	No	GLO	capacitor, SMD type, surface-mounting, at plant	2.00E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
film type capacitors				electronics	component	No	GLO	capacitor, film, through-hole mounting, at plant	2.00E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
small electrolyt capacitors				electronics	component	No	GLO	capacitor, electrolyte type, < 2cm height, at plant	2.00E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
big electrolyt capacitors				electronics	component	No	GLO	capacitor, electrolyte type, > 2cm height, at plant	2.00E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
Tantalum capacitors				electronics	component	No	GLO	capacitor, Tantalum-, through-hole mounting, at plant	2.00E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
				electronics	component	No	GLO	capacitor, unspecified, at plant	1.00E+00	kg			1	

↑ unspecified capacitor

7 Inductors

An **inductor** is a passive electrical device employed in electrical circuits for its property of inductance. An inductor can take many forms. Synonyms for the term “inductor” that are in use are “coil” and “choke”.⁴

Most inductors are constructed as a coil out of conducting material (e.g. copper wire) that is wrapped around a core of air or of ferromagnetic material. The higher the permeability of the core material, the higher is the inductance of the component. One part of the inductors is constructed as enamel coated wire wrapped around a ferrite bobbin with wire exposed on the outside. Some inductors have an adjustable core, which enables changing of the inductance. Those inductors that are used to block very high frequencies can contain a wire passing through a ferrite cylinder or bead.

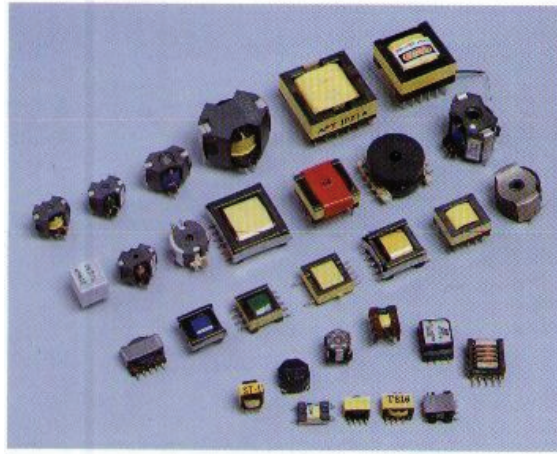


Fig. 7.1 Some forms of Inductors (Picture taken from www.isomatic.co.uk).

While a capacitor opposes changes in voltage, an inductor opposes changes in current. An ideal inductor would offer no resistance to a constant direct current; however, only superconducting inductors have truly zero electrical resistance.

7.1 Characterisation, Use & Market

Inductors are used extensively in analog circuits and signal processing. Inductors in conjunction with capacitors and other components form tuned circuits which can emphasize or filter out specific signal frequencies. This can range from the use of large inductors as chokes in power supplies, now obsolete, which in conjunction with filter capacitors remove residual hum or other fluctuations from the direct current output, to such small inductances as generated by a ferrite bead or torus around a cable to prevent radio frequency interference from being transmitted down the wire. Smaller inductor/capacitor combinations provide tuned circuits used in radio reception and broadcasting, for instance.

Currently, there is a huge variety of different types of inductors on the market – a variety that can not be covered 100% by the here described ecoinvent datasets. Instead a choice has been made according to the data sources available – resulting in three different types of inductors.

⁴ Wikipedia: Inductor (<http://en.wikipedia.org/wiki/Inductor>), accessed March 14, 2007.

7.2 System Characterisation

While about the composition of the here chosen types of inductors, a variety of data sources can be found, it is very hard to get any information concerning the actual production efforts for the production of an inductor. Hence, the composition is much more accurate and in details shown in this study than the production itself. Thus – for the production efforts of inductors, no distinction between the different here examined types of inductors is made – instead, one average dataset representing the “production efforts” is established; and all types of inductors will then have this dataset as one of their input values. All in all this ends up with the following scheme for the representation of the electronic components “Inductors” in the framework of the ecoinvent database.

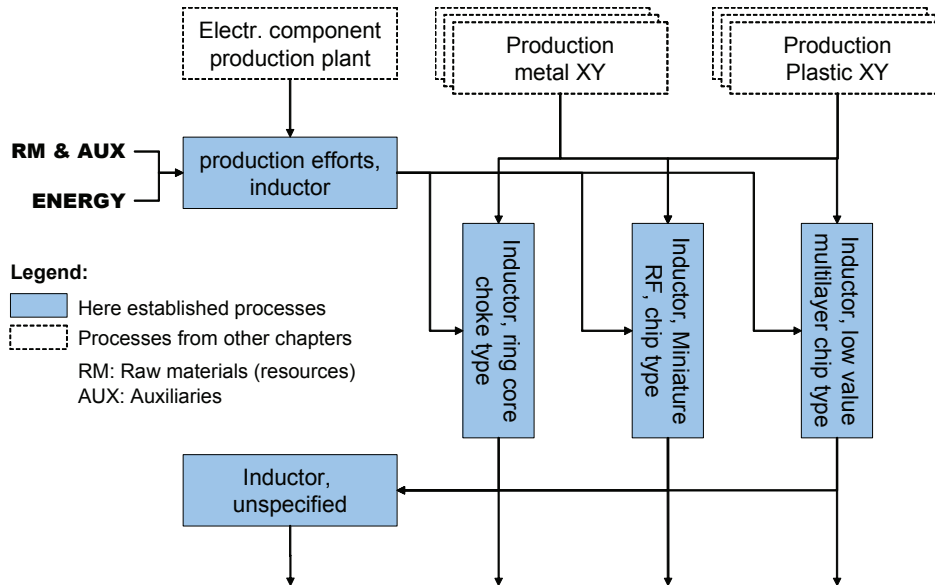


Fig. 7.2 Scheme of the representation of the electronic components „Inductors“ in the ecoinvent database

All these datasets are based on information representing current models of inductors. The used data sources are mainly Behrendt et al. (1998), CPM (2006), EPCOS (2006) and Venkel (2006), representing material specification sheets from industry as well as further data sources.

They represent mainly current models of inductors. The companies behind this information are producing on a global scale – hence, the here developed data can be considered as “global” valid data.

The respective meta information for the various datasets of inductors can be found in Tab. 7.1.

Tab. 7.1 Ecoinvent meta information for the processes for inductors within the database.

Name	production efforts, inductor	inductor, unspecified, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This module includes auxiliary and energy consumption, emissions (air, water) and waste amounts, as well as an estimation of the efforts for transportation and of the infrastructure for the production of inductors.	This dataset covers raw material input and production efforts for the production of current inductors.
Amount	1	1
Local Name	Induktoren, Herstellaufwände	Induktor, unspezifisch, ab Werk
Synonyms	-	-
General Comment	Data are a rough estimation of the average production efforts over all types of inductors. Data for auxiliaries, energy, emissions and waste are out of an international LCI database plus estimations based on the production efforts of capacitors. Infrastructure and transportation efforts are based on own assumptions.	This module represents an unweighted average of all specific inductor datasets represented in the ecoinvent database.
Start Date	1998	1994
End Date	2005	2007
Data valid for entire period	1	1
Geography Text	Data from literature and estimation based on capacitor production efforts – here representing the global situation for inductor production.	Data represent global valuable composition data for inductors.
Technology Text	average production comprising the following steps: core mounting; epoxy-curing; winding; welding; molding; marking; bending; testing and final control.	average production comprising the following steps: core mounting; epoxy-curing; winding; welding; molding; marking; bending; testing and final control.
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature & own estimations	own estimation
Extrapolations	see Geography and General Comments	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 7.1 (Cont.) Ecoinvent meta information for the processes for inductors within the database.

Name	inductor, ring core choke type, at plant	inductor, Miniature RF chip type, MRFI, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used ring core choke type inductors.	This dataset covers raw material input and production efforts for the production of currently used miniature RF chip type inductors.
Amount	1	1
Local Name	Induktor, Ringkerndrossel-Typ, ab Werk	Induktor, Miniatur-RF-Chip-Typ, ab Werk
Synonyms	coil	coil
General Comment	The data represent a typical ring core choke type inductor used in the information and communication technology. Material data are taken from datasheets of producers of such inductors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of inductors with an average weight of 6.0 gram/unit.	The data represent a typical miniature RF chip type inductor, used in the information and communication technology. Material data are taken from datasheets of producers of such inductors. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of inductors with an average weight of 16.8 miligram/unit.
Start Date	2005	2005
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of ring core choke type inductors.	Data represent a global valuable composition of miniature RF chip type inductors.
Technology Text	average production comprising the following steps: core mounting; epoxoy-curing; winding; welding; molding; marking; bending; testing and final control.	average production comprising the following steps: core mounting; epoxoy-curing; winding; welding; molding; marking; bending; testing and final control.
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature	Literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 7.1 (Cont.) Ecoinvent meta information for the processes for inductors within the database.

Name	inductor, low value multilayer chip type, LMCI, at plant
Location	GLO
Infrastructure Process	0
Unit	kg
Dataset Version	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used low value multilayer chip type inductors.
Amount	1
Local Name	Induktor, Tiefwertiger Mehrschichtenchip-Typ, ab Werk
Synonyms	coil
General Comment	The data represent a typical low value multilayer chip type inductor, used in the information and communication technology. Material data are taken from data-sheets of producers of such inductors. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of inductors with an average weight of 2.1 miligram/unit.
Start Date	2005
End Date	2007
Data valid for entire period	1
Geography Text	Data represent a global valuable composition of low value multilayer chip type inductors.
Technology Text	average production comprising the following steps: core mounting; epoxy-curing; winding; welding; molding; marking; bending; testing and final control.
Representativeness	see General Comments
Production Volume	-
Sampling Procedure	Literature
Extrapolations	see Geography & Technology
Uncertainty Adjustments	None

7.3 Dataset “production efforts, inductor”

This dataset describes the production efforts for the production of the various types of inductors or coils examined within ecoinvent.

7.3.1 System Characterization

The dataset established here comprises the input of auxiliaries and energy as well as the output of emissions and waste due to these auxiliaries in the framework of the inductor production. The input of the used infrastructure, as well as the transport efforts for the auxiliaries and the raw materials of the various inductors are included as well. Main data source for this information is the information about

the inductor production in CPM (2006). Furthermore, estimations, based on the capacitor production (reported in chapter 6.3 of this report here) are used in case of lack of information.

7.3.2 Life Cycle inventory

As the raw material input is taken into account in the respective inductor datasets – hence this dataset has no input of raw materials. Instead, this dataset here contains all other input and output flows – i.e. auxiliaries, energy, transport amount (for auxiliaries and raw materials!), infrastructure as well as the emissions and waste amounts produced. In details these various aspects are taken into account based on the following information:

- **Auxiliaries:** From CPM (2006) no information about the use of auxiliaries can be taken; hence, the amount is estimated based on the average input of about 0.99 kg of auxiliaries per kg of produced capacitors. Like for capacitors the overall amount is shown as:

Acetone	10%	-
Ethyl Acetate	40%	-
Methyl ethyl ketone	40%	-
Hardener	10%	as “chemicals organic, at plant”

- **Energy:** In CPM (2006) an average input of electricity of 48 kWh per kg of produced inductors is reported and used here. Furthermore, from the emissions of CO₂, an input of fuels can be calculated. Due to a lack of information, a natural gas heating system is assumed – resulting in an input of 0.0875 MJ of heat from natural gas.
- **Emissions to air:** In CPM (2006) a total amount of 5.36E-03 kg of emissions to air is reported. Thereof, more than 90% is reported as carbon dioxide. This amount, together with the amount of SO₂ reported, is already included into the dataset used for the fuel input (see above) and hence, they are not further taken into account here. For the actual process emissions, due to a lack of more detailed information in the data about inductors, the average output of 0.34 kg of air emissions per kg of produced capacitors is used as a proxy – similar like for the auxiliary input. Hence, this output amount is split up in the following way as emissions to air:

Acetone, to air	10%	-
Ethyl Acetate, to air	40%	-
Methyl ethyl ketone, to air	40%	-
Hardener, to air	10%	as “NMVOC, ”

Furthermore, it was assumed, that 100% of the electricity consumed, i.e. 48 kWh per kg inductor produced, is converted to waste heat. In accordance with the general rules for ecoinvent, it was assumed that 100% of the waste heat is released to the air.

- **Emissions to water & Water input:** Due to a lack of data, the information from EPCOS (2007) for capacitors, representing an average cooling water amount of 0.13 m³ resp. a process water amount of 0.15 m³ per kg, is used here as well. Due to a lack of respective emission information, the outflow of the latter amount is approximated by the use of the dataset “treatment, sewage, to waste water treatment, class 2”.
- **Waste:** No information is given in CPM (2006) concerning waste of the inductor production. Assuming that the auxiliaries are emitted either as emissions (air, water, soil) or as waste, it can be concluded that those parts that are not emitted to air, are obviously here assumed to be disposed of as waste (using the dataset “disposal, solvent mixture, to HWT”). Due to a lack of inductor-specific information, the amount of 0.48 kg non-used raw materials from the capacitor dataset is used here

as well. It is assumed as 50% plastic wastes to sanitary landfilling and 50% anorganic waste to landfilling (using the dataset “disposal, waste, Si waferprod., to residual material landfill”).

- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here. The amount takes into account the consumed auxiliaries as well as the raw materials for the production of any type of capacitor. The amount of raw materials used equals the material in the inductor (i.e. 1 kg) as well as the waste out of raw materials (i.e. 0.48 kg).
- **Infrastructure and land-use:** Due to a lack of more specific information, the infrastructure module established in chapter 2, representing an electronic component production plant, is used also for the production of inductors.

7.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is only a rough estimation of the actual inductor production efforts, based on information from one dataset out of the Swedish LCI database (CPM (2006)) and on estimations from the capacitor production. Hence, the respective uncertainty values are accordingly high.

The resulting unit process “production efforts, inductor” is shown in Tab. 7.5 (on page 133).

7.4 Dataset “inductor, ring core choke type, at plant”

This dataset here describes the production efforts for the production of ring core type chokes or inductors. The dataset here has a functional unit of “1 kg of inductors”, based on the assumption that a typical ring core choke / inductor in electronic modules has an average weight of **6.0 g**.

7.4.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 6.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Behrendt et al. (1998; EPCOS (2006).

7.4.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, inductor”, the production of 1 kg of inductors lead to a waste amount of 1.13 kg – whereof about 0.48 kg are unused raw materials. Hence, this dataset here has an input of 1.48 kg of raw materials per kg of inductors. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998) and EPCOS (2006) and is summarized in Tab. 7.2. In this table the respective ecoinvent datasets used are also shown.

Tab. 7.2 Material specifications of ring core chokes resp. inductors (data sources: see text)

		ring core	Used ecoinvent dataset
		inductor	
Weight	g	6.000	
[i] Metals			
Iron	%	0.69	pig iron, at plant (GLO)
Copper	%	6.29	copper, primary, at plant (RER)
Lead	%	2.55	lead, at regional storage (RER)
Tin	%	2.55	tin, at regional storage (RER)
[ii] Others			
plastics, unspecified	%	58.00	as 100% "polycarbonate, at plant (RER)"
Glass (Pb containing)	%	26.93	funnel glass, CRT screen, at plant (RER)
Ceramics (Zr, Ti)	%	1.29	as 100% "zircon, 50% zirconium, at plant (RER)"
Epoxy resin	%	1.69	epoxy resin, liquid, at plant (RER)

7.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical ring core choke or inductor, based on composition information from literature and a global producer. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific inductor type and have thus, accordingly higher uncertainties.

The resulting unit process “inductor, ring core choke type, at plant” is shown in Tab. 7.6 (on page 135).

7.5 Dataset “inductor, Miniature RF chip type, MRFI, at plant”

This dataset here describes the material composition of the production of miniature RF chip type inductors. The dataset here has a functional unit of “1 kg of inductors”, based on the assumption that a typical MRFI inductor in electronic modules has an average weight of **16.8 milligram**.

7.5.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 6.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data source for the input of raw materials in this dataset here is Venkel (2006), describing several different types of this kind of inductor.

7.5.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, inductor”, the production of 1 kg of inductors lead to a waste amount of 1.13 kg – whereof about 0.48 kg are unused raw materials. Hence, this dataset here has an input of 1.48 kg of raw materials per kg of inductors. The split to the different raw materials (metals, plastics) is based on

the material specifications reported in Venkel (2006) and is summarized in Tab. 7.3. In this table the respective ecoinvent datasets used are also shown.

Tab. 7.3 Material specifications of miniature RF chip inductors, MRFI (data sources: see text)

		MRFI	Used ecoinvent dataset
		inductor	
Weight	mg	16.8	
[i] Metals			
Copper	%	10.88	copper, primary, at plant (RER)
Nickel	%	2.22	nickel, 99.5%, at plant (GLO)
Gold	%	0.08	gold, at regional storage (RER)
[ii] Others			
aluminium oxide	%	67.28	aluminium oxide, at plant (RER)
silicon oxide	%	5.35	silica sand, at plant (RER)
calcium oxide	%	1.53	quicklime milled, packed, at plant (RER)
Epoxy resin	%	12.66	epoxy resin, liquid, at plant (RER)

7.5.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical MRFI inductor, based on composition information from a global producer. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific inductor type and have thus, accordingly higher uncertainties.

The resulting unit process “inductor, Miniature RF chip type, MRFI, at plant” is shown in Tab. 7.7 (on page 136).

7.6 Dataset “inductor, low value multilayer chip type, LMCI, at plant”

This dataset here describes the material composition of the production of low value multilayer chip type inductors. The dataset here has a functional unit of “1 kg of inductors”, based on the assumption that a typical LMCI inductor in electronic modules has an average weight of **2.1 miligram**.

7.6.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 6.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data source for the input of raw materials in this dataset here is Venkel (2006), describing several different types of this kind of inductor.

7.6.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, inductor”, the production of 1 kg of inductors lead to a waste amount of 1.13 kg –

whereof about 0.48 kg are unused raw materials. Hence, this dataset here has an input of 1.48 kg of raw materials per kg of inductors. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Venkel (2006) and is summarized in Tab. 7.4. In this table the respective ecoinvent datasets used are also shown.

Tab. 7.4 Material specifications of low value multilayer chip inductors, LMCI (data sources: see text)

		LMCI	Used ecoinvent dataset
		inductor	
Weight	mg	2.1	
[i] Metals			
Silver	%	30.22	silver, at regional storage (RER)
Nickel	%	1.66	nickel, 99.5%, at plant (GLO)
Tin	%	3.89	tin, at regional storage (RER)
[ii] Others			
aluminium oxide	%	27.89	aluminium oxide, at plant (RER)
glass (B-containing)	%	34.09	glass tube, borosilicate, at plant (DE)
glass (Pb containing)	%	2.11	funnel glass, CRT screen, at plant (RER)
cupric oxide (CuO)	%	0.14	copper oxide, at plant (RER)

7.6.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical LMCI inductor, based on composition information from a global producer. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific inductor type and have thus, accordingly higher uncertainties.

The resulting unit process “inductor, low value multilayer chip type, LMCI, at plant” is shown in Tab. 7.8 (on page 137).

7.7 Dataset “inductor, unspecified, at plant”

Last but not least, for all those cases where no information is given about the inductor type used, an unspecified dataset is established as well. This latter one is an unweighted mean value of all three above described inductor types. The dataset has as functional unit also “1 kg of inductors”.

7.7.1 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an unweighted average of the various inductor types shown in this database. Hence, the uncertainty values are accordingly high.

The resulting unit process “inductor, unspecified, at plant” is shown in Tab. 7.9 (on page 138).

7. Inductors

Tab. 7.5 Unit process inventory of the dataset “Production efforts, inductor (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
Water (process)	production efforts, inductor			resource	in water			Water, unspecified natural origin	150E-01	m3	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.5,12); average from literature & CER
Water (cooling)				resource	in water			Water, cooling, unspecified natural origin	130E-01	m3	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.5,12); assumption from conclusion by analogy (using capacitors)
acetone				chemicals	organics	No	RER	acetone, liquid, at plant	9.90E-02	kg	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.5,3); assumption from conclusion by analogy (using capacitors)
ethyl acetate				chemicals	organics	No	RER	ethyl acetate from butane, at plant	3.96E-01	kg	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.5,3); assumption from conclusion by analogy (using capacitors)
methyl ethyl ketone				chemicals	organics	No	RER	methyl ethyl ketone from butane, at plant	3.96E-01	kg	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.5,3); assumption from conclusion by analogy (using capacitors)
hardener			dataset "chemical organic" as proxy used	chemicals	organics	No	GLO	chemicals organic, at plant	9.90E-02	kg	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.5,3); assumption from conclusion by analogy (using capacitors)
electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	4.80E+01	kWh	data reported in a dataset in CPM (2006)	1	1.27	(2.4.3,3.15,2); value from literature
natural gas (heat)				natural gas	heating systems	No	RER	heat, natural gas, at industrial furnace >100kW	8.75E-02	MJ	calculated from information in a dataset in CPM (2006)	1	1.27	(2.4.3,3.15,1); calculated from value in literature
Transport (part by lorry)				transport systems	road	No	RER	transport, lorry >16t, fleet average	2.47E-01	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,na,na,na,5); standard distances
Transport (part by train)				transport systems	train	No	RER	transport, freight, rail	1.48E+00	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,na,na,na,5); standard distances
Infrastructure (inductor plant)			electronics	component	Yes	GLO	electronic component production plant	2.00E-08	unit	rough estimation	1	3.77	(4.5,13.5,4.9); rough estimation of infrastructure	

continued ▼

7. Inductors

Tab. 7.5 (Cont.) Unit process inventory of the dataset “Production efforts, inductor (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
▼ continued	production efforts, inductor	↑ auxiliaries to waste	due to a lack of data, the dataset "disposal, solvent mixture, to HW" is used as a proxy	waste management	hazardous waste incineration	No	CH	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	6.50E-01	kg	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.56); average from literature & CER
		↑ raw materials to waste	due to a lack of information, 50% are assumed as plastics waste to sanitary landfill, ...	waste management	sanitary landfill	No	CH	disposal, plastics, mixture, 15.3% water, to sanitary landfill	2.40E-01	kg	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.56); assumption, based on literature & CER
		↑ raw materials to waste	...and the remaining 50% are assumed to be anorganic waste to landfill (proxy of Si wafer waste used)	waste management	residual material landfill	No	CH	disposal, waste, Si waferprod., inorg, 9.4 % water, to residual material landfill	2.40E-01	kg	approximation, based on capacitor production efforts	1	1.61	(3.4.3,14.56); assumption, based on literature & CER
		↑ waste water to treatment	due to a lack of knowledge of content, dataset "treatment, sewage, WWTP class 2" is used as proxy	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 2	150E-01	m3	calculated as 100% of process water input ...	1	1.61	(3.4.3,14.56); average from literature & CER
		↑ waste heat	100% of electricity input (according to quality guidelines of ecoinvent)	air	high population density			Heat, waste	173E+02	MJ	calculated from electricity input	1	2.08	(2.4.3,15.23); calculated from electricity input
		↑ acetone, to air		air	high population density			Acetone	3.40E-02	kg	calculated from input of auxiliaries	1	2.32	(3.4.3,14.523); calculated from input of auxiliaries
		↑ ethyl acetate, to air		air	high population density			Ethyl acetate	136E-01	kg	calculated from input of auxiliaries	1	2.32	(3.4.3,14.523); calculated from input of auxiliaries
		↑ methyl ethyl ketone, to air		air	high population density			Methyl ethyl ketone	136E-01	kg	calculated from input of auxiliaries	1	2.32	(3.4.3,14.523); calculated from input of auxiliaries
		↑ hardener, to air		air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	3.40E-02	kg	calculated from input of auxiliaries	1	2.32	(3.4.3,14.523); calculated from input of auxiliaries
		↑ Production efforts for capacitors		electronics	component	No	GLO	production efforts, inductor	100E+00	kg		1		

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Tab. 7.6 Unit process inventory of the dataset “Inductor, ring core choke type, at plant (GLO)”

General Flow information					Representation in ecoinvent						Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	inductor, ring core choke type, at plant	ring core choke inductor		electronics	component	No	GLO	production efforts, inductor	100E+00	kg	average inductor production from CPM (2006) & own assumptions	1	165	(3,4,4,114,5,11); rough estimation
iron				metals	extraction	No	GLO	pig iron, at plant	102E-02	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	9.31E-02	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
lead				metals	extraction	No	RER	lead, at regional storage	3.77E-02	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	3.77E-02	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
Plastics			assumption 100% polycarbonate	plastics	polymers	No	RER	polycarbonate, at plant	8.88E-01	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
Glass, Pb containing			dataset "funel glass, at plant" as proxy used	electronics	component	No	GLO	funel glass, CRT screen, at plant	3.99E-01	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
Ceramics (Zr, Ti)			dataset "zircon, 50% zirkonium, at plant" as proxy used	chemicals	inorganics	No	A U	zircon, 50%zirkonium, at plant	1.91E-02	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	2.50E-02	kg	Data according to Beherendt (1998) and EPCOS (2007)	1	126	(14,4,114,3); average from literature & company information
				electronics	component	No	GLO	inductor, ring core choke type, at plant	100E+00	kg		1		

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Tab. 7.7 Unit process inventory of the dataset “inductor, miniature RF chip type, MRFI, at plant, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	inductor, miniature RF chip type, MRFI, at plant	MRFI inductor		electronics	component	No	GLO	production efforts, inductor	100E+00	kg	average inductor production from CPM (2006) & own assumptions	1	165	(3,4,4,1,4,5,1); rough estimation
copper				metals	extraction	No	RER	copper, primary, at refinery	161E-01	kg	Data according to Venkel (2006)	1	126	(1,4,4,1,1,4,3); average from company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	3.29E-02	kg	Data according to Venkel (2006)	1	126	(1,4,4,1,1,4,3); average from company information
gold				metals	extraction	No	RER	gold, at regional storage	1.8E-03	kg	Data according to Venkel (2006)	1	126	(1,4,4,1,1,4,3); average from company information
aluminium oxide				chemicals	inorganics	No	RER	aluminium oxide, at plant	9.96E-01	kg	Data according to Venkel (2006)	1	126	(1,4,4,1,1,4,3); average from company information
silicon oxide				construction materials	additives	No	DE	silica sand, at plant	7.92E-02	kg	Data according to Venkel (2006)	1	126	(1,4,4,1,1,4,3); average from company information
calcium oxide				construction materials	additives	No	CH	quicklime, milled, packed, at plant	2.26E-02	kg	Data according to Venkel (2006)	1	126	(1,4,4,1,1,4,3); average from company information
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	187E-01	kg	Data according to Venkel (2006)	1	126	(1,4,4,1,1,4,3); average from company information
				electronics	component	No	GLO	inductor, miniature RF chip type, MRFI, at plant	100E+00	kg		1		

7. Inductors

Tab. 7.8 Unit process inventory of the dataset “inductor, low value multilayer chip type, LMCI, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
production efforts	inductor, low value multilayer chip type, LMCI, at plant	LMCI inductor		electronics	component	No	GLO	production efforts, inductor	100E+00	kg	average inductor production from CPM (2006) & own assumptions	1	165	(3,4,4,1,4,5,1); rough estimation
silver			metals	extraction	No	RER	silver, at regional storage	4.47E-01	kg	Data according to Venkel (2006)	1	126	(14,4,1,14,3); average from company information	
nickel			metals	extraction	No	GLO	nickel 99.5% at plant	2.46E-02	kg	Data according to Venkel (2006)	1	126	(14,4,1,14,3); average from company information	
tin			metals	extraction	No	RER	tin, at regional storage	5.76E-02	kg	Data according to Venkel (2006)	1	126	(14,4,1,14,3); average from company information	
aluminium oxide			chemicals	inorganics	No	RER	aluminium oxide, at plant	4.13E-01	kg	Data according to Venkel (2006)	1	126	(14,4,1,14,3); average from company information	
glass (B containing)			glass	construction	No	DE	glass tube, borosilicate, at plant	5.05E-01	kg	Data according to Venkel (2006)	1	126	(14,4,1,14,3); average from company information	
glass (Pb containing)			electronics	component	No	GLO	funnel glass, CRT screen, at plant	3.12E-02	kg	Data according to Venkel (2006)	1	126	(14,4,1,14,3); average from company information	
cupric oxide			chemicals	inorganics	No	RER	copper oxide, at plant	2.07E-03	kg	Data according to Venkel (2006)	1	126	(14,4,1,14,3); average from company information	
				electronics	component	No	GLO	inductor, low value multilayer chip type, LMCI, at plant	100E+00	kg		1		

Tab. 7.9 Unit process inventory of the dataset “Inductor, unspecified, at plant (GLO)”

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
Ring core choke inductor	Inductor, unspecified, at plant	unspecified inductor		electronics	component	No	GLO	Inductor, ring core choke type, at plant	3.33E-01	kg	assumption	1	189	(5.4.4, 1.4.5.3); assumption, based on specific data
miniature RF chip inductor				electronics	component	No	GLO	Inductor, miniature RF chip type, MRFI, at plant	3.33E-01	kg	assumption	1	189	(5.4.4, 1.4.5.3); assumption, based on specific data
low value multilayer chip inductor				electronics	component	No	GLO	Inductor, low value multilayer chip type, LMCI, at plant	3.33E-01	kg	assumption	1	189	(5.4.4, 1.4.5.3); assumption, based on specific data
				electronics	component	No	GLO	Inductor, unspecified, at plant	100E+00	kg		1		

8 Resistors & Potentiometer

A **resistor** is a two-terminal electrical or electronic component that resists an electric current by producing a voltage drop between its terminals in accordance with Ohm's law – the electrical resistance is equal to the voltage drop across the resistor divided by the current through the resistor. In general, a resistor is used to create a known voltage-to-current ratio in an electric circuit.⁵

Hence, by knowing the current in a circuit, a resistor can be used to create a known potential difference that is proportional to this current. This means then also that if the potential difference between two points in a circuit is known, a resistor can be used to create a known current proportional to that difference. Resistors can also be used to limit the current that passes through other components – e.g. through light-emitting diodes (LED). Further usages of resistors cover e.g. the regulation of CD motors, the voltage reduction of signals (-> attenuator) or the minimization of signal reflections (-> line terminator).

The **potentiometer** then is “a variable resistor that can be used as a voltage divider”.⁶ Hence, it describes an electrical device which has a user-adjustable resistance. Usually, this is a three-terminal resistor with a sliding contact in the center. If all three terminals are used, it can act as a variable voltage divider. If only two terminals are used (one side and the wiper), it acts as a variable resistor.

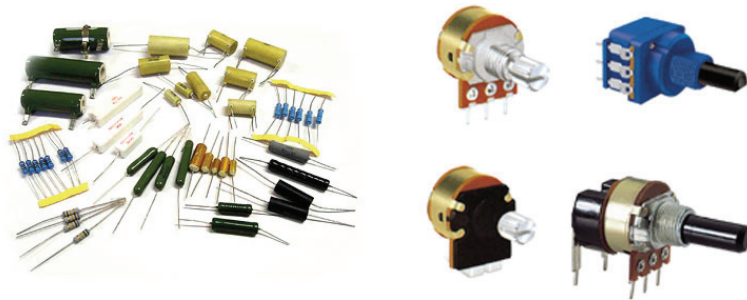


Fig. 8.1 Different types of resistors (left – picture taken from Germes Online (2007)) and typical potentiometers (right – picture taken from Alibaba.com (2007)).

8.1 Characterisation, Use & Market

According to Hobbs (2007), resistors “are the most common and well known of all electrical components”. Their uses are many, they are used to drop voltage, limit current, attenuate signals, act as heaters, act as fuses, furnish electrical loads and divide voltages.

Thereby resistors have a variety of characteristics which determine their accuracy when used. Each of these points influences the accuracy to a greater or lesser extent depending on the application. Among these characteristics are “Tolerance at DC”, “Temperature Coefficient of Resistance (TCR)”, “Frequency Response”, “Voltage Coefficient”, “Noise”, “Stability with Time and Load”, “Temperature Rating”, “Power Rating”, “Physical Size”, “Mounting Characteristics”, “Thermocouple Effect”, and “Reliability”.

There is a huge variety of different types of resistors on the market. There are, according to Hobbs (2007), for example Precision Wirewound, NIST Standards, Power Wirewound, Fuse Resistors, Carbon Composition, Carbon Film, Metal Film, Foil, Filament Wound, and Power Film Resistors – and each single of these resistors has a useful purpose. It is obvious that such a variety can not be covered

⁵ Wikipedia: resistor (<http://en.wikipedia.org/wiki/resistor>), accessed March 14, 2007.

⁶ Wikipedia: potentiometer (<http://en.wikipedia.org/wiki/potentiometer>), accessed March 14, 2007.

100% by the here described ecoinvent datasets. Main difference is the way how the film of the resistive material is applied to the surface. While e.g. in thick film resistors the film is applied by the use of traditional screen-printing technologies – thin film resistors are made by sputtering the resistive material onto the surface of the resistor.⁷ Due to the actual availability of information, the following types of resistors are covered in this database:

- **Metal-film type resistor:** They represent according to Hobbs (2007) the best compromise of all resistors. They are not as accurate and have a higher temperature coefficient of resistance and are not as stable as Precision Wirewounds. But they are more accurate, do not have a voltage coefficient, and have a lower temperature coefficient than Carbon Film. This type of resistor has a very low noise level when properly manufactured.
- **SMD type resistor:** This type of resistors is generally associated to chip or resistors that are mounted by solder reflow techniques. This consists of a resistive element of a flat ceramic substrate (or a cylindrical ceramic core) with a solder pad on each end (Hobbs (2007)).
- **Wirewound type resistor:** According to Hobbs (2007) this type “is a highly accurate resistor with a very low TCR and can be accurate within 0.005%. A temperature coefficient of resistance (TCR) of as little as 3 part per million per degree Celsius (3ppm/°C) can be achieved. The frequency response of this type is not good. Their power handling capability is very small. However these components are too expensive for general use and are normally used only in highly accurate DC measuring equipment, and as reference resistors for voltage regulators and decoding networks.
- **Potentiometer:** As an example of a variable resistor, an average/typical trim potentiometer, like used in the information and communication technology (e.g. television production), is represented here as the fourth type of resistors.

8.2 System Characterisation

Similar like for capacitors, a variety of data sources can be found about the composition of various types of resistors (including potentiometer as a specific type of resistor), but it is very hard to get any information concerning the actual production efforts for the production of such an electronic component. Hence, also for resistors, the composition is much more accurate and in details shown in this study than the production itself. Thus – for the production efforts of resistors, no distinction between the different here examined types of resistors is made – instead, one average dataset representing the “production efforts” is established; and all types of resistors will then have this dataset as one of their input values. All in all this ends up with the following scheme for the representation of the electronic components “resistors”, including “potentiometers” in the framework of the ecoinvent database.

All these datasets are based on information representing current models of resistors. The used data sources are mainly Alibaba.com (2007), Behrendt et al. (1998), CPM (2006), exact (1998), Germes Online (2007), US DoE (2007), and ZVEI (2006), representing material specification sheets from industry as well as further data sources.

⁷ Wikipedia: resistor (<http://en.wikipedia.org/wiki/resistor>), accessed March 14, 2007.

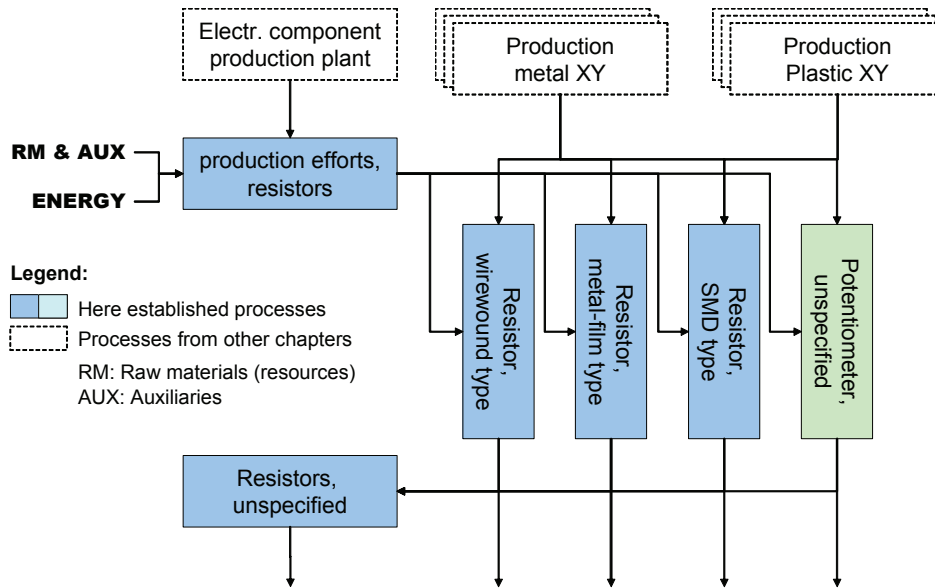


Fig. 8.2 Scheme of the representation of electronic components „resistor” and “potentiometer” in the ecoinvent database

They represent resistors and potentiometers that have been in use in the last about 10 years – as well as current models. The companies behind this information are producing on a global scale – hence, the here developed data can be considered as “global” valid data.

The respective meta information for the various datasets of capacitors can be found in Tab. 8.1.

Tab. 8.1 Ecoinvent meta information for the processes for resistors within the database.

Name	production efforts, resistors	resistor, unspecified, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This module includes auxiliary and energy consumption, emissions (air, water) and waste amounts, as well as an estimation of the efforts for transportation and of the infrastructure for the production of resistors.	This dataset covers raw material input and production efforts for the production of current resistors.
Amount	1	1
Local Name	Widerstände, Herstelleraufwände	Widerstand, unspezifisch, ab Werk
Synonyms	-	-
General Comment	Data are a rough estimation of the average production efforts over all types of resistors. Data for auxiliaries, energy, emissions and waste are taken from several datasets about resistors and potentiometers out of an international LCI database. Infrastructure and transportation efforts are based on own assumptions.	This module represents an unweighted average of all specific resistor datasets represented in the ecoinvent database.
Start Date	1998	1994
End Date	2005	2005
Data valid for entire period	1	1
Geography Text	Data from literature, used to represent the global situation.	Data represent global valuable composition data for resistors.
Technology Text	Average technology for the production of resistors and potentiometers - comprising termination printing; resistive body printing; glass coat printing; laser trimming; substrate stripping (SMT) resp. ceramic rods; deposition processes; ageing; capping (THT) resp. printing; curing; punching; revetting; molding (potentiometer.)	Average technology for the production of resistors and potentiometers - comprising termination printing; resistive body printing; glass coat printing; laser trimming; substrate stripping (SMT) resp. ceramic rods; deposition processes; ageing; capping (THT) resp. printing; curing; punching; revetting; molding (potentiometer.)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature & own estimations	own estimation
Extrapolations	see Geography and General Comments	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 8.1 (Cont.) Ecoinvent meta information for the processes for resistors within the database.

Name	resistor, SMD type, for surface mounting, at plant	resistor, metal film type, for through-hole mounting, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used SMD resistors.	This dataset covers raw material input and production efforts for the production of currently used metal film resistors.
Amount	1	1
Local Name	Widerstand, SMD-Typ, für surface-mounting, ab Werk	Widerstand, Metallfilm-, für through-hole mounting, ab Werk
Synonyms	-	-
General Comment	The data represent a typical SMD resistor for surface mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such resistors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of resistors with an average weight of 9.8 miligram/unit.	The data represent a typical metal-film type capacitor for through-hole mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such resistors and from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of resistors with an average weight of 0.48 gram/unit.
Start Date	1994	1994
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of SMD resistors.	Data represent a global valuable composition of metal-film resistors.
Technology Text	Average technology for the production of resistors and potentiometers - comprising termination printing; resistive body printing; glass coat printing; laser trimming; substrate stripping (SMT) resp. ceramic rods; deposition processes; ageing; capping (THT) resp. printing; curing; punching; revetting; molding (potentiometer.)	Average technology for the production of resistors and potentiometers - comprising termination printing; resistive body printing; glass coat printing; laser trimming; substrate stripping (SMT) resp. ceramic rods; deposition processes; ageing; capping (THT) resp. printing; curing; punching; revetting; molding (potentiometer.)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	literature	Literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 8.1 (Cont.) Ecoinvent meta information for the processes for resistors within the database.

Name	resistor, wirewound type, for through-hole mounting, at plant	potentiometer, unspecified, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	1.0	1.0
Included Processes	This dataset covers raw material input and production efforts for the production of currently used wirewound resistors.	This dataset covers raw material input and production efforts for the production of currently used potentiometers.
Amount	1	1
Local Name	Widerstand, Draht-, through-hole mounting, ab Werk	Potentiometer, unspezifisch, ab Werk
Synonyms	-	-
General Comment	The data represent a typical wirewound type resistor for through-hole mounting, used in the information and communication technology. Material data are taken from datasheets of producers of such resistors. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of resistors with an average weight of 3.8 gram/unit.	The data represent a typical potentiometer, used in the information and communication technology. Material data are taken from the literature. Infrastructure and production efforts are based on own assumptions. The dataset represents 1 kg of potentiometers with an average weight of 6.1 gram/ unit.
Start Date	1994	1994
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of wirewound resistors.	Data represent a global valuable composition of a potentiometer.
Technology Text	Average technology for the production of resistors and potentiometers - comprising termination printing; resistive body printing; glass coat printing; laser trimming; substrate stripping (SMT) resp. ceramic rods; deposition processes; ageing; capping (THT) resp. printing; curing; punching; revetting; molding (potentiometer.)	Average technology for the production of resistors and potentiometers - comprising termination printing; resistive body printing; glass coat printing; laser trimming; substrate stripping (SMT) resp. ceramic rods; deposition processes; ageing; capping (THT) resp. printing; curing; punching; revetting; molding (potentiometer.)
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature	literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

8.3 Dataset “production efforts, resistor”

This dataset describes the material composition of the production of the various types of resistors examined within ecoinvent. The dataset has the functional unit “efforts for the production of 1 kg of resistors”.

8.3.1 System Characterization

The dataset established here comprises the input of auxiliaries and energy as well as the output of emissions and waste due to these auxiliaries in the framework of the resistor production. The input of

the used infrastructure, as well as the transport efforts for the auxiliaries and the raw materials of the various resistors are included as well. Main data sources for this information are the information about resistors and potentiometers in CPM (2006), exact (1998), and US DoE (2007).

8.3.2 Life Cycle inventory

As the raw material input is taken into account in the respective resistor datasets – hence this dataset has no input of raw materials. Instead, this dataset here contains all other input and output flows – i.e. auxiliaries, energy, transport amount (for auxiliaries and raw materials!), infrastructure as well as the emissions and waste amounts produced. In details these various aspects are taken into account based on the following information:

- **Auxiliaries:** From CPM (2006) an average input of about 0.016 kg of auxiliaries per kg of produced resistors can be calculated. Based on the information in CPM (2006), the following substances are those with the highest amount – and hence, the overall amount is shown as:

Ethyl cellulose	20%	as “carboxymethyl cellulose, powder, at plant”
Solvents, unspecified	40%	as “solvents, organic, unspecified, at plant”
Hardener	40%	as “chemicals organic, at plant”

- **Energy:** From CPM (2006) and exact (1998) an average input of electricity of 38.3 kWh, and an input of oil (as light fuel oil) of 10.4 MJ per kg of produced resistors can be calculated.
- **Emissions to air:** According to CPM (2006), ethyl cellulose and solvents are emitted a 100% into air. For the hardener, no such information is available – hence it is assumed here that only 50% of it is emitted into air.

Furthermore, it was assumed, that 100% of the electricity consumed, i.e. 38.3 kWh per kg resistor produced, is converted to waste heat. In accordance with the general rules for ecoinvent, it was assumed that 100% of the waste heat is released to the air.

- **Emissions to water & Water input:** According to CPM (2006), a total water consumption of 0.041 m³ per kg of resistors takes place. It is assumed due to lack of more precise information, that 50% of this is used as cooling water – and 50% as process water. Due to a lack of respective emission information, the outflow of the latter amount is approximated by the use of the dataset “treatment, sewage, to waste water treatment, class 2”.
- **Waste:** According to CPM (2006), a total of 0.23 kg of non-used raw materials can be found in the production process. Assuming that the auxiliaries are emitted either as emissions (air, water, soil) or as waste, it can be concluded that only the remaining 50% of hardener have to be considered as waste. Due to a lack of data they are shown as “disposal, solvent mixture, to HWI”.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here. The amount takes into account the consumed auxiliaries as well as the raw materials for the production of any type of resistor. The amount of raw materials used equals the material in the resistor (i.e. 1 kg) as well as the waste out of raw materials (i.e. 0.23 kg).
- **Infrastructure and land-use:** Due to a lack of more specific information, the infrastructure module established in chapter 2, representing an electronic component production plant, is used also for the production of resistors.

8.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset is only a rough estimation of the actual resistor production efforts, based on information from three datasets out of the Swedish LCI database (CPM (2006)) and from two datasets of Swiss resistor producing companies (exact (1998)). Hence, the respective uncertainty values are accordingly high.

The resulting unit process “production efforts, resistors” is shown in Tab. 8.6 (on page 152).

8.4 Dataset “resistor, SMD type, for surface mounting, at plant”

This dataset here describes the material composition of the production of SMD type resistors for the surface mounting technology. The dataset has a functional unit of “1 kg of resistors”, based on the assumption that a typical SMD resistor in electronic modules has an average weight of **9.8 mg**.

8.4.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 8.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Behrendt et al. (1998) and ZVEI (2006).

8.4.2 Life Cycle inventory

The ecoinvent dataset includes only the raw materials used (see above). According to the dataset “production efforts, resistors”, the production of 1 kg of resistors leads to a waste amount of 0.233 kg – whereof about 0.23 kg are unused raw materials. Hence, this dataset here has an input of 1.23 kg of raw materials per kg of resistor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998) and ZVEI (2006), and is summarized in Tab. 8.2. In this table the respective ecoinvent datasets used are also shown.

8.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical SMD resistor for surface mounting, based on composition information from various worldwide resistor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific resistor type and have thus, accordingly higher uncertainties.

The resulting unit process “resistor, SMD type, for surface mounting, at plant” is shown in Tab. 8.7 (on page 154).

Tab. 8.2 Material specifications of SMD type resistors (data sources: see text)

		SMD type	Used ecoinvent dataset
		resistor	
Weight	g	0.0098	
[i] Metals			
lead	%	0.44	lead, at regional storage (RER)
iron	%	2.35	pig iron, at plant (GLO)
nickel	%	1.65	nickel, 99.5%, at plant (GLO)
copper	%	0.37	copper, primary, at plant (RER)
tin	%	1.17	tin, at regional storage (RER)
chromium	%	0.09	chromium, at regional storage (RER)
silicium	%	0.01	MG-silicon, at plant (NO)
palladium	%	0.14	palladium, at regional storage (RER)
silver	%	1.00	silver, at regional storage (RER)
gold	%	0.51	gold, at regional storage (RER)
[ii] Others			
Al ₂ O ₃	%	69.95	aluminium oxide, at plant (RER)
SiO ₂	%	20.02	silica sand, at plant (RER)
CaCO ₃	%	0.05	limestone, milled, packed, at plant (CH)
Glass	%	0.66	glass tube, borosilicate, at plant (DE)
Epoxy resin	%	0.94	epoxy resin, liquid, at plant (RER)
Polyacrylate	%	0.64	polymethyl methacrylate, beads, at plant (RER)

8.5 Dataset “resistor, metal film type, for through-hole mounting, at plant”

This dataset here describes the material composition of the production of metal film type resistors for the through-hole mounting technology. The dataset has a functional unit of “1 kg of resistors”, based on the assumption that a typical metal film resistor in electronic modules has an average weight of 0.48 g.

8.5.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 8.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are Behrendt et al. (1998; ZVEI (2006).

8.5.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, resistors”, the production of 1 kg of resistors leads to a waste amount of 0.233 kg – whereof about 0.23 kg are unused raw materials. Hence, this dataset here has an input of 1.23 kg of raw materials per kg of resistor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998; ZVEI (2006) and is summarized in Tab. 8.3. In this table the respective ecoinvent datasets used are also shown.

Tab. 8.3 Material specifications of metal film type resistors (data sources: see text)

		metal	Used ecoinvent dataset
		film	
Weight	g	0.480	
[i] Metals			
lead	%	5.03	lead, at regional storage (RER)
Iron	%	10.27	pig iron, at plant (GLO)
Steel	%	6.00	steel, low-alloyed, at plant (RER)
Copper	%	37.20	copper, primary, at plant (RER)
Aluminium	%	0.03	aluminium, primary, at plant (RER)
Nickel	%	2.36	nickel, 99.5%, at plant (GLO)
Tin	%	8.75	tin, at regional storage (RER)
[ii] Others			
Al ₂ O ₃	%	10.96	aluminium oxide, at plant (RER)
SiO ₂	%	12.50	silica sand, at plant (RER)
CaCO ₃	%	2.41	limestone, milled, packed, at plant (CH)
TiO ₂	%	0.06	titanium dioxide, production mix, at plant (RER)
MgSiO ₃	%	0.42	proxy "magnesium oxide, at plant" used !
Epoxy resin	%	1.68	epoxy resin, liquid, at plant (RER)
acrylate resin	%	0.01	proxy "methyl acrylate, at plant" used !
silicon resin	%	0.16	proxy "silicon product, at plant" used !
pigments	%	2.05	proxy "PET, amorphous, at plant" used !
polyurethane	%	0.12	polyurethane, flexible foam, at plant (RER)

8.5.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical metal film resistor for through-hole mounting, based on composition information from various worldwide resistor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific resistor type and have thus, accordingly higher uncertainties.

The resulting unit process “resistor, metal film type, for through-hole mounting, at plant” is shown in Tab. 8.8 (on page 156).

8.6 Dataset “resistor, wirewound type, for through-hole mounting, at plant”

This dataset here describes the material composition of the production of wirewound type resistors for the through-hole mounting technology. The dataset here has a functional unit of “1 kg of resistors”, based on the assumption that a typical wirewound resistor in electronic modules has an average weight of 3.8 g.

8.6.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 8.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as

well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are various fact sheets from ZVEI (2006).

8.6.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, resistors”, the production of 1 kg of resistors leads to a waste amount of 0.233 kg – whereof about 0.23 kg are unused raw materials. Hence, this dataset here has an input of 1.23 kg of raw materials per kg of resistor. The split to the different raw materials (metals, plastics) is based on the material specifications reported in the fact sheets from ZVEI (2006) and is summarized in Tab. 8.4. In this table the respective ecoinvent datasets used are also shown.

Tab. 8.4 Material specifications of wirewound type resistors (data sources: see text)

		wire-wound	Used ecoinvent dataset
Weight	g	3.8	
[i] Metals			
Iron	%	5.20	pig iron, at plant (GLO)
Steel	%	11.10	steel, low-alloyed, at plant (RER)
Copper	%	3.30	copper, primary, at plant (RER)
Aluminium	%	0.40	aluminium, primary, at plant (RER)
Nickel	%	1.10	nickel, 99.5%, at plant (GLO)
Tin	%	2.20	tin, at regional storage (RER)
[ii] Others			
Al ₂ O ₃	%	24.05	aluminium oxide, at plant (RER)
SiO ₂	%	40.68	silica sand, at plant (RER)
CaCO ₃	%	9.83	limestone, milled, packed, at plant (CH)
(Fibre) glass	%	1.20	glass tube, borosilicate, at plant (DE)
silicon resin	%	0.93	proxy "silicon product, at plant" used !

8.6.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents a typical wirewound resistor for through-hole mounting, based on composition information from various worldwide resistor producers. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific resistor type and have thus, accordingly higher uncertainties.

The resulting unit process “resistor, wirewound type, for through-hole mounting, at plant” is shown in Tab. 8.9 (on page 158).

8.7 Dataset “potentiometer, unspecified, at plant”

This dataset here describes the material composition of the production of a typical potentiometer. The dataset here has a functional unit of “1 kg of potentiometer”, based on the assumption that a typical potentiometer in electronic modules has an average weight of 6.1 g.

8.7.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) and a link to the above described dataset about the production efforts (see chapter 8.3). This latter one contains all further inputs and outputs – i.e. the consumption of energy or auxiliaries, the used infrastructure as well as the produced emissions and waste amounts. Main data sources for the input of raw materials in this dataset here are the information found in Behrendt et al. (1998).

8.7.2 Life Cycle inventory

The dataset for ecoinvent includes only the raw materials used (see above). According to the dataset “production efforts, resistors”, the production of 1 kg of resistors leads to a waste amount of 0.233 kg – whereof about 0.23 kg are unused raw materials. Hence, this dataset here has an input of 1.23 kg of raw materials per kg of potentiometer. The split to the different raw materials (metals, plastics) is based on the material specifications reported in Behrendt et al. (1998) and is summarized in Tab. 8.5. In this table the respective ecoinvent datasets used are also shown.

Tab. 8.5 Material specifications of potentiometers in this database (data sources: see text)

		Potentiometer	Used ecoinvent dataset
Weight	g	6.1	
[i] Metals			
steel	%	6.84	steel, low-alloyed, at plant (RER)
copper	%	5.22	copper, primary, at plant (RER)
zinc	%	0.71	zinc, for coating, at plant (RER)
nickel	%	0.46	nickel, 99.5%, at plant (GLO)
tin	%	0.46	tin, at regional storage (RER)
[ii] Others			
Al ₂ O ₃	%	8.01	aluminium oxide, at plant (RER)
Paper	%	2.24	kraftpaper, unbleached, at plant (RER)
plastic housing	%	72.70	100% shown as "polycarbonate, at plant (RER)"
Pigments	%	3.37	proxy "PET, amorphous, at plant" used !

8.7.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents typical potentiometers, based on composition information from one literature source at End of 90s. Hence, the uncertainty values are estimated accordingly. All production efforts (summed up in a separate dataset) represent not a specific capacitor type and have thus, accordingly higher uncertainties.

The resulting unit process “potentiometer, unspecified, at plant” is shown in Tab. 8.10 (on page 159).

8.8 Dataset “resistor, unspecified, at plant”

Last but not least, for all those cases where no information is given about the resistor type used, an unspecified dataset is established as well. This latter one is an unweighted mean value of all four above described resistor and potentiometer types. The dataset has as functional unit also “1 kg of resistors”.

8.8.1 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an unweighted average of the various resistor types shown in this database. Hence, the uncertainty values are accordingly high.

The resulting unit process “resistor, unspecified, at plant” is shown in Tab. 8.11 (on page 160).

Tab. 8.6 Unit process inventory of the dataset “Production efforts, resistor (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	STdv 95%	General Comment
Water (process)	production efforts, resistors (Input-Part)			resource	in water			Water, unspecified natural origin	2.05E-02	m3	calculated from 4 dataset in CPM (2006)	1	134	(3,4,4,1,15,12); assumption, based on literature
Water (cooling)				resource	in water			Water, cooling, unspecified natural origin	2.05E-02	m3	calculated from 4 dataset in CPM (2006)	1	134	(3,4,4,1,15,12); assumption, based on literature
ethyl cellulose			dataset "carboxymethyl cellulose" as proxy used	washing agents	auxiliary agents	No	RER	carboxymethyl cellulose, powder, at plant	3.20E-03	kg	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,15,3); average from literature
solvents, unspecified				chemicals	organics	No	GLO	solvents, organic, unspecified, at plant	6.40E-03	kg	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,15,3); average from literature
hardener			dataset "chemical organic" as proxy used	chemicals	organics	No	GLO	chemicals organic, at plant	6.40E-03	kg	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,15,3); average from literature
electricity				electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	3.83E+01	kWh	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,15,2); average from literature
light fuel oil (heat)				oil	heating systems	No	RER	heat, light fuel oil, at industrial furnace 11MW	104E+01	MJ	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,15,1); average from literature
Transport (part by lorry)				transport systems	road	No	RER	transport, lorry >16t, fleet average	125E-01	tkm	calculated from 4 dataset in CPM (2006)	1	2.09	(4,5,nanana,5); standard distances
Transport (part by train)				transport systems	train	No	RER	transport, freight, rail	7.48E-01	tkm	calculated from 4 dataset in CPM (2006)	1	2.09	(4,5,nanana,5); standard distances
Infrastructure (capacitor plant)				electronics	component	Yes	GLO	electronic component production plant	2.00E-08	unit	own assumption	1	3.77	(4,5,13,5,4,9); rough assumption

Tab. 8.6 (Cont.) Unit process inventory of the dataset “Production efforts, resistor (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment	
production efforts, resistors (Output-Part)		↑	↑	due to a lack of data, the dataset "disposal, solvent mixture, to HW" is used as a proxy	waste management	hazardous waste incineration	No	CH	disposal, solvents mixture, 16.5%water, to hazardous waste incineration	3.20E-03	kg	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,156); calculated frominput of auxiliaries
		↑	raw materials to waste	due to a lack of information, 50%are assumed as plastics waste to sanitary landfill ...	waste management	sanitary landfill	No	CH	disposal, plastics, mixture, 15.3%water, to sanitary landfill	1.15E-01	kg	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,156); assumption, based on literature
		↑	raw materials to waste	...and the remaining 50%are assumed to be anorganic waste to landfill (proxy of Si wafer waste used)	waste management	residual material landfill	No	CH	waferprod., inorg, 9.4% water, to residual material landfill	1.15E-01	kg	calculated from 4 dataset in CPM (2006)	1	127	(2,4,3,3,156); assumption, based on literature
		↑	waste water to treatment	due to a lack of knowledge of content, dataset "treatment, sewage, WWTP class 2" is used as proxy	waste management	wastewater treatment	No	CH	treatment, sewage, to wastewater treatment, class 2	2.05E-02	m3	calculated as 100%of process water input ...	1	127	(2,4,3,3,156); assumption, based on literature
		↑	waste heat		air	high population density			Heat, waste	138E+02	MJ	calculated fromthe electricity input	1	127	(2,4,3,3,15,13); calculated, from electricity input
		↑	ethyl cellulose, to air		air	high population density			Ethyl cellulose	3.20E-03	kg	calculated frominput of auxiliaries	1	2.08	(2,4,3,3,1523); calculated frominput of auxiliaries
		↑	solvents, to air	proxy "NM VOC, to air" used	air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	6.40E-03	kg	calculated frominput of auxiliaries	1	2.08	(2,4,3,3,1523); calculated frominput of auxiliaries
		↑	hardener, to air	proxy "NM VOC, to air" used	air	high population density			NM VOC, non-methane volatile organic compounds, unspecified origin	3.20E-03	kg	calculated frominput of auxiliaries	1	2.08	(2,4,3,3,1523); calculated frominput of auxiliaries
		↑	Production efforts for capacitors		electronics	component	No	GLO	production efforts, resistors	100E+00	kg		1		

Tab. 8.7 Unit process inventory of the dataset “Resistor, SMD type, surface mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	SDv 95%	General Comment
production efforts	resistor, SMD type, surface mounting, at plant			electronics	component	No	GLO	production efforts, resistors	100E+00	kg	average resistor production from CPM (2006)	1	1.61	(3,4,3,1,4,5,1); rough estimation
lead				metals	extraction	No	RER	lead, at regional storage	5.41E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	2.89E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	2.03E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	4.55E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	1.44E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
chromium				metals	extraction	No	RER	chromium, at regional storage	1.11E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
silicium				metals	extraction	No	NO	M-G-silicon, at plant	123E-04	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information
palladium				metals	extraction	No	RER	palladium, at regional storage	1.72E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14,2,1,14,3); average from literature & company information

continued ▶

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Tab. 8.7 (Cont.) Unit process inventory of the dataset “Resistor, SMD type, surface mounting, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
silver	resistor, SMD type, surface mounting, at plant	continued		metals	extraction	No	RER	silver, at regional storage	123E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
gold				metals	extraction	No	RER	gold, at regional storage	6.27E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
Aluminiumoxide				chemicals	Inorganics	No	RER	aluminium oxide, at plant	8.60E-01	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
Silicium oxide				construction materials	additives	No	DE	silica sand, at plant	2.46E-01	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
calcium carbonate				construction materials	others	No	CH	limestone, milled, packed, at plant	6.15E-04	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
glass				glass	construction	No	DE	glass tube, borosilicate, at plant	8.12E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	1.16E-02	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
polyacrylate				plastics	polymers	No	RER	polymethyl methacrylate, beads, at plant	7.87E-03	kg	Average according to Beherendt (1998) and ZVEI (2006)	1	1.16	(14.2, 114.3); average from literature & company information
				electronics	component	No	GLO	resistor, SMD type, surface mounting, at plant	100E+00	kg		1		

Tab. 8.8 Unit process inventory of the dataset “Resistor, metal film type, through-hole mounting, at plant (GLO)”

General Flow information					Representation in ecoinvent						Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	resistor, metal film type, through-hole mounting, at plant			electronics	component	No	GLO	production efforts, resistors	100E+00	kg	average resistor production from CPM (2006)	1	1.61	(3,4,3,1,4,5,1); rough estimation
lead				metals	extraction	No	RER	lead, at regional storage	6.16E-02	kg	Average according to Behrendt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
iron				metals	extraction	No	GLO	pig iron, at plant	126E+01	kg	Average according to Behrendt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
steel				metals	extraction	No	RER	steel, low-alloyed, at plant	738E-02	kg	Average according to Behrendt (1998) and ZVEI (2006)	1	1.16	(14,2,114,3); average from literature & company information
copper				metals	extraction	No	RER	copper, primary, at refinery	4.58E+01	kg	Average according to Behrendt (1998) and Venkel (2006)	1	1.16	(14,2,114,3); average from literature & company information
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	3.69E-04	kg	Average according to Behrendt (1998) and Venkel (2006)	1	1.16	(14,2,114,3); average from literature & company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	2.90E-02	kg	Average according to Behrendt (1998) and Venkel (2006)	1	1.16	(14,2,114,3); average from literature & company information
tin				metals	extraction	No	RER	tin, at regional storage	108E+01	kg	Average according to Behrendt (1998) and Venkel (2006)	1	1.16	(14,2,114,3); average from literature & company information
aluminium oxide				chemicals	inorganics	No	RER	aluminium oxide, at plant	135E+01	kg	Average according to Behrendt (1998) and Venkel (2006)	1	1.16	(14,2,114,3); average from literature & company information

continued ▼

Tab. 8.8 (Cont.) Unit process inventory of the dataset “Resistor, metal film type, through-hole mounting, at plant (GLO)”

General Flow information					Representation in ecoinvent						Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
silicium dioxide	resistor, metal film type, through-hole mounting, at plant	continued		construction materials	additives	No	DE	silica sand, at plant	154E-01	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
calcium carbonate				construction materials	others	No	CH	limestone, milled, packed, at plant	2.96E-02	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
titanium dioxide				chemicals	inorganics	No	RER	titanium dioxide, production mix, at plant	7.38E-04	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
magnesium silicate				chemicals	inorganics	No	RER	magnesium oxide, at plant	5.17E-03	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
epoxy resin				plastics	monomers	No	RER	epoxy resin, liquid, at plant	2.07E-02	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
acrylate resin				chemicals	organics	No	GLO	methyl acrylate, at plant	123E-04	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
silicon resin				chemicals	inorganics	No	RER	silicone product, at plant	197E-03	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
pigments				plastics	polymers	No	RER	polyethylene terephthalate, granulate, amorphous, at plant	2.52E-02	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
polyurethane				plastics	polymers	No	RER	polyurethane, flexible foam, at plant	148E-03	kg	Average according to Beherendt (1998) and Venkel (2006)	1	1.16	(14.2,114.3); average from literature & company information
		metal-film resistor		electronics	component	No	GLO	resistor, metal film type, through-hole mounting, at plant	100E+00	kg		1		

Tab. 8.9 Unit process inventory of the dataset “Resistor, wirewound, through-hole mounting, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Module name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	resistor, wirewound, through-hole mounting, at plant	↑	↑	electronics	component	No	GLO	production efforts, resistors	100E+00	kg	average resistor production from CPM (2006)	1	1.61	(3,4,3,1,4,5,1); rough estimation
iron				metals	extraction	No	GLO	pig iron, at plant	6.40E-02	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
steel				metals	extraction	No	RER	steel, low-alloyed, at plant	137E-01	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
copper				metals	extraction	No	RER	copper, primary, at refinery	4.06E-02	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
aluminium				metals	extraction	No	RER	aluminium, primary, at plant	4.92E-03	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
nickel				metals	extraction	No	GLO	nickel, 99.5% at plant	135E-02	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
tin				metals	extraction	No	RER	tin, at regional storage	2.71E-02	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
aluminium oxide				chemicals	inorganics	No	RER	aluminium oxide, at plant	2.96E-01	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
silicium dioxide				construction materials	additives	No	DE	silica sand, at plant	5.00E-01	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
calcium carbonate				construction materials	others	No	CH	limestone, milled, packed, at plant	121E-01	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
(fibre) glass				glass	construction	No	DE	glass tube, borosilicate, at plant	148E-02	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
silicon resin				chemicals	inorganics	No	RER	silicone product, at plant	114E-02	kg	Typical data according to ZVEI(2006)	1	1.16	(14,2,1,14,3); company information
		↑	↑	electronics	component	No	GLO	resistor, wirewound, through-hole mounting, at plant	100E+00	kg		1		

Tab. 8.10 Unit process inventory of the dataset “Potentiometer, unspecified, at plant (GLO)”

General Flow information				Representation in ecoinvent						Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
production efforts	↑ potentiometer, unspecified, at plant	↑		electronics	component	No	GLO	production efforts, resistors	100E+00	kg	average resistor production from CPM (2006)	1	161	(3,4,3,14,5,1); rough estimation
steel		↑		metals	extraction	No	RER	steel, low-alloyed, at plant	8.41E-02	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
copper		↑		metals	extraction	No	RER	copper, primary, at refinery	6.42E-02	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
zinc		↑		metals	extraction	No	RER	zinc, primary, at regional storage	8.73E-03	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
nickel		↑		metals	extraction	No	GLO	nickel, 99.5% at plant	5.66E-03	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
tin		↑		metals	extraction	No	RER	tin, at regional storage	5.66E-03	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
aluminium oxide		↑		chemicals	inorganics	No	RER	aluminium oxide, at plant	9.85E-02	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
paper		↑		paper & cardboard	packaging papers	No	RER	kraft paper, unbleached, at plant	2.76E-02	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
plastic housing		↑		plastics	polymers	No	RER	polycarbonate, at plant	8.94E-01	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
pigments		↑		plastics	polymers	No	RER	polyethylene terephthalate, granulate, amorphous, at plant	4.15E-02	kg	Data taken from Behrendt (1998)	1	126	(14,4,114,3); literature data
		↑ potentiometer		electronics	component	No	GLO	potentiometer, unspecified, at plant	100E+00	kg		1		

Tab. 8.11 Unit process inventory of the dataset “Resistor, unspecified, at plant (GLO)”

General Flow information					Representation in ecoinvent							Uncertainty information		
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
SMD resistors	resistor, unspecified, at plant	↑	↑	electronics	component	No	GLO	resistor, SMD type, surface mounting, at plant	2.50E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
metal film resistors				electronics	component	No	GLO	resistor, metal film type, through-hole mounting, at plant	2.50E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
wirewound resistors				electronics	component	No	GLO	resistor, wirewound, through-hole mounting, at plant	2.50E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
potentiometers				electronics	component	No	GLO	potentiometer, unspecified, at plant	2.50E-01	kg	assumption	1	189	(5.4.4,1.4.5.3); assumption, based on specific data
		↑ unspecified resistors		electronics	component	No	GLO	resistor, unspecified, at plant	100E+00	kg		1		

↑ unspecified resistors

9 Transformers

A **transformer** is an electrical device that transfers energy from one circuit to another by magnetic coupling, without requiring relative motion between its parts. A transformer comprises two or more coupled windings, and, in most cases, a magnetic core to concentrate magnetic flux. A changing voltage applied to one winding creates a time-varying magnetic flux in the core, which induces a voltage in the other windings.⁸



Fig. 9.1 Impression of the very different size (and forms) of current transformers (Picture from www.big-bit.com [left], www.strategosinc.com [middle] resp. www.electricityforum.com [right]).

Transformers come in a range of sizes from a thumbnail-sized coupling transformer hidden inside a stage microphone to huge gigawatt units used to interconnect large portions of national power grids. All operate with the same basic principles and with many similarities in their parts.

9.1 Characterisation, Use & Market

According to Professor Calvert⁹, most transformers with iron cores can be considered as *ideal* when in use. An ideal transformer has no losses, an aim that is closely attained in practice, so the energy transfer from the primary circuit to the secondary circuit is perfect. In transformer design it is before all important to design a transformer that is economical, efficient and makes the best use of available materials. However, as in all design processes, there are numerous trade-offs between competing requirements, and some sort of optimum is sought. On the other hand, the properties of available materials rigidly limit the possibilities. Aspects that have to be defined are size, shape and material of the core – as well as the number of turns of the primary and secondary windings.

The main usage of transformers is the power supply from an alternating current power grid to equipment which uses a different voltage (e.g. in power supply units). However, other types of transformers can also be used for the transmission of power over long distances, transferring signals (e.g. in the VHS video head system) – or large forms are in use in steelmaking (electric arc furnace). Hence, there is a huge variety of different types of transformers on the market – a variety that can not be covered 100% by the here described ecoinvent datasets.

9.2 System Characterisation

Very few sources with actual information concerning the material composition of current transformers used in the information and communication technology are available. Hence, it is not possible to es-

⁸ Wikipedia: Transformer (<http://en.wikipedia.org/wiki/Transformer>), accessed March 14, 2007.

⁹ Calvert, J.B.: Inside Transformers (<http://mysite.du.edu/~jcalvert/tech/transfor.htm>), accessed March 21, 2007.

establish a specific dataset concerning the production efforts of transformers, but just two datasets representing smaller and bigger such devices used in the ICT domain.

Both datasets use as main data source Behrendt et al. (1998), representing material specification sheets from proper measurements of transformers out of a television device. As this information can not be attributed to one specific producer of this kind of devices, the here developed data can be considered as “global” valid data. The respective meta information for the various datasets of inductors can be found in Tab. 9.1.

Tab. 9.1 Ecoinvent meta information for the processes for transformers within the database.

Name	transformer, low voltage use, at plant	transformer, high voltage use, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input and production efforts for the production of power transformers used in the low voltage area, i.e. in an area of 5 up to about 25 V.	This dataset covers raw material input and production efforts for the production of high voltage type transformers, i.e. for an area of 200 up to 380 V.
Amount	1	1
Local Name	Transformator, Niederspannungsbereich, ab Werk	Transformator, Hochspannungsbereich, ab Werk
Synonyms	-	-
General Comment	The data represent a typical low voltage transformer, used in the information and communication technology. Material data are taken from the literature. Production efforts are approximated by using material specific treatment processes. Infrastructure is included in this production efforts approximation. The dataset represents 1 kg of transformers with a weight of 30 gram/unit.	The data represent a typical high voltage transformer, used in the information and communication technology. Material data are taken from the literature. Production efforts are approximated by using material specific treatment processes. Infrastructure is included in this production efforts approximation. The dataset represents 1 kg of transformers with a weight of 280 gram/unit.
Start Date	1994	1994
End Date	2007	2007
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of low voltage transformers.	Data represent a global valuable composition of high voltage transformers.
Technology Text	average transformer production technology, based on literature and assumptions.	average transformer production technology, based on literature and assumptions.
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature	Literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

9.3 Dataset “transformer, low voltage use, at plant”

This dataset here describes the production efforts for the production of power transformers for the low voltage area. The dataset has a functional unit of “1 kg of transformer”, based on the assumption that a

typical low voltage use transformer in electronic modules has an assumed weight of about **30 g** (Behrendt et al. (1998)).

9.3.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) as well as the respective processing steps, waste amounts, the infrastructure and the transport services required for the production of 1 kg of transformer. Main data sources for the input of raw materials in this dataset here are the information found in Behrendt et al. (1998). Tab. 9.2 gives an overview of the material composition as reported in Behrendt et al. (1998). In this table the respective ecoinvent datasets used are also shown.

Tab. 9.2 Material specifications of low voltage use transformers in this database (data sources: see text)

		low voltage transformer	Used ecoinvent dataset
Weight	g	30	
Copper	%	28.79	copper, primary, at plant (RER)
Ferrite	%	56.55	ferrite, at plant (GLO)
plastics, unspecified	%	13.02	as 100% "polycarbonate, at plant (RER)"
remaining materials	%	1.63	as 100% "epoxy resin, liquid, at plant (RER)"

9.3.2 Life Cycle inventory

- **Raw materials:** For the total amount, the mean values from Behrendt et al. (1998) as shown in Tab. 9.2, taking into account an overall process efficiency of 90%, are used.
- **Auxillaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** the energy consumption is indirectly taken into account by using specific processing datasets for the further treatment of the various above mentioned raw materials – i.e. wire drawing (in case of copper) resp. injection moulding (plastics, remaining materials).
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here.
- **Infrastructure and land-use:** no separate infrastructure included due to that fact that the various material processing datasets used have already the respective infrastructure included – hence, an additional infrastructure would be, at least partly, a double counting.
- **Emissions to air & water, Waste:** Due to a lack of respective information, no emission flows are taken into account. Concerning waste streams, all metals are supposed to go for recycling and thus are not further taken into account here in accordance with Frischknecht et al. (2007). For the remaining waste amount the dataset “disposal, plastic mixture, to municipal incineration” is used as a proxy.

9.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an estimation for a

small power transformer, based on the composition information in one literature source (Behrendt et al. (1998)). Hence, the uncertainty values are accordingly high. For the production efforts data for the processing of the various materials are used as a first approximation – with an accordingly high uncertainty. The resulting unit process “transformer, type small power transformer, at plant” is shown in Tab. 9.4 (on page 166).

9.4 Dataset “transformer, high voltage use, at plant”

This dataset here describes the production efforts for the production of transformers for the high voltage area. The dataset has a functional unit of “1 kg of transformer”, based on the assumption that a typical high voltage use transformer in electronic modules has an average weight of **280 g** (Behrendt et al. (1998)).

9.4.1 System Characterization

The dataset established here comprises the input of raw materials (metallic and plastics) as well as the respective processing steps, waste amounts, the infrastructure and the transport services required for the production of 1 kg of transformer. Main data sources for the input of raw materials in this dataset here are the information found in Behrendt et al. (1998). Tab. 9.3 gives an overview of the material composition as reported in Behrendt et al. (1998). In this table the respective ecoinvent datasets used are also shown.

Tab. 9.3 Material specifications of high voltage use transformers in this database (data sources: see text)

		high voltage transformer	Used ecoinvent dataset
Weight	g	280	
Copper	%	9.46	copper, primary, at plant (RER)
steel	%	0.45	steel, low-alloyed, at plant (RER)
Ferrite	%	45.05	ferrite, at plant (GLO)
plastics, unspecified	%	12.38	as 100% "polycarbonate, at plant (RER)"
remaining materials	%	32.67	as 100% "epoxy resin, liquid, at plant (RER)"

9.4.2 Life Cycle inventory

- **Raw materials:** For the total amount, the mean values from Behrendt et al. (1998) as shown in Tab. 9.3, taking into account an overall process efficiency of 90%, are used.
- **Auxillaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** the energy consumption is indirectly taken into account by using specific processing datasets for the further treatment of the various above mentioned raw materials – i.e. wire drawing (in case of copper) resp. injection moulding (plastics, remaining materials).
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here.
- **Infrastructure and land-use:** no separate infrastructure included due to that fact that the various material processing datasets used have already the respective infrastructure included – hence, an additional infrastructure would be, at least partly, a double counting.

- **Emissions to air & water, Waste:** Due to a lack of respective information, no emission flows are taken into account. Concerning waste streams, all metals are supposed to go for recycling and thus are not further taken into account here in accordance with Frischknecht et al. (2007). For the remaining waste amount the dataset “disposal, plastic mixture, to municipal incineration” is used as a proxy.

9.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an estimation for a high voltage transformer, based on the composition information in one literature source (Behrendt et al. (1998)). Hence, the uncertainty values are accordingly high. For the production efforts data for the processing of the various materials are used as a first approximation – with an accordingly high uncertainty. The resulting unit process “transformer, high voltage use, at plant” is shown in Tab. 9.5 (on page 167).

Tab. 9.4 Unit process inventory of the dataset “Transformer, low voltage use, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment	
copper	transformer, low voltage use, at plant	low voltage transformer		metals	extraction	No	RER	copper, primary, at refinery	3.20E-01	kg	Data according to Beherndt (1998)	1	126	(14.4,114.3); averaged from literature	
wire production			metals	processing	No	RER	wire drawing, copper	3.20E-01	kg	Own assumption - based on Cu input	1	146	(4.53,3.3,5.3); own estimation		
ferrite			electronics	component	No	GLO	ferrite, at plant	6.28E-01	kg	Data according to Beherndt (1998)	1	126	(14.4,114.3); averaged from literature		
Plastics				plastics	polymers	No	RER	polycarbonate, at plant	1.48E-01	kg	Data according to Beherndt (1998)	1	126	(14.4,114.3); averaged from literature	
remaining materials				plastics	monomers	No	RER	epoxy resin, liquid, at plant	189E-02	kg	Data according to Beherndt (1998)	1	126	(14.4,114.3); averaged from literature	
treatment of plastics and remaining mat.				plastics	processing	No	RER	injection moulding	1.66E-01	kg	Own assumption - based on material input	1	146	(4.53,3.3,5.3); own estimation	
Transport (part by lorry)				transport systems	road	No	RER	transport, lorry >16t, fleet average	1.1E-01	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nan,nan,nan,5); standard distances	
Transport (part by train)				transport systems	train	No	RER	transport, freight, rail	6.69E-01	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,nan,nan,nan,5); standard distances	
waste disposal residues (plastics, remaining mat.)				waste management	incineration of plastic in MSW as proxy used	municipal incineration	No	CH	disposal, plastics, mixture, 15.3% water, to municipal incineration	1.63E-02	kg	Own assumption - assuming a yield of 90%	1	146	(4.53,3.3,5.3); own estimation
				electronics	component	No	GLO	transformer, low voltage use, at plant	1.00E+00	kg		1			

Tab. 9.5 Unit process inventory of the dataset “Transformer, high voltage use, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StdV 95%	General Comment
copper	transformer, high voltage use, at plant	high voltage transformer		metals	extraction	No	RER	copper, primary, at refinery	105E-01	kg	Data according to Beherendt (1998)	1	126	(14.4, 114.3); average from literature
steel				metals	extraction	No	RER	steel, low-alloyed, at plant	500E-03	kg	Data according to Beherendt (1998)	1	126	(14.4, 114.3); average from literature
wire production				metals	processing	No	RER	wire drawing, copper	1.10E-01	kg	Own assumption - based on Cu input	1	146	(4.53, 3.3, 5.3); own estimation
ferrite				electronics	component	No	GLO	ferrite, at plant	50.1E-01	kg	Data according to Beherendt (1998)	1	126	(14.4, 114.3); average from literature
Plastics			assumption 100% polycarbonate	plastics	polymers	No	RER	polycarbonate, at plant	1.40E-01	kg	Data according to Beherendt (1998)	1	126	(14.4, 114.3); average from literature
remaining materials			assumption 100% epoxy resin	plastics	monomers	No	RER	epoxy resin, liquid, at plant	3.70E-01	kg	Data according to Beherendt (1998)	1	126	(14.4, 114.3); average from literature
treatment of plastics and remaining mat.			injection moulding as proxy used	plastics	processing	No	RER	injection moulding	5.11E-01	kg	Own assumption - based on material input	1	146	(4.53, 3.3, 5.3); own estimation
Transport (part by lorry)				transport systems	road	No	RER	transport, lorry >16t, fleet average	1.12E-01	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,na,na,na,na,5); standard distances
Transport (part by train)				transport systems	train	No	RER	transport, freight, rail	6.73E-01	tkm	standard distances according to Frischknecht et al. (2004)	1	2.09	(4.5,na,na,na,na,5); standard distances
waste disposal residues (plastics, remaining mat.)			incineration of plastic in MSW as proxy used	waste management	municipal incineration	No	CH	disposal, plastics, mixture, 15.3% water, to municipal incineration	5.01E-02	kg	Own assumption - assuming a yield of 90%	1	146	(4.53, 3.3, 5.3); own estimation
				electronics	component	No	GLO	transformer, high voltage use, at plant	100E+00	kg		1	#NV	

10 Connectors and Switches

The last two types of passive elements reported are connectors and switches. Both types of components are rather simple concerning their material composition (mixture of metals and plastics); hence they are of an ancillary importance concerning the environmental impact of electronics.

Within literature, pretty few information can be found concerning these elements – hence, this chapter here describes rather rough datasets for the ecoinvent database. The datasets cover the (raw) material input and the energy consumption for the production – auxiliaries, emissions (air, water, soil) as well as waste are not reported in the used sources and thus, are neither taken into account in the datasets here. The used data sources are mainly various information from producers (collected by internet search), representing product specification sheets from industry, few information from literature () and own estimations and calculations based upon all these other sources.

In order to have a broad variety of different types of connectors covered within the ecoinvent database, the following three datasets for connectors are established here:

- | | |
|--|--------------|
| - Connector, clamp connection, at plant | chapter 10.1 |
| - Connector, PCI bus, at plant | chapter 10.2 |
| - Connector, computer, peripheric type, at plant | chapter 10.3 |

Last but not least, concerning the various types of switches, only the toggle type switch is examined here (see chapter 10.4). The respective meta information for the various datasets of connectors and switches can be found in Tab. 10.1.

Tab. 10.1 Ecoinvent meta information for the processes for transformers within the database.

Name	connector, clamp connection, at plant	connector, PCI bus, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input, energy consumption, infrastructure and transport efforts for the production of clamp type connectors.	This dataset covers raw material input, energy consumption, infrastructure and transport efforts for the production of PCI bus type connectors.
Amount	1	1
Local Name	Anschluss, Klemmverbindung, ab Werk	Anschluss, PCI-Bus, ab Werk
Synonyms	-	-
General Comment	The data represent a current clamp type connector for three wires used in the electronics industry in general. Material data are taken from product sheets of a producer. The energy demand for the production is from literature. Infrastructure and transport efforts rough estimations. The dataset represents 1 kg of clamp type connectors with a weight of 9 gram/unit.	The data represent a current PCI bus type connector like used in current information and communication technology (i.e. computers). Material data are taken from product sheets of producers. The energy demand is estimated based on literature. Infrastructure and transport efforts rough estimations. The dataset represents 1 kg of PCI bus type connectors with a weight of 11.6 gram/unit.
Start Date	2004	2004
End Date	2006	2006
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of 3-wired clamp type connectors.	Data represent a global valuable composition of a 64-bit PCI slot connector.
Technology Text	average connector production technology, based on literature and assumptions.	average connector production technology, based on literature and assumptions.
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature	Literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

Tab. 10.1 (Cont.) Ecoinvent meta information for the processes for transformers within the database.

Name	connector, computer, peripheral type, at plant	switch, toggle type, at plant
Location	GLO	GLO
Infrastructure Process	0	0
Unit	kg	kg
Dataset Version	2.0	2.0
Included Processes	This dataset covers raw material input, energy consumption, infrastructure and transport efforts for the production of computer peripheral type connectors.	This dataset covers raw material input, energy consumption, infrastructure and transport efforts for the production of toggle type switches.
Amount	1	1
Local Name	Anschluss, Computer, Peripheriestecker, ab Werk	Schalter, Kippschalter, ab Werk
Synonyms	-	-
General Comment	The data represent a current computer peripheral type connector, as used in current computers. Material data are taken from product sheets of producers. The energy demand is estimated based on literature. Infrastructure and transport efforts rough estimations. The dataset represents 1 kg of computer peripheral type connectors with a weight of 10 gram/unit.	The data represent a current toggle type switch, as applied in current information and communication technology (ICT). Material data are taken from product sheets of producers. The energy demand is estimated based on literature. Infrastructure and transport efforts rough estimations. The dataset represents 1 kg of toggle type switches with a weight of 29 gram/unit.
Start Date	2004	2004
End Date	2006	2006
Data valid for entire period	1	1
Geography Text	Data represent a global valuable composition of a serial computer port.	Data represent a global valuable composition of a toggle type switch.
Technology Text	average connector production technology, based on literature and assumptions.	average switch production technology, based on literature and assumptions.
Representativeness	see General Comments	see General Comments
Production Volume	-	-
Sampling Procedure	Literature	Literature
Extrapolations	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None

10.1 Dataset “connector, clamp connection, at plant”

This dataset here describes the production efforts for the production of clamp connection type connectors for electronics in general. The dataset has a functional unit of “1 kg of connector”, based on the assumption that a typical clamp type connector for 3 wires has an average weight of **9.0 g**.

10.1.1 System Characterization

The datasets represents a clamp connection, used e.g. for the fixation of cables on a printed wiring board or for the fixation of power cables on electronic components. Main data source are various product description schemes from the global producing company ABB (ABB (2006)). The data thus represent the global situation for a 3 wire clamp connection. It can be assumed that for more than 3 wires, the amount of clamp connection is linear proportional to the here reported amount.

10.1.2 Life Cycle Inventory

The dataset for ecoinvent includes the raw materials used as well as the energy consumption for the production of the connector. The different parts of the inventory are based on the following information:

- **Raw materials:** For the total amount, the above mentioned value – based on ABB (2006) – is used here. Due to a lack of respective information, the following split to individual materials is assumed here: 1/3 plastics (thereof 50% as polycarbonate and 50% as polyamide – according to the information in ABB (2006)) and 2/3 metals (100% copper).
- **Auxiliaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** One dataset in a LCI database (CPM (2006)) reports a value for the electricity consumption of 6.51 kWh per kg of produced connectors. Due to a lack of further data sources, this value is used here.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here.
- **Infrastructure and land-use:** Due to the fact that for the production of connectors a less complex infrastructure is requested than for the various components described so far in this report here, the dataset of the printed wiring board mounting plant (see chapter 2 in part II of this report) is used here as proxy.
- **Emissions to air & water. Waste:** Due to a lack of respective information, no such flows are taken into account.

10.1.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an estimation for a 3 wire clamp connection, based on qualitative information in one literature source (ABB (2006)). Hence, the uncertainty values are accordingly high. For the production efforts the data for the production of a not further specified connection from a LCI database (CPM (2006)) is used with an accordingly high uncertainty. The resulting unit process “Connector, clamp connection, at plant” is shown in Tab. 10.2 (on page 175).

10.2 Dataset “connector, PCI bus, at plant”

This dataset here describes the production efforts for the production of internal connectors for computer technology, i.e. the so-called PCI¹⁰ bus. The dataset has a functional unit of “1 kg of connector”, based on the assumption that a typical 64-bit PCI bus connector has an average weight of **11.6 g**.

10.2.1 System Characterization

The datasets represents a 64-bit PCI slot, used for the connection of specific boards (e.g. graphic adapter, sound card) to the main board of a computer. Main data source are own calculations based on the product description schemes of such a 64-bit PCI slot – taken as an average connector of this type here. The data represent thereby the global situation for PCI connections.

¹⁰ **PCI** = Peripheral Component Interconnect

10.2.2 Life Cycle Inventory

The dataset for ecoinvent includes the raw materials used as well as the energy consumption for the production of the connector. The different parts of the inventory are based on the following information:

- **Raw materials:** The total weight (reported above) as well as the weight of the different parts here is based on own calculations. Starting point therefore is the technical specification of the here examined 64-bit PCI slot – containing general information about the used material as well the geometrical dimensions. For this dataset, the material composition is chosen as the following: nylon 66 as insulator – contacts out of brass – coated with 10 um gold.
- **Auxiliaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** One dataset in a LCI database (CPM (2006)) reports a value for the electricity consumption of 6.51 kWh per kg of produced connectors. Due to a lack of further data sources, this value is used here.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here.
- **Infrastructure and land-use:** Due to the fact that for the production of connectors a less complex infrastructure is requested than for the various components described so far in this report here, the dataset of the printed wiring board mounting plant (see chapter 2 in part II of this report) is used here as proxy.
- **Emissions to air & water. Waste:** Due to a lack of respective information, no such flows are taken into account.

10.2.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an estimation of a 64-bit PCI bus connection as typical internal connection type in computers, through calculation based on qualitative information from various producers. Hence, the uncertainty values are accordingly high. For the production efforts the data for the production of a not further specified connection from a LCI database (CPM (2006)) is used with an accordingly high uncertainty. The resulting unit process “Connector, PCI bus, at plant” is shown in Tab. 10.3 (on page 176).

10.3 Dataset “connector, computer, peripheral type, at plant”

This dataset here describes the production efforts for the production of connectors used to plug external device onto a computer (e.g. USB port, serial port, parallel port, etc.). The dataset has a functional unit of “1 kg of connector”, based on the assumption that a peripheral type connector of a computer has a typical weight of **10.0 g**.

10.3.1 System Characterization

The datasets represents the various connectors used currently for the connection of external devices to a personal computer. Main data source are own calculations based on the product description schemes of a serial port – taken as an average connector of this type here – out of ITT (2005a). The data represent thereby the global situation for such a serial port connection.

10.3.2 Life Cycle Inventory

The dataset for ecoinvent includes the raw materials used as well as the energy consumption for the production of the connector. The different parts of the inventory are based on the following information:

- **Raw materials:** The total weight (reported above) as well as the weight of the different parts here is based on own calculations. Starting point therefore is the technical specification of the here examined serial port – containing general information about the used material as well the geometrical dimensions (ITT (2005a)). For this dataset, the material composition is chosen as the following: polyethylene as insulator – contacts out of copper – remaining part out of steel.
- **Auxiliaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** One dataset in a LCI database (CPM (2006)) reports a value for the electricity consumption of 6.51 kWh per kg of produced connectors. Due to a lack of further data sources, this value is used here.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here.
- **Infrastructure and land-use:** Due to the fact that for the production of connectors a less complex infrastructure is requested than for the various components described so far in this report here, the dataset of the printed wiring board mounting plant (see chapter 2 in part II of this report) is used here as proxy.
- **Emissions to air & water. Waste:** Due to a lack of respective information, no such flows are taken into account.

10.3.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an estimation of a serial port connection as a mean peripheral connection, through calculation based on qualitative information from ITT (2005a). Hence, the uncertainty values are accordingly high. For the production efforts the data for the production of a not further specified connection from a LCI database (CPM (2006)) is used with an accordingly high uncertainty. The resulting unit process “Connector, computer, peripheral type, at plant” is shown in Tab. 10.4 (on page 177).

10.4 Dataset “switch, toggle type, at plant”

This dataset here describes the production efforts for the production of toggle switches used in ICT equipment. The dataset has a functional unit of “1 kg of switches”, based on the assumption that a toggle switch has a typical weight of 29 g.

10.4.1 System Characterization

The datasets represents a typical toggle switch. Main data source are own calculations based on the product description scheme of toggle switches in ITT (2005b) – taken as an average switch of this type here. The data represent thereby the global situation for such toggle switches.

10.4.2 Life Cycle Inventory

The dataset for ecoinvent includes the raw materials used as well as the energy consumption for the production of the switch. The different parts of the inventory are based on the following information:

- **Raw materials:** The total weight (reported above) as well as the weight of the different parts here is based on own calculations. Starting point therefore is the technical specification of the here examined toggle switch (ITT (2005b)) – containing general information about the used material as well the geometrical dimensions. For this dataset, the material composition is chosen as the following: case out of glass filled nylon 6/6, housing out of steel, remaining parts out of brass – end contacts further coated with gold and nickel.
- **Auxiliaries & Packaging materials:** Due to a lack of respective information, no such flows are taken into account.
- **Energy:** Due to a lack of respective information, the value for the electricity consumption used for connectors of 6.51 kWh per kg of produced connectors is used here as a first approximation.
- **Transport:** Due to a lack of specific information, standard transport distances and means according to the ecoinvent methodology (see Frischknecht et al. (2007)) are used here.
- **Infrastructure and land-use:** Due to the fact that for the production of switches a less complex infrastructure is requested than for the various components described so far in this report here, the dataset of the printed wiring board mounting plant (see chapter 2 in part II of this report) is used here as proxy.
- **Emissions to air & water. Waste:** Due to a lack of respective information, no such flows are taken into account.

10.4.3 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset represents an estimation of a toggle switch, through calculation based on qualitative information in ITT (2005b). Hence, the uncertainty values are accordingly high. For the production efforts the data for the production of a not further specified connection from a LCI database (CPM (2006)) is used as an estimation with an accordingly high uncertainty. The resulting unit process “Switch, toggle type, at plant” is shown in Tab. 10.5 (on page 178).

Tab. 10.2 Unit process inventory of the dataset “Connector, clamp connection, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
energy for production	connector, clamp connection, at plant	clamp connection	only the electricity consumption reported and included into dataset	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.5E+00	kWh	production information from CPM (2006)	1	1.61	(3,4,3,3,4,5,10); literature value as first approximation used
copper				metals	extraction	No	RER	copper, primary, at refinery	6.67E-01	kg	Data estimated, based on ABB (2006)	1	1.38	(4,5,13,15,3); estimation, based on total weight & qualitative information
Plastics				plastics	polymers	No	RER	polycarbonate, at plant	1.67E-01	kg	Data estimated, based on ABB (2006)	1	1.38	(4,5,13,15,3); estimation, based on total weight & qualitative information
Plastics (II)				plastics	monomers	No	RER	epoxy resin, liquid, at plant	1.67E-01	kg	Data estimated, based on ABB (2006)	1	1.38	(4,5,13,15,3); estimation, based on total weight & qualitative information
Transport (part by lorry)				transport systems	road	No	RER	transport, lorry >4t, fleet average	1.00E-01	t.km	standard distances according to Frischknecht et al. (2007)	1	2.09	(4,5,na,na,na,na,5); standard distances
Transport (part by train)				transport systems	train	No	RER	transport, freight, rail	6.00E-01	t.km	standard distances according to Frischknecht et al. (2007)	1	2.09	(4,5,na,na,na,na,5); standard distances
Infrastructure					printed wiring board mounting plant used as a proxy	electronics	module	Yes	GLO	printed wiring board mounting plant	2.08E-07	unit	rough estimation	1
				electronics	component	No	GLO	connector, clamp connection, at plant	1.00E+00	kg		1	1.05	

Tab. 10.3 Unit process inventory of the dataset “connector, PCI bus, at plant (GLO)”

General Flow information					Representation in ecoinvent						Uncertainty information					
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment		
energy for production	connector, PCI bus, at plant	PCI slot connector	only the electricity consumption reported and included into dataset	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.5E+00	kWh	production information from CPM (2006)	1	1.61	(3,4,3,3,4,5,11); literature value as first approximation used		
brass				metals	extraction	No	CH	brass, at plant	2.03E-01	kg	Data calculated from technical plan of PCI slot	1	138	(4,5,2,3,15,3); estimation, calculated weight, based on qualitative information		
gold				metals	extraction	No	RER	gold, at regional storage	2.05E-03	kg	Data calculated from technical plan of PCI slot	1	138	(4,5,2,3,15,3); estimation, calculated weight, based on qualitative information		
insulation material					assumed to be 100% nylon 66	plastics	polymers	No	RER	nylon 66, at plant	7.95E-01	kg	Data calculated from technical plan of PCI slot	1	138	(4,5,2,3,15,3); estimation, calculated weight, based on qualitative information
Transport (part by lorry)						transport systems	road	No	RER	transport, lorry >6t, fleet average	100E-01	tkm	standard distances according to Frischknecht et al. (2007)	1	2.09	(4,5,na,na,na,na,5); standard distances
Transport (part by train)						transport systems	train	No	RER	transport, freight, rail	6.00E-01	tkm	standard distances according to Frischknecht et al. (2007)	1	2.09	(4,5,na,na,na,na,5); standard distances
infrastructure					printed wiring board mounting plant used as a proxy	electronics	module	Yes	GLO	printed wiring board mounting plant	2.08E-07	unt	rough estimation	1	3.77	(4,5,13,5,4,9); rough estimation of infrastructure

Tab. 10.4 Unit process inventory of the dataset “Connector, computer, peripheral type, at plant (GLO)”

General Flow information					Representation in ecoinvent					Uncertainty information				
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
energy for production	connector, computer, peripheral type, at plant	computer peripheral connector	only the electricity consumption reported and included into dataset	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.5E+00	kWh	production information from CPM (2006)	1	1.61	(3,4,3,3,4,5,11); literature value as first approximation used
copper				metals	extraction	No	RER	copper, primary, at refinery	8.70E-02	kg	Data calculated, based on ITT (2005a)	1	1.38	(4,5,2,3,15,3); estimation, calculated weight, based on qualitative information
steel				metals	extraction	No	RER	steel, low-alloyed, at plant	7.3E-01	kg	Data calculated, based on ITT (2005a)	1	1.38	(4,5,2,3,15,3); estimation, calculated weight, based on qualitative information
Insulation material				plastics	polymers	No	RER	polyethylene, HDPE, granulate, at plant	183E-01	kg	Data calculated, based on ITT (2005a)	1	1.38	(4,5,2,3,15,3); estimation, calculated weight, based on qualitative information
Transport (part by lorry)				transport systems	road	No	RER	transport, lorry >8t, fleet average	100E-01	tkm	standard distances according to Frischknecht et al. (2007)	1	2.09	(4,5,na,na,na,na,5); standard distances
Transport (part by train)				transport systems	train	No	RER	transport, freight, rail	6.0E-01	tkm	standard distances according to Frischknecht et al. (2007)	1	2.09	(4,5,na,na,na,na,5); standard distances
Infrastructure				electronics	module	Yes	GLO	printed wiring board mounting plant	2.08E-07	unit	rough estimation	1	3.77	(4,5,13,5,4,9); rough estimation of infrastructure

Tab. 10.5 Unit process inventory of the dataset “Switch, toggle type, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
energy for production	switch, toggle type, at plant	toggle switch	estimation, from the connector production	electricity	production mix	No	UCTE	electricity, medium voltage, production UCTE, at grid	6.5E+00	kWh	rough estimation	1	168	(4.5,13.4,5.1); estimation, based on literature value for a different type of component
brass			metals	extraction	No	CH	brass, at plant	4.49E-01	kg	Data calculated, based on ITT (2005b)	1	138	(4.52,3,15,3); estimation, calculated weight, based on qualitative information	
steel			metals	extraction	No	RER	steel, low-alloyed, at plant	2.52E-01	kg	Data calculated, based on ITT (2005b)	1	138	(4.52,3,15,3); estimation, calculated weight, based on qualitative information	
copper			metals	extraction	No	RER	copper, primary, at refinery	4.02E-02	kg	Data calculated, based on ITT (2005b)	1	138	(4.52,3,15,3); estimation, calculated weight, based on qualitative information	
nickel			metals	extraction	No	GLO	nickel, 99.5% at plant	4.10E-04	kg	Data calculated, based on ITT (2005b)	1	138	(4.52,3,15,3); estimation, calculated weight, based on qualitative information	
gold			metals	extraction	No	RER	gold, at regional storage	4.10E-04	kg	Data calculated, based on ITT (2005b)	1	138	(4.52,3,15,3); estimation, calculated weight, based on qualitative information	
Insulation material			plastics	polymers	No	RER	nylon 66, glass-filled, at plant	2.58E-01	kg	Data calculated, based on ITT (2005b)	1	138	(4.52,3,15,3); estimation, calculated weight, based on qualitative information	
Transport (part by lorry)			transport systems	road	No	RER	transport, lorry >8t, fleet average	100E-01	tkm	standard distances according to Frischknecht et al. (2007)	1	2.09	(4.5,na,na,na,na,5); standard distances	
Transport (part by train)			transport systems	train	No	RER	transport, freight, rail	6.00E-01	tkm	standard distances according to Frischknecht et al. (2007)	1	2.09	(4.5,na,na,na,na,5); standard distances	
Infrastructure			electronics	module	Yes	GLO	printed wiring board mounting plant	2.08E-07	unt	rough estimation	1	3.77	(4.5,13.5,4.9); rough estimation of infrastructure	1

11 Unspecified datasets for electronic components

For all those cases where not enough information is given about the type of electronic components used, a serie of unspecified datasets is established here as well. Depending on the degree of information available, the most appropriate dataset out of this serie can ben chosen. All these datasets have as functional unit “1 kg of electronic components”.

Starting point for the various datasets established in this chapter here are the data about the composition of printed wiring boards in various electronic devices – i.e. a television (data taken from Behrendt et al. (1998)), a motherboard of a desktop computer and a motherboard of a laptop computer (in both cases the here developed respective datasets – see chapter 2, part II fo this report – are used as starting point). Tab. 11.1 shows all these here used data.

Tab. 11.1 Data used as starting point for the calculation of the unspecified datasets for electronic components (sources: see text)

TV (Behrendt, 1998)	[%]	Data Desktop PC	[%]	Data Laptop PC	[%]
		connectors, PCI bus type	28.0%	connectors, PCI bus type	8.4%
		connectors, peripheral type	9.3%	connectors, peripheral type	11.3%
		tantalum capacitor	7.0%	tantalum capacitor	0.0%
		electrolyt capacitor	27.8%	electrolyt capacitor	0.0%
		capacitors, SMD type	1.7%	capacitors, SMD type	7.1%
capacitors, unspecified	21.0%				
transistors	2.6%	wired transistors	7.0%	transistors, SMD type	2.1%
inductors / transformers	65.5%	coils	1.7%	coils	0.0%
IC, unspecified	5.3%	IC, logic type	12.4%	IC, logic type	37.3%
		IC, memory type	3.1%	IC, memory type	27.8%
resistors, unspecified	4.7%	resistors, SMD type	1.7%	resistors, SMD type	5.0%
diodes, unspecified	0.9%	diodes, SMD type	0.3%	diodes, SMD type	0.8%
		LED	0.1%	LED	0.3%

In detail, the following unspecified datasets have been established in the framework of this study here:

- **electronic component, passive, unspecified, at plant:** unweighted average between the two following datasets of passive components: connectors, capacitors, inductors and resistors – for all those cases, where passive components are not further specified.
- **electronic component, active, unspecified, at plant:** unweighted average between the two following datasets of active components: diodes, transistors and integrated circuits – for all those cases, where active components are not further specified.
- **electronic component, unspecified, at plant:** unweighted average between the two other, here described datasets about unspecified active and unspecified passive components – for all those cases, where no information at all about the type of components is available.

The respective meta information for these three datasets can be found in Tab. 11.2.

Tab. 11.2 Ecoinvent meta information for the processes for transformers within the database.

Name	electronic component, unspecified, at plant	electronic component, passive, unspecified, at plant	electronic component, active, unspecified, at plant
Location	GLO	GLO	GLO
Infrastructure Process	0	0	0
Unit	kg	kg	kg
Dataset Version	2.0	2.0	2.0
Included Processes	This dataset covers material input (and within this the respective production efforts) for unspecified electronic components.	This dataset covers material input (and within this the respective production efforts) for unspecified passive electronic components.	This dataset covers material input (and within this the respective production efforts) for unspecified active electronic components.
Amount	1	1	1
Local Name	Elektronisches Bauteil, unspezifisch, ab Werk	Passives elektronisches Bauteil, unspezifisch, ab Werk	Aktives elektronisches Bauteil, unspezifisch, ab Werk
Synonyms	-	-	-
General Comment	This module represents an unweighted average of all components on a desktop motherboard, a laptop mainboard and a CRT TV board.	This module represents an unweighted average of all passive components on a desktop motherboard, a laptop mainboard and a CRT TV board.	This module represents an unweighted average of all active components on a desktop motherboard, a laptop mainboard and a CRT TV board.
Start Date	1994	1994	1994
End Date	2007	2007	2007
Data valid for entire period	1	1	1
Geography Text	Data represent a global valuable average composition of active and passive electronic components.	Data represent a global valuable average composition of passive electronic components.	Data represent a global valuable average composition of active electronic components.
Technology Text	see general comments	see general comments	see general comments
Representativeness	-	-	-
Production Volume	-	-	-
Sampling Procedure	Literature & own estimations	Literature & own estimations	Literature & own estimations
Extrapolations	see Geography & Technology	see Geography & Technology	see Geography & Technology
Uncertainty Adjustments	None	None	None

11.1 Dataset “electronic component, passive, unspecified, at plant”

This dataset here represents an average composition of passive electronic components, based on the composition of three different types of printed wiring board (see Tab. 11.1) for all those cases, where passive components are not further specified. The dataset has a functional unit of “1 kg of passive components”.

11.1.1 System Characterization & Life Cycle Inventory

The datasets represents an average composition of passive components as it can be found on printed wiring boards in television devices, desktop computers and laptop computers. The following in Tab. 11.1 listed components are passive components and thus are taken into account here: connectors, capacitors, inductors and resistors.

Calculating the average from the three printed wiring boards in Tab. 11.1, results in the distribution for the passive components that is used here in order to establish this unspecified dataset (the used specific datasets for the respective components are mentioned here as well) – shown in Tab. 11.3.

Tab. 11.3 Composition of the unspecified dataset for passive components (data represent average composition of PWB from television, desktop computer and laptop computer)

Used dataset	[%]
connectors, PCI bus, at plant	18.2%
connectors, computer, peripheral type, at plant	10.3%
capacitor, Tantalum-, through-hole mounting, at plant	3.5%
capacitor, electrolyte type, < 2cm height, at plant	13.9%
capacitors, SMD type, surface-mounting, at plant	4.4%
capacitors, unspecified, at plant	10.5%
inductor, ring core choke type, at plant	33.6%
resistor, unspecified, at plant	5.7%

11.1.2 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset here represents a weighted average of various passive components shown in this database – weighted on the basis of three different types of printed wiring board. Nevertheless, the uncertainty values are rather high.

The resulting unit process “electronic component, passive, unspecified, at plant” is shown in Tab. 11.5 (on page 183).

11.2 Dataset “electronic component, active, unspecified, at plant”

This dataset here represents an average composition of active electronic components, based on the composition of three different types of printed wiring board (see Tab. 11.1) for all those cases, where active components are not further specified. The dataset has a functional unit of “1 kg of active components”.

11.2.1 System Characterization & Life Cycle Inventory

The datasets represents an average composition of active components as it can be found on printed wiring boards in television devices, desktop computers and laptop computers. The following in Tab. 11.1 listed components are active components and thus are taken into account here: transistors, integrated circuits, and diodes (including light emitting diode, LED).

Calculating the average from the three printed wiring boards in Tab. 11.1, results in a distribution for the active components that is used here in order to establish this unspecified dataset (the used specific datasets for the respective components are mentioned here as well) – shown in Tab. 11.4.

Tab. 11.4 Composition of the unspecified dataset for active components (data represent average composition of PWB from television, desktop computer and laptop computer)

Used dataset	[%]
transistor, unspecified, at plant	11.7%
integrated circuit, IC, logic type, at plant	55.0%
integrated circuit, IC, memory type, at plant	30.9%
diode, unspecified, at plant	2.0%
light emitting diode, LED, at plant	0.4%

11.2.2 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset here represents a weighted average of various active components shown in this database – weighted on the basis of three different types of printed wiring board. Nevertheless, the uncertainty values are rather high.

The resulting unit process “electronic component, active, unspecified, at plant” is shown in Tab. 11.6 (on page 184).

11.3 Dataset “electronic component, unspecified, at plant”

This dataset here represents an average composition of active and passive electronic components, based on the composition of three different types of printed wiring board (see Tab. 11.1) for all those cases, where these components are not further specified. The dataset has a functional unit of “1 kg of components”.

The datasets represents an average composition of active and passive components as it can be found on printed wiring boards in television devices, desktop computers and laptop computers. From the data in Tab. 11.1 results a mixture of 67% passive and 33% active components – mixture that is used here together with the two preceding datasets (for unspecific passive resp. active components).

11.3.1 Data Uncertainty and Input Data for ecoinvent Database

The uncertainty scores established according to the method used in the ecoinvent project (see Frischknecht et al. (2007)) include reliability, completeness, temporal correlation, geographical correlation, further technological correlation and sample size. The dataset here represents a weighted average of various components shown in this database – weighted on the basis of three different types of printed wiring board. Nevertheless, the uncertainty values are rather high.

The resulting unit process “electronic component, unspecified, at plant” is shown in Tab. 11.7 (on page 184).

11. Unspecified datasets for electronic components

Tab. 11.5 Unit process inventory of the dataset “electronic component, passive, unspecified, at plant (GLO)”

General Flow information					Representation in ecoinvent						Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Location	Modul name in ecoinvent	Mean value	Unit	Source	Type	StDv 95%	General Comment
connector, PCI bus type	electronic component, passive, unspecified, at plant	passive component, unspecified		electronics	component	No	GLO	connector, PCI bus, at plant	182E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
connector, peripheral type				electronics	component	No	GLO	connector, computer, peripheral type, at plant	103E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
tantalum capacitor				electronics	component	No	GLO	capacitor, Tantalum-, through-hole mounting, at plant	3.48E-02	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
electrolyt capacitor				electronics	component	No	GLO	capacitor, electrolyte type, < 2cm height, at plant	139E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
SMD type capacitor				electronics	component	No	GLO	capacitor, SMD type, surface-mounting, at plant	4.38E-02	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
unspecified capacitor				electronics	component	No	GLO	capacitor, unspecified, at plant	105E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
coils & inductors				electronics	component	No	GLO	inductor, ring core choke type, at plant	3.36E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
resistor mix				electronics	component	No	GLO	resistor, unspecified, at plant	5.67E-02	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
				electronics	component	No	GLO	electronic component, passive, unspecified, at plant	100E+00	kg		1		

Tab. 11.6 Unit process inventory of the dataset “electronic component, active, unspecified, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	Stdv 95%	General Comment
transistor mix	electronic component, active, unspecified, at plant	active component, unspecified		electronics	component	No	GLO	transistor, unspecified, at plant	1.70E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
IC, logic type				electronics	component	No	GLO	integrated circuit, IC, logic type, at plant	5.50E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
IC, memory type				electronics	component	No	GLO	integrated circuit, IC, memory type, at plant	3.09E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
Diodes mix				electronics	component	No	GLO	diode, unspecified, at plant	1.96E-02	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
LEDs				electronics	component	No	GLO	light emitting diode, LED, at plant	3.75E-03	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
				electronics	component	No	GLO	electronic component, active, unspecified, at plant	1.00E+00	kg		1		

Tab. 11.7 Unit process inventory of the dataset “electronic component, unspecified, at plant (GLO)”

General Flow information				Representation in ecoinvent							Uncertainty information			
Input	Process Name	Output	Remarks	Category	Sub-category	Infra-structure	Loca-tion	Modul name in ecoinvent	Mean value	Unit	Source	Type	STdv 95%	General Comment
passive component	electronic component, unspecified, at plant	unspecified component		electronics	component	No	GLO	electronic component, passive, unspecified, at plant	6.70E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
active component				electronics	component	No	GLO	electronic component, active, unspecified, at plant	3.30E-01	kg	average of PC motherboard, laptop motherboard and a TV board	1	2.29	(53.2,155.3); calculated from Desktop PC, Laptop and TV mainboards
				electronics	component	No	GLO	electronic component, unspecified, at plant	1.00E+00	kg		1		

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